Enterprise-scale DBMS and SQL basics

Week 8

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Week 8 Outline

1. Introduction
2. Enterprise-scale DBMS
3. Cloud Models (IaaS, PaaS, SaaS)
4. Provisioning Azure SQL
5. SQL Basics
6. Security in Azure – RBAC (Role-Based Access Control)
7. Week 8 Tutorial Activity (Step-by-Step)
8. Review & Reflection
9. Discussion & Role-Play Questions

Introduction

- This week, we move from small-scale (like Access) to enterprise-scale databases (like SQL Server & Azure SQL).
- Analogy: *Think of Access like a family car (good for small trips), and Azure SQL like a train network (handles thousands of passengers daily).*
- By the end, you should:
  1. Understand enterprise DBMS basics.
  2. Provision a relational database in Azure.
  3. Write simple SQL queries.

Enterprise-scale DBMS

- Enterprise DBMS = handles very large data, many users, high security
- Used in banks, airlines, hospitals, universities
- MS Access → simple, desktop-based
- Azure SQL → cloud-based, scalable

Enterprise-scale DBMS

Comparison Table

Feature	MS Access (Small-scale)	Azure SQL (Enterprise)
Data Size	Small (MBs–GBs)	Huge (TBs–PBs)
Users	1–10	1000s
Security	Basic	Advanced (RBAC + encryption)
Cost	One-time	Subscription (monthly)
Example (Aus)	Café loyalty program	Qantas booking system

Enterprise-scale DBMS

- **Access DB:** You'll see the file size is small and usually only supports a handful of users.
- **Scaling to Woolworths:** Problems would include slow performance, frequent crashes, limited security, and no ability to handle thousands of users or massive data.
- **Discussion point:** Woolworths would never run on Access because it needs a system that can handle millions of transactions daily, strict security, and high availability — things only enterprise systems like Azure SQL or Oracle can provide.

Cloud Models (IaaS, PaaS, SaaS)

IaaS, PaaS, SaaS – Simple Explanation

- **IaaS:** You rent a server. You manage everything yourself.
- **PaaS:** You get a managed platform. Example: **Azure SQL** (Microsoft handles backups & security).
- **SaaS:** Software delivered to you, ready to use. Example: **Gmail, Office 365**.

Cloud Models (IaaS, PaaS, SaaS)

- IaaS = Infrastructure as a Service
- PaaS = Platform as a Service
- SaaS = Software as a Service

Cloud Models (IaaS, PaaS, SaaS)

IaaS (Infrastructure as a Service)

You rent the basic building blocks (servers, storage, network)

You control almost everything

- You manage: OS, apps, data
- Provider manages: hardware

Think: “I get the computer, I do the rest”

Cloud Models (IaaS, PaaS, SaaS)

PaaS (Platform as a Service)

You get a ready-made environment to build/run apps

- You manage: your app + data
- Provider manages: OS, runtime, updates

Think: “I just build my app”

Cloud Models (IaaS, PaaS, SaaS)

SaaS (Software as a Service)

You just use the software

- You manage: nothing technical
- Provider manages: everything

Think: “I just use it”

Cloud Models (IaaS, PaaS, SaaS)

Easy memory trick

- IaaS → Infrastructure → lowest level
- PaaS → Platform → middle
- SaaS → Software → highest (user level)

Cloud Models (IaaS, PaaS, SaaS)

IaaS = control everything

PaaS = build apps only

SaaS = just use the software

Cloud Models (IaaS, PaaS, SaaS)

Analogy

- **IaaS =** renting a block of land & building your own house.
- **PaaS =** renting an apartment with services included.
- **SaaS =** Airbnb – just move in and use it, no management.

Cloud Models (IaaS, PaaS, SaaS)

Discussion Question)

Classify these: Office 365, Azure SQL, Google Drive → IaaS, PaaS, SaaS

Model Answer:

- Office 365 → SaaS (ready-to-use software)
- Azure SQL → PaaS (managed database platform)
- Google Drive → SaaS (software/storage service you just use)

Provisioning Azure SQL

Provisioning = setting up a cloud service.

Azure SQL steps (overview):

- Choose subscription (student/edu).
- Pick/create a resource group.
- Database name = StudentIDName.
- Admin login + password.
- Region = *Australia East (Sydney)*.

Example

- DB Name: A00127870_SajanDB
- Resource Group: StudentRG2025

SQL Basics

SQL = Structured Query Language

- The common language all databases understand.
- You use it to ask for information.

Example Queries

- SELECT * FROM Customers; → shows *all* details of all customers.
- SELECT FirstName, LastName FROM Employees WHERE Department='IT'; → shows only names of employees in IT.

SQL Basics

Analogy

SQL is like ordering food in a restaurant:

- You ask (query) → “I want a pizza.”
- The kitchen (database) prepares and delivers your order.

Security in Azure – RBAC (Role-Based Access Control)

- Security Principal = Who → e.g., user, student, staff
- Role = What they can do → read, write, delete
- Scope = Where → database, table, or resource

Example

A bank teller can *view customer accounts* but cannot *change system passwords*.

Security in Azure – RBAC (Role-Based Access Control)

Hands-on Debate

Question: Should students have full admin rights on Uni DBs?

Answer:

No — giving students full admin rights is risky. They might:

- Accidentally delete or change important data
- Access private information
- Cause security or system issues

Security in Azure – RBAC (Role-Based Access Control)

Better approach:

- Students should only get the role and scope they need (e.g., read-only access for learning).
- Admin rights stay with IT staff to keep the system safe.

Week 8 Tutorial Activity (Step-by-Step)

Week 8+ uses Virtual Lab in browser, no install needed.

- **Topic:** Exploring Azure SQL Database
- **Focus:** How to create (provision), secure, and query a cloud database

Week 8 Tutorial Activity (Step-by-Step)

- **Why important?**
  - Real-world companies use cloud DBs (banks, airlines, supermarkets)
  - Same SQL language works in Access *and* Azure
  - Learn how to scale from small desktop DBs → enterprise cloud systems

Week 8 Tutorial Activity (Step-by-Step)

Learning Goals for Week 8

- Understand how to provision an Azure SQL Database
- Practice SQL queries in the cloud
- Compare Access vs Azure results
- Apply security basics (RBAC roles & scope)

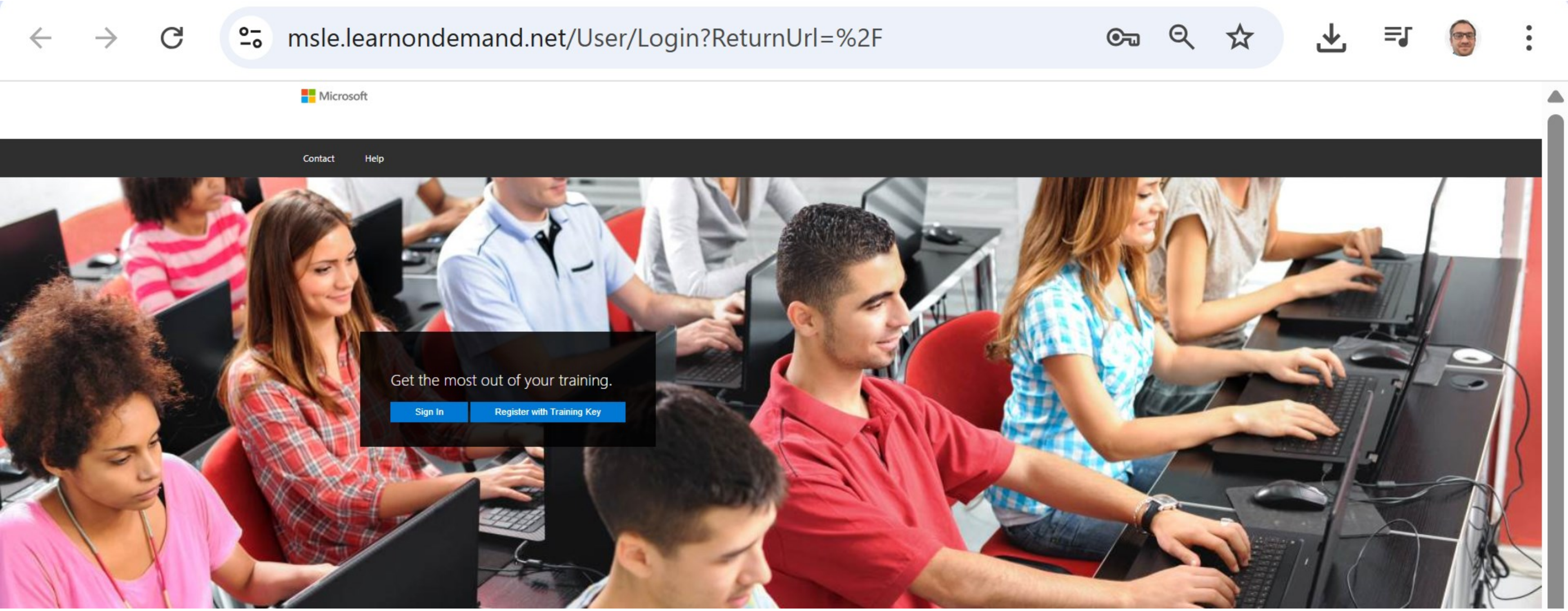
Analogy

Running queries in Access is like using a calculator.

Running queries in Azure is like using a supercomputer that many people can share safely.

Week 8 Tutorial Activity (Step-by-Step)

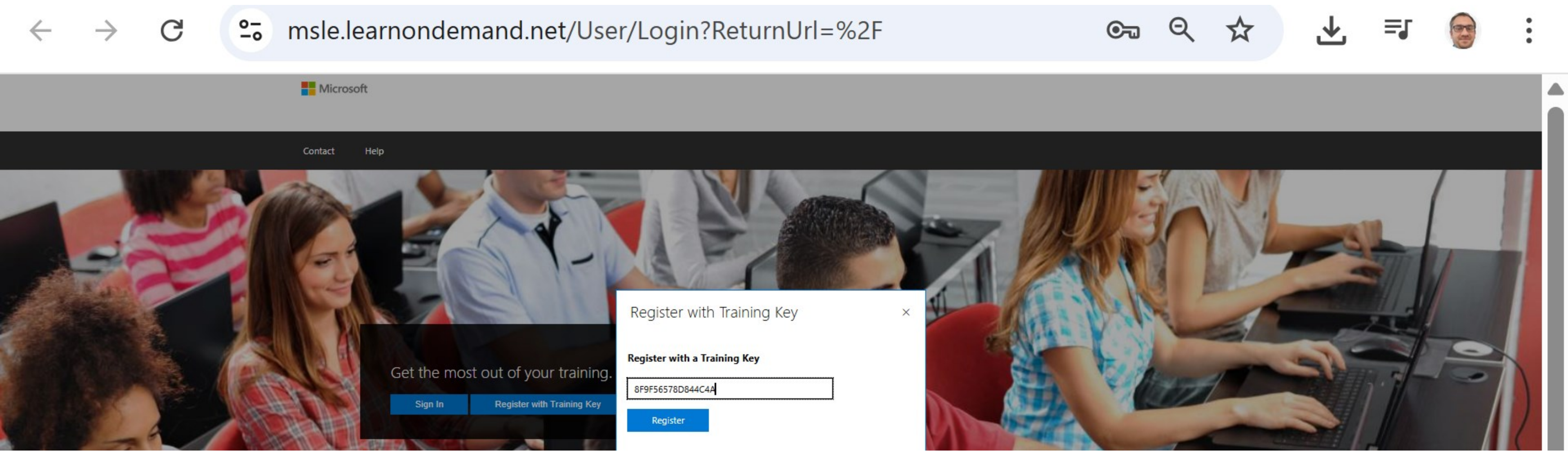
- Go to https://msle.learnondemand.net/



Open this page in your browser

Week 8 Tutorial Activity (Step-by-Step)

- Register with key: 39595D4F29DCCDC7



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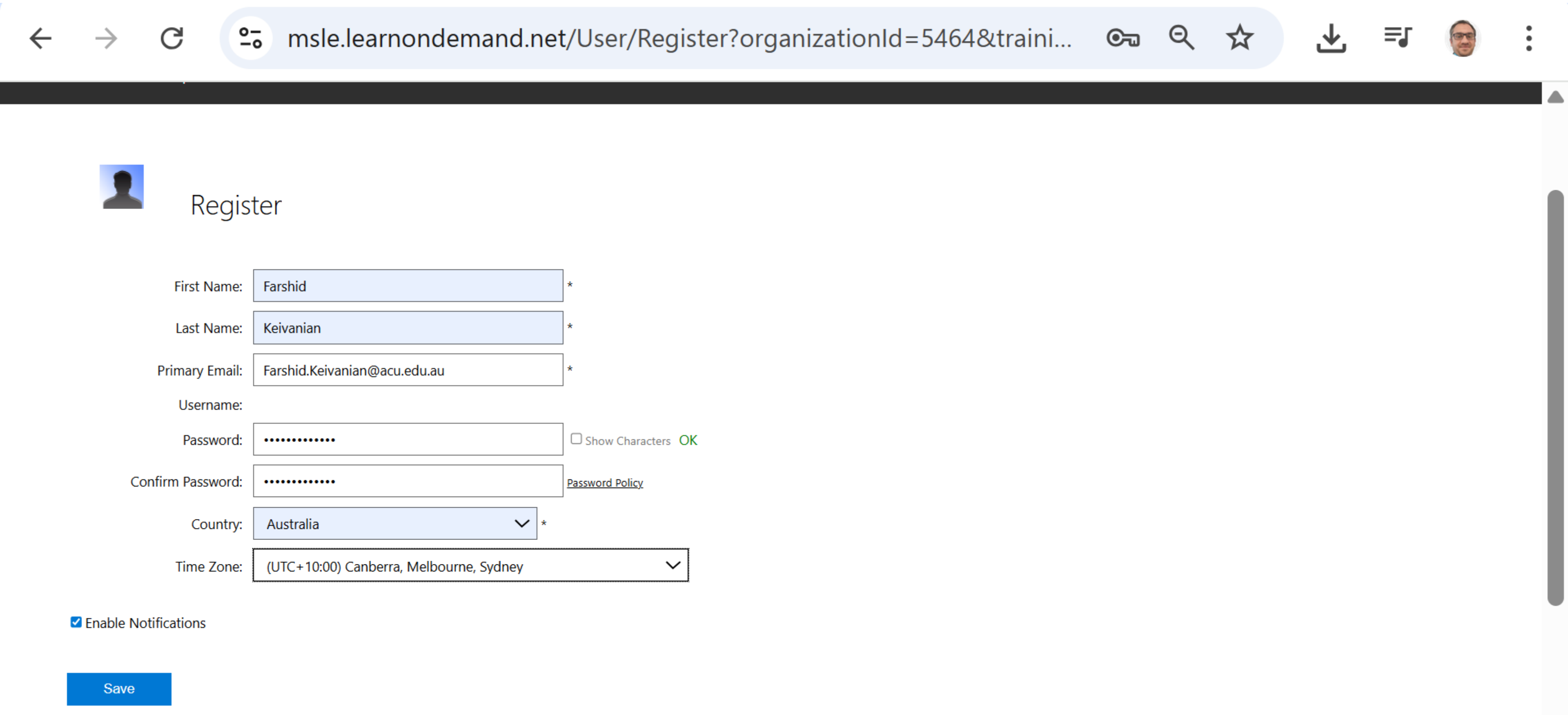
Week 8 Tutorial Activity (Step-by-Step)

- Fill the form

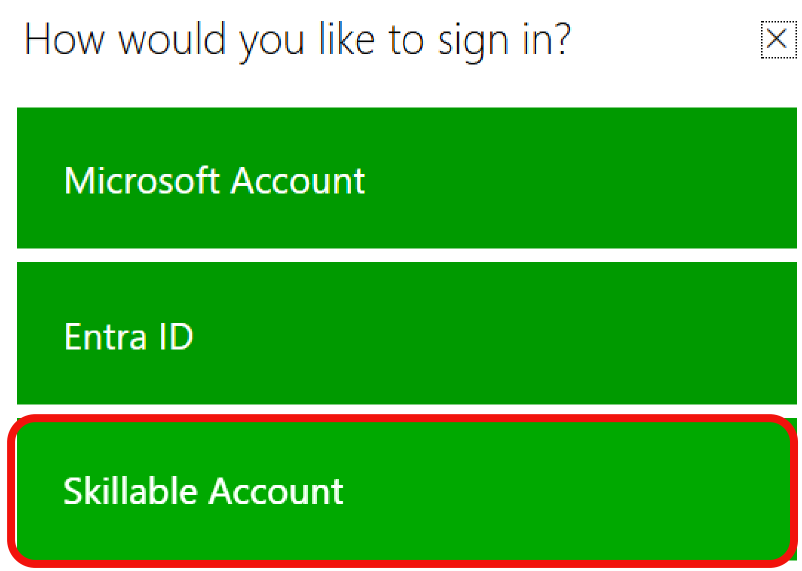


Open this page in your browser

Week 8 Tutorial Activity (Step-by-Step)

- If already registered and you want to log in

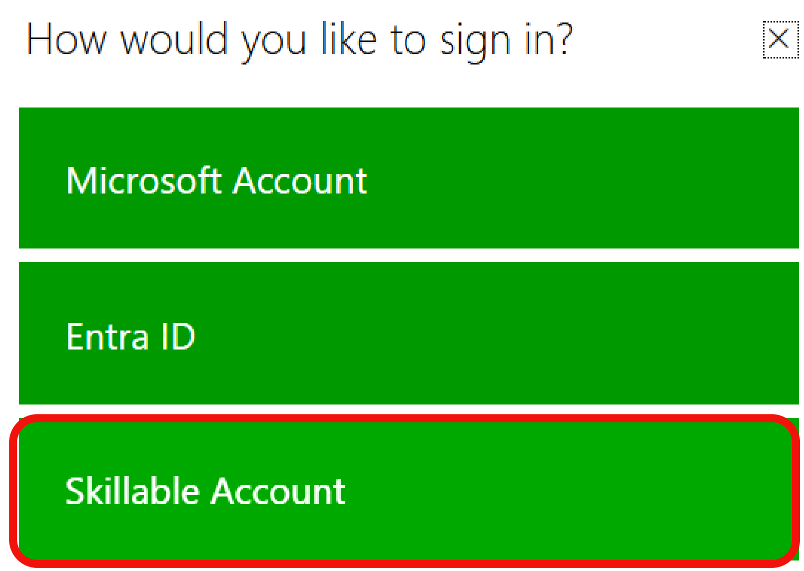
1. On the popup (as in your screenshot), choose Skillable Account.
2. Enter the username/email and password you created when you first registered with the key.
3. Once logged in, you'll see your training dashboard → then launch Explore Azure SQL Database.



Week 8 Tutorial Activity (Step-by-Step)

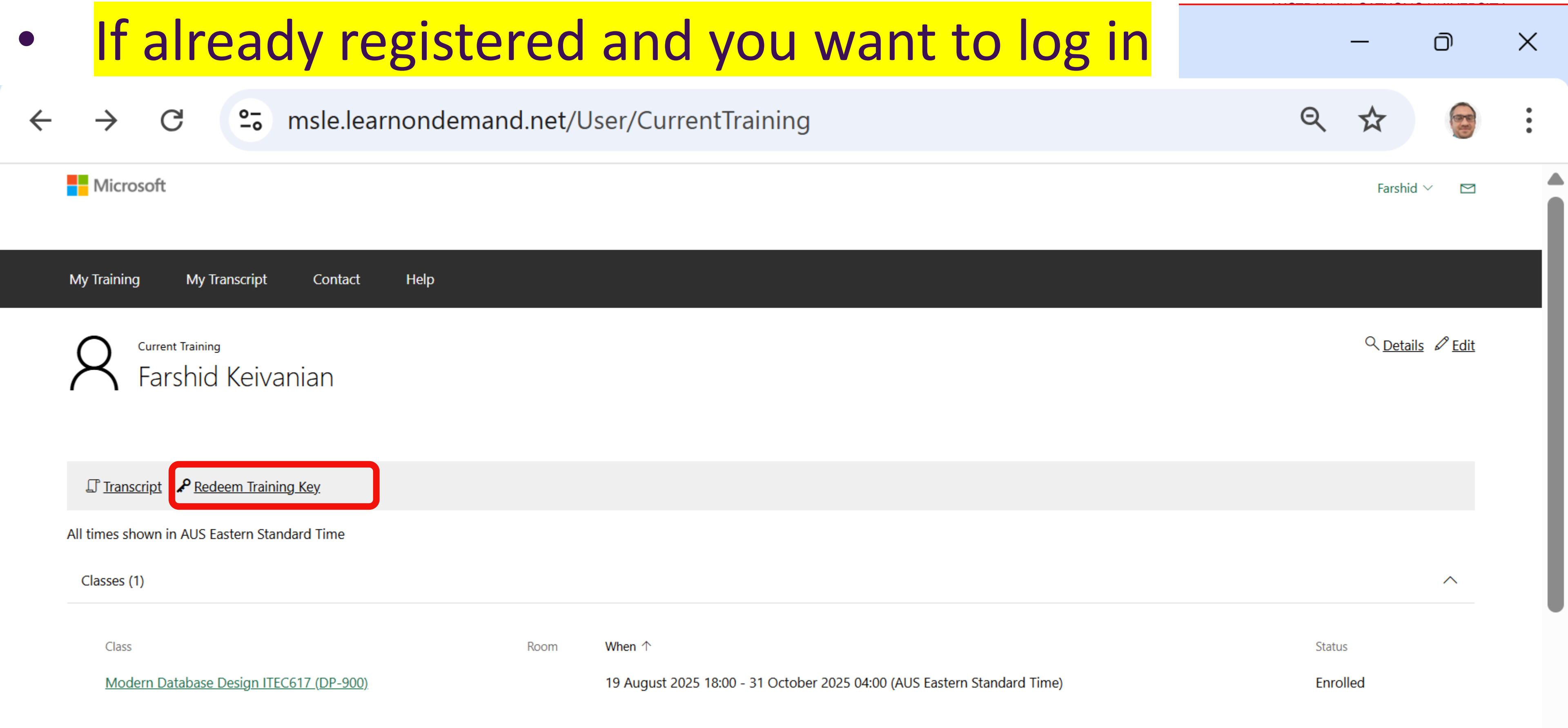
- If already registered and you want to log in

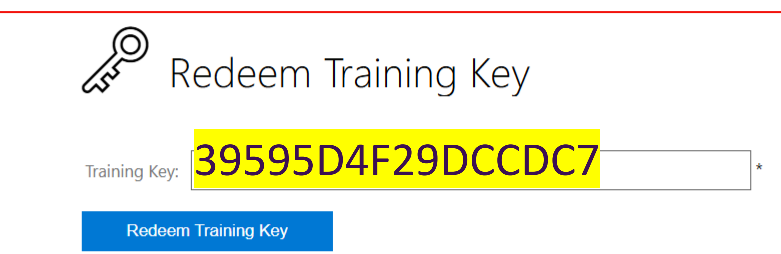
  Don’t pick Microsoft Account or Entra ID unless your lab instructions specifically require it — those are for enterprise logins.



Week 8 Tutorial Activity (Step-by-Step)

- If already registered and you want to log in

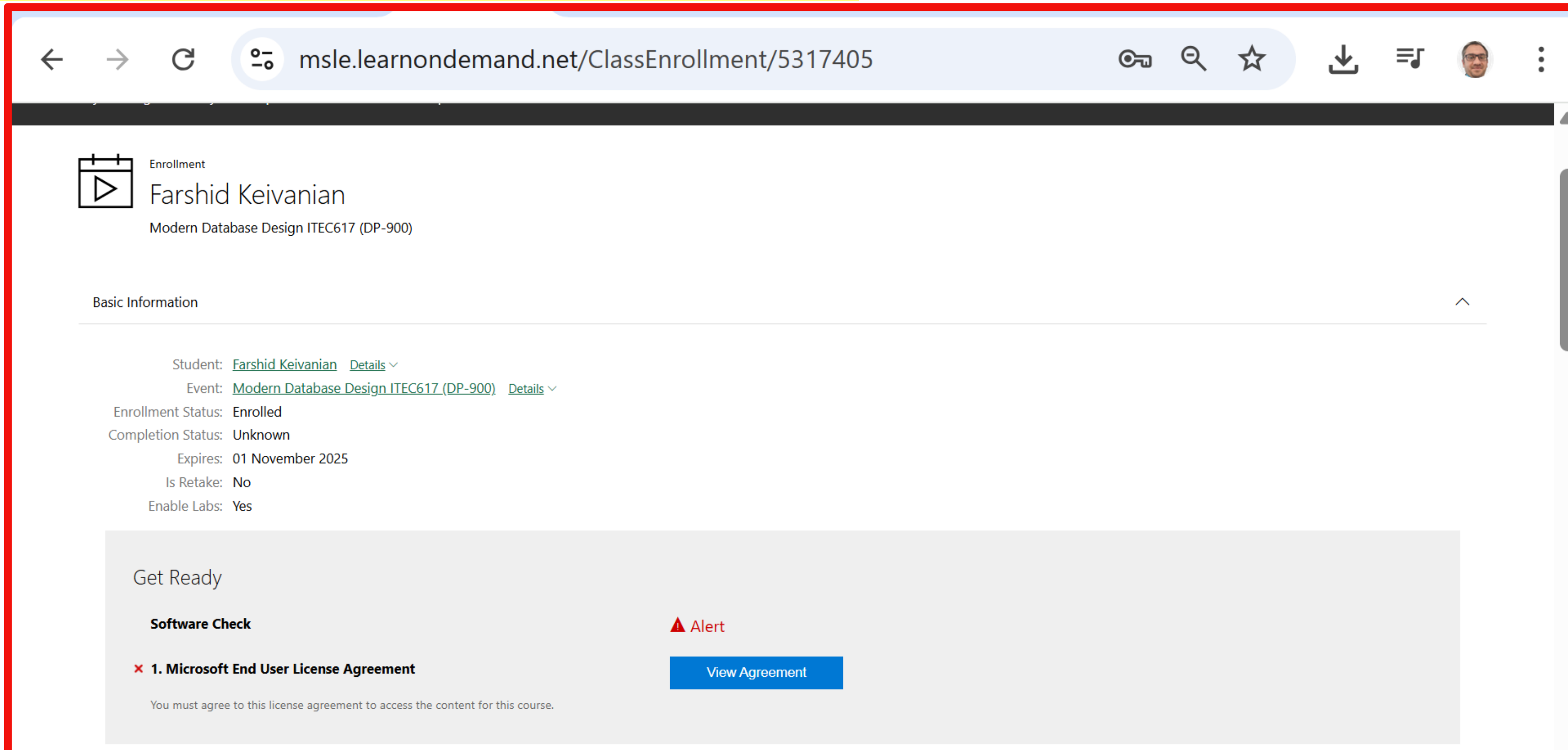




Open this page in your browser

Week 8 Tutorial Activity (Step-by-Step)

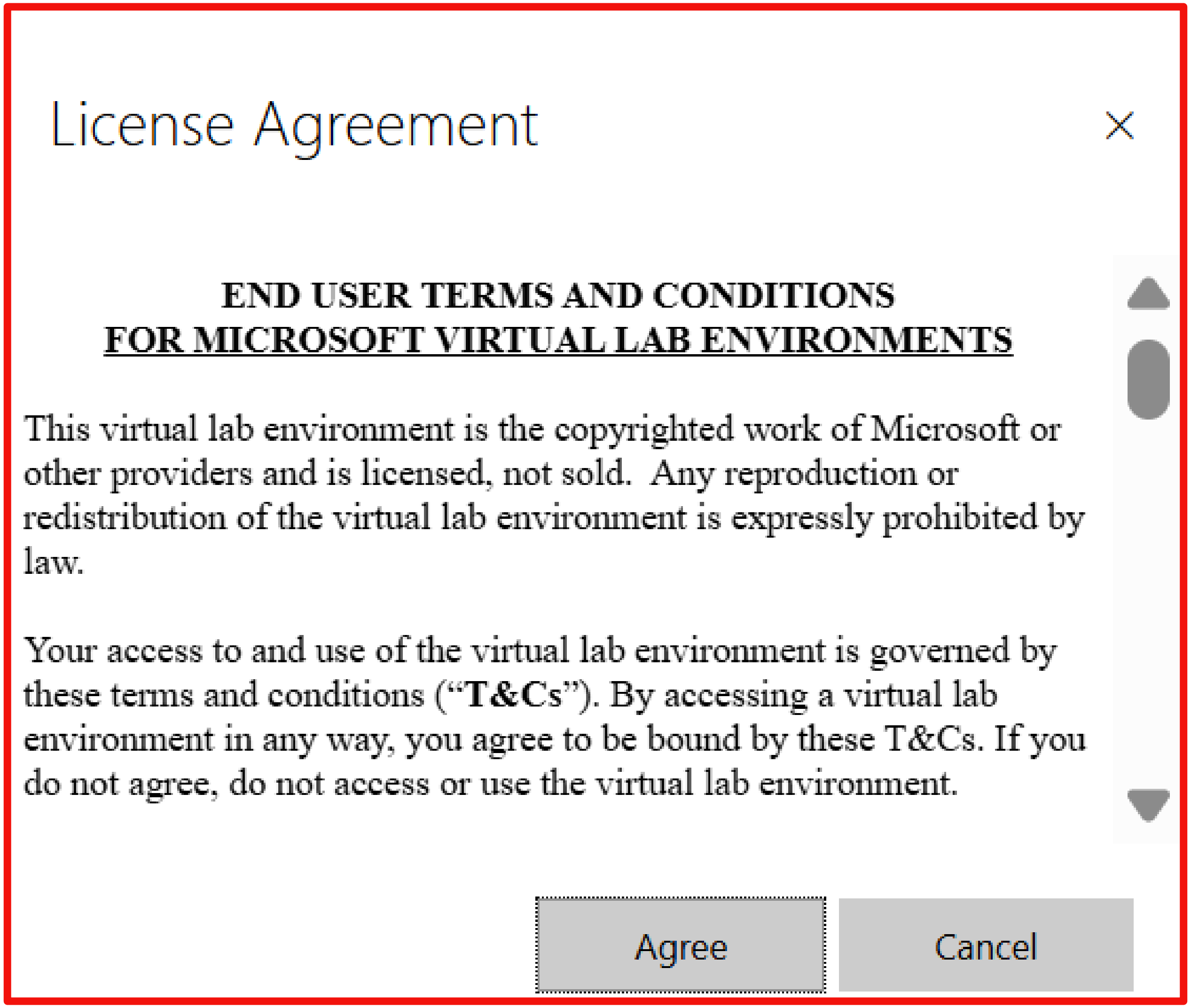
- View Agreement (either you just registered or logged in, you will see this page)



Open this page in your browser

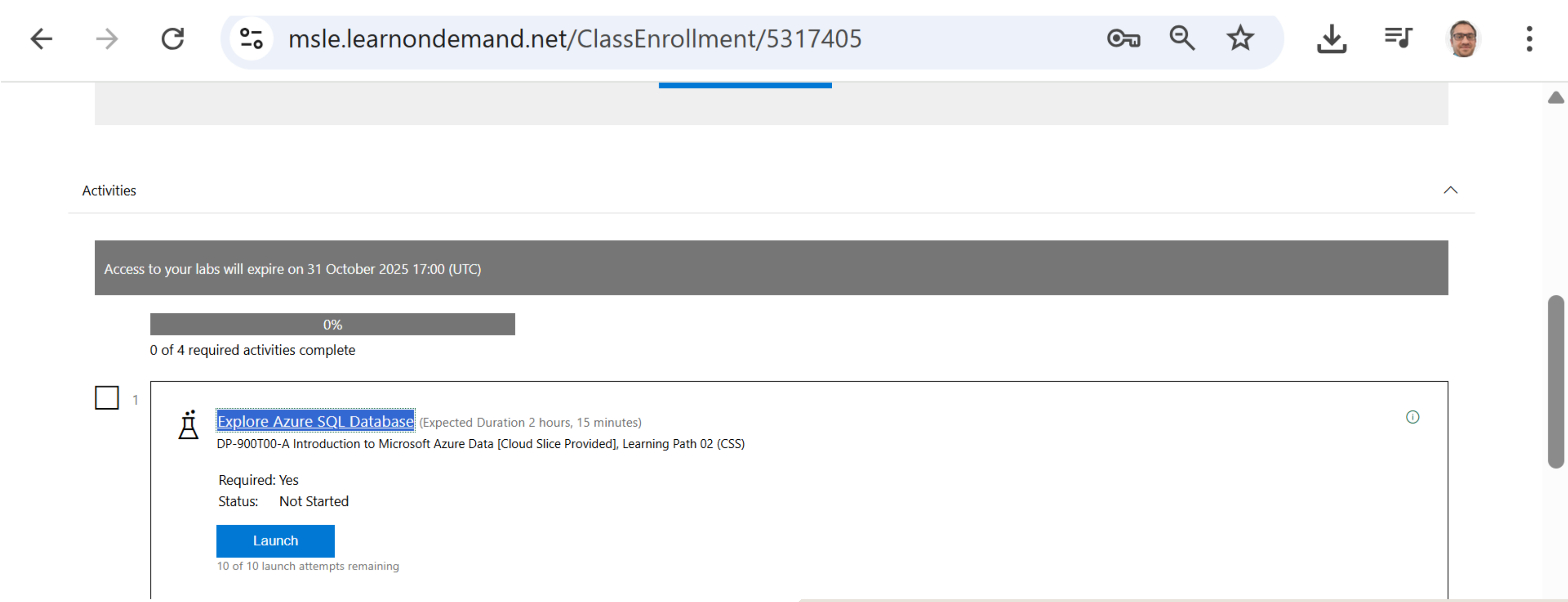
Week 8 Tutorial Activity (Step-by-Step)

- Accept the license (Agree)

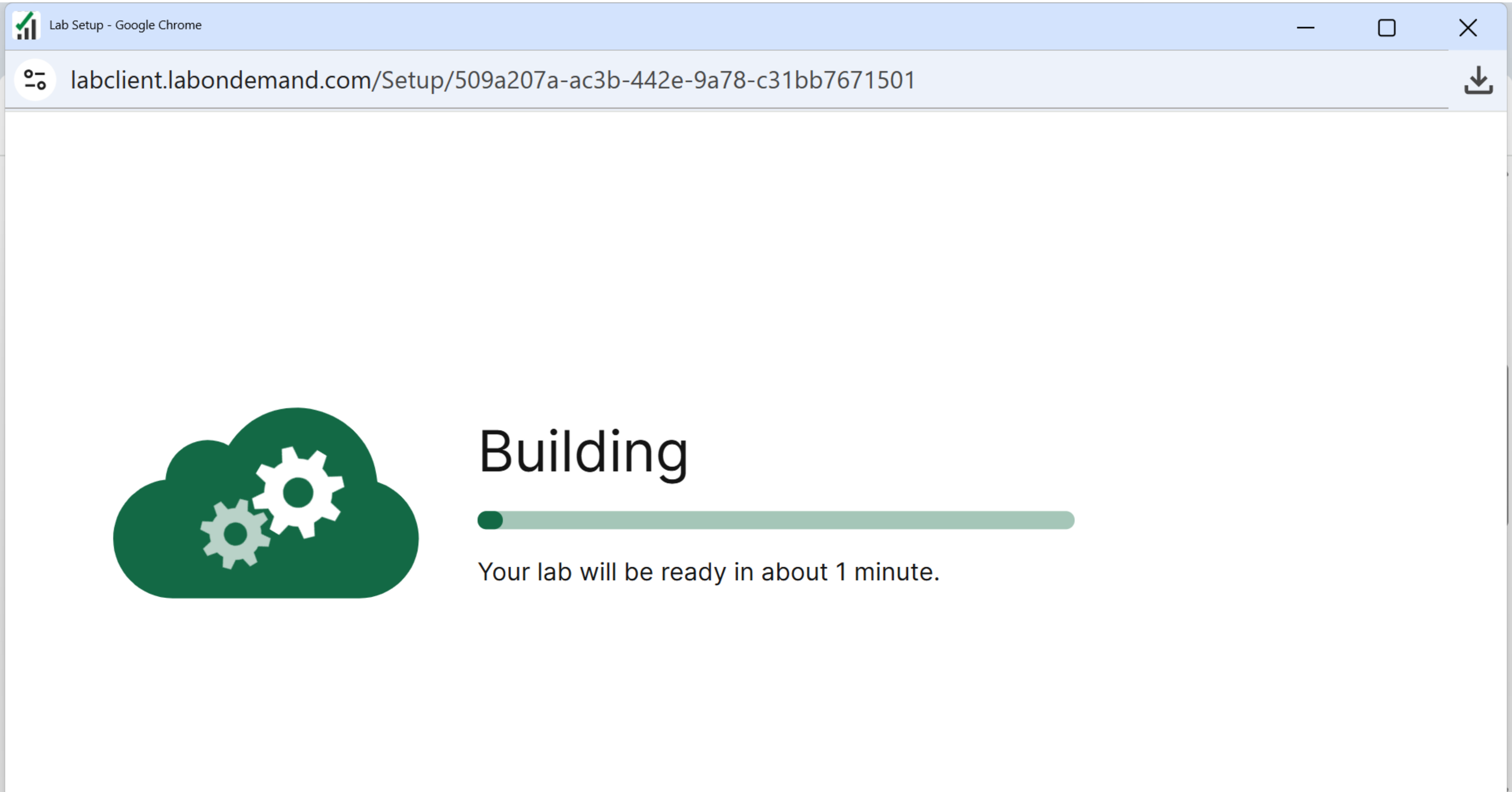


Week 8 Tutorial Activity (Step-by-Step)

- Launch the lab “Explore Azure SQL Database”
- Use the lab’s Azure username/password to sign in to the Azure Portal.
- Click Start → Open Azure Portal and sign in with those credentials.



Week 8 Tutorial Activity (Step-by-Step)



Week 8 Tutorial Activity (Step-by-Step)

Important:

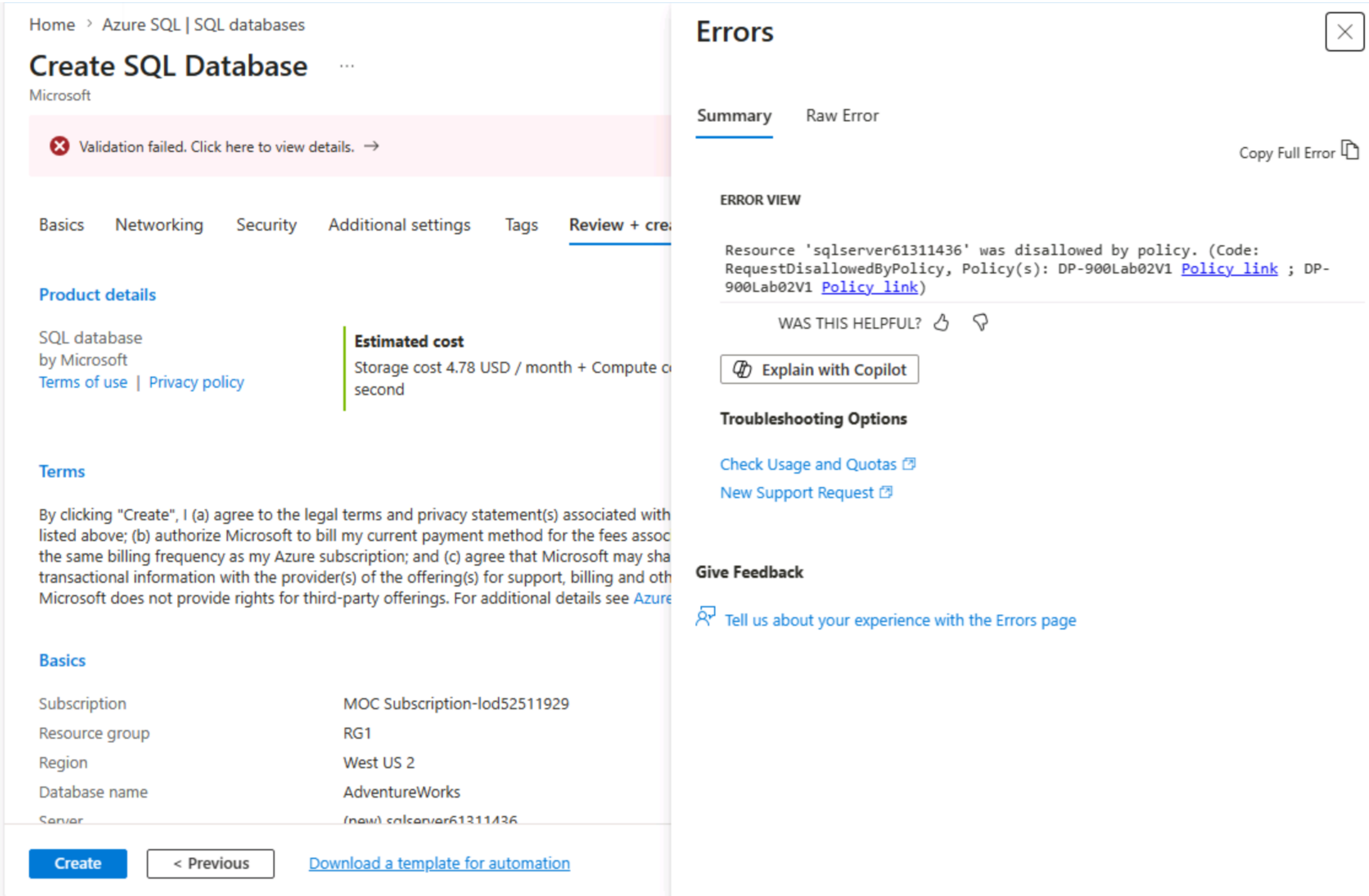
Follow the on-screen instructions in the right-hand side window of the virtual lab to log in and complete the tasks as instructed.

Week 8 Tutorial Activity (Step-by-Step)

Important: Do not open Azure Portal on your browser, instead run the virtual machine and then open the Azure Portal on the virtual machine's browser

Week 8 Tutorial Activity (Step-by-Step)

Common Lab Issue: Azure Policy Validation Error



Week 8 Tutorial Activity (Step-by-Step)

What does it mean?

- You are working in a controlled lab environment
- Azure applies policies (rules) to restrict configurations
- If settings don't match lab requirements → validation fails

Week 8 Tutorial Activity (Step-by-Step)

Why this happens

- Incorrect region (e.g., West US 2 instead of West US)
- Wrong or random server name
- Not using the required resource group (RG1)
- Ignoring instructions on the right-hand lab panel

Week 8 Tutorial Activity (Step-by-Step)

How to fix it

- ✓ Use server name provided on the right side window
- ✓ Avoid double click on server name (only one click)
- ✓ Match region EXACTLY as shown in lab instructions
- ✓ Use RG1 as the resource group
- ✓ Follow lab panel settings (not defaults)
- ✓ If needed → go back and re-enter details or restart lab

Week 8 Tutorial Activity (Step-by-Step)

Key takeaway

Small configuration differences in cloud environments can cause errors —

Always follow lab instructions exactly.

“This is not a mistake — it’s how real cloud systems enforce security and consistency.”

Week 8 Tutorial Activity (Step-by-Step)

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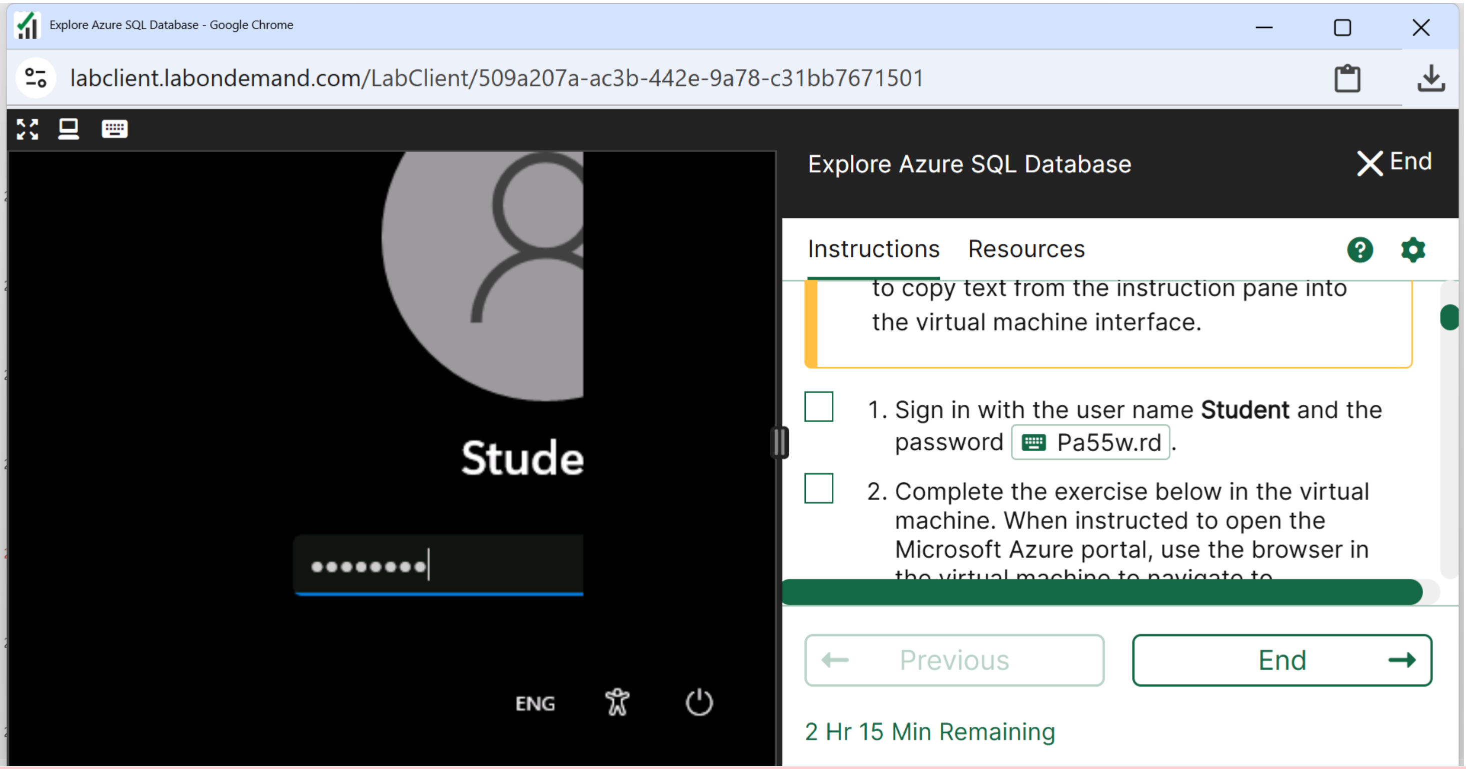
Click on Password and Press Enter



Important: Do not open Azure Portal on your browser, instead run the virtual machine and then open the Azure Portal on the virtual machine's browser

Week 8 Tutorial Activity (Step-by-Step)

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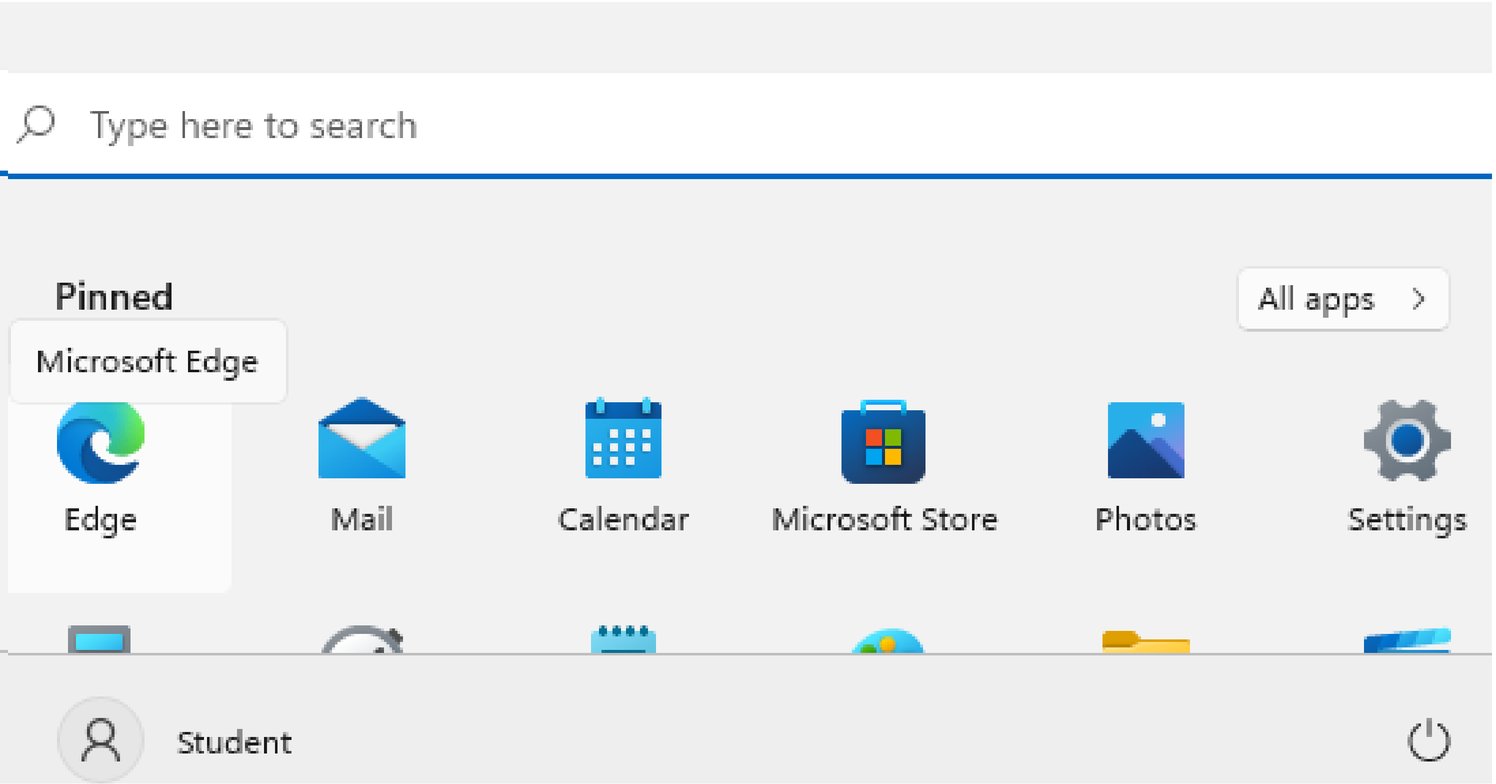
Click Start and Go to Edge Browser

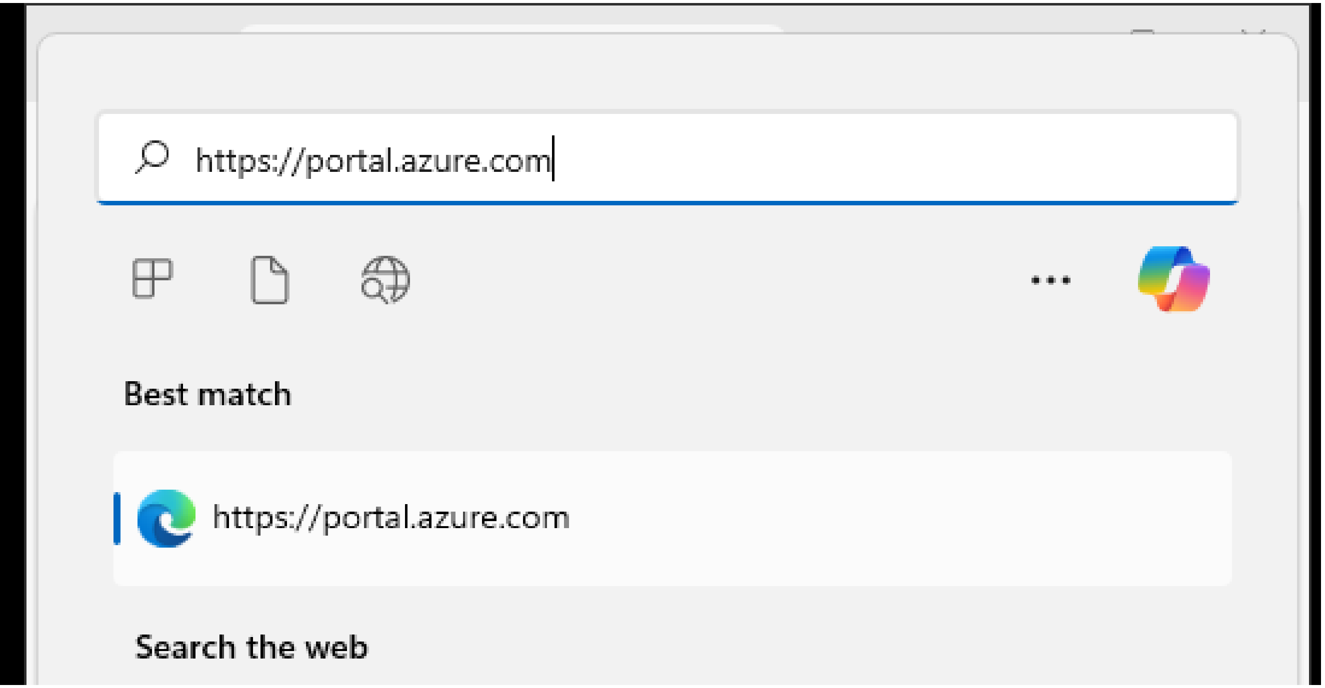


Important: Do not open Azure Portal on your browser, instead run the virtual machine and then open the Azure Portal on the virtual machine’s browser

Week 8 Tutorial Activity (Step-by-Step)

Click on Website (Portal); it is on right sidebar

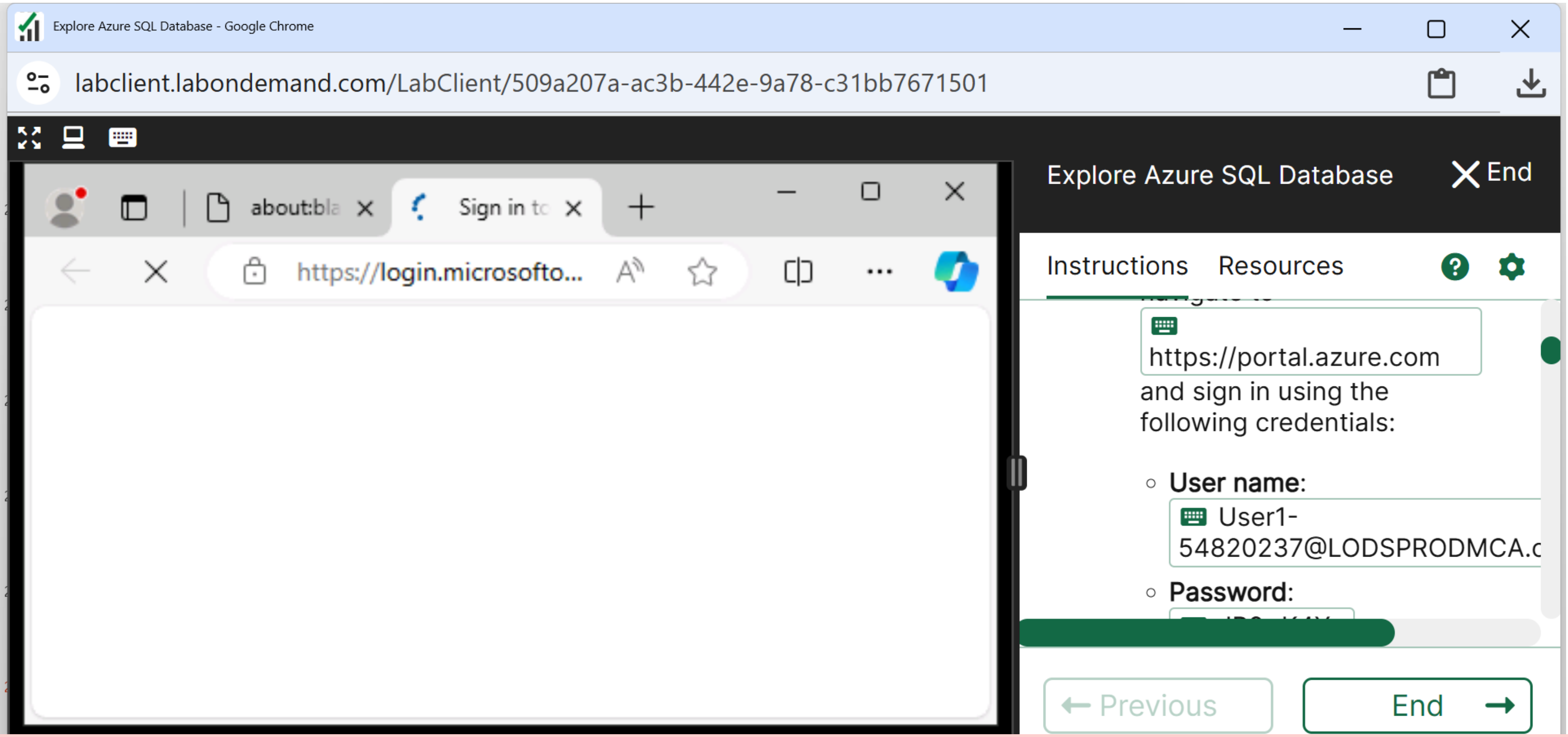
You will see this



Important: Do not open Azure Portal on your browser, instead run the virtual machine and then open the Azure Portal on the virtual machine's browser

Week 8 Tutorial Activity (Step-by-Step)

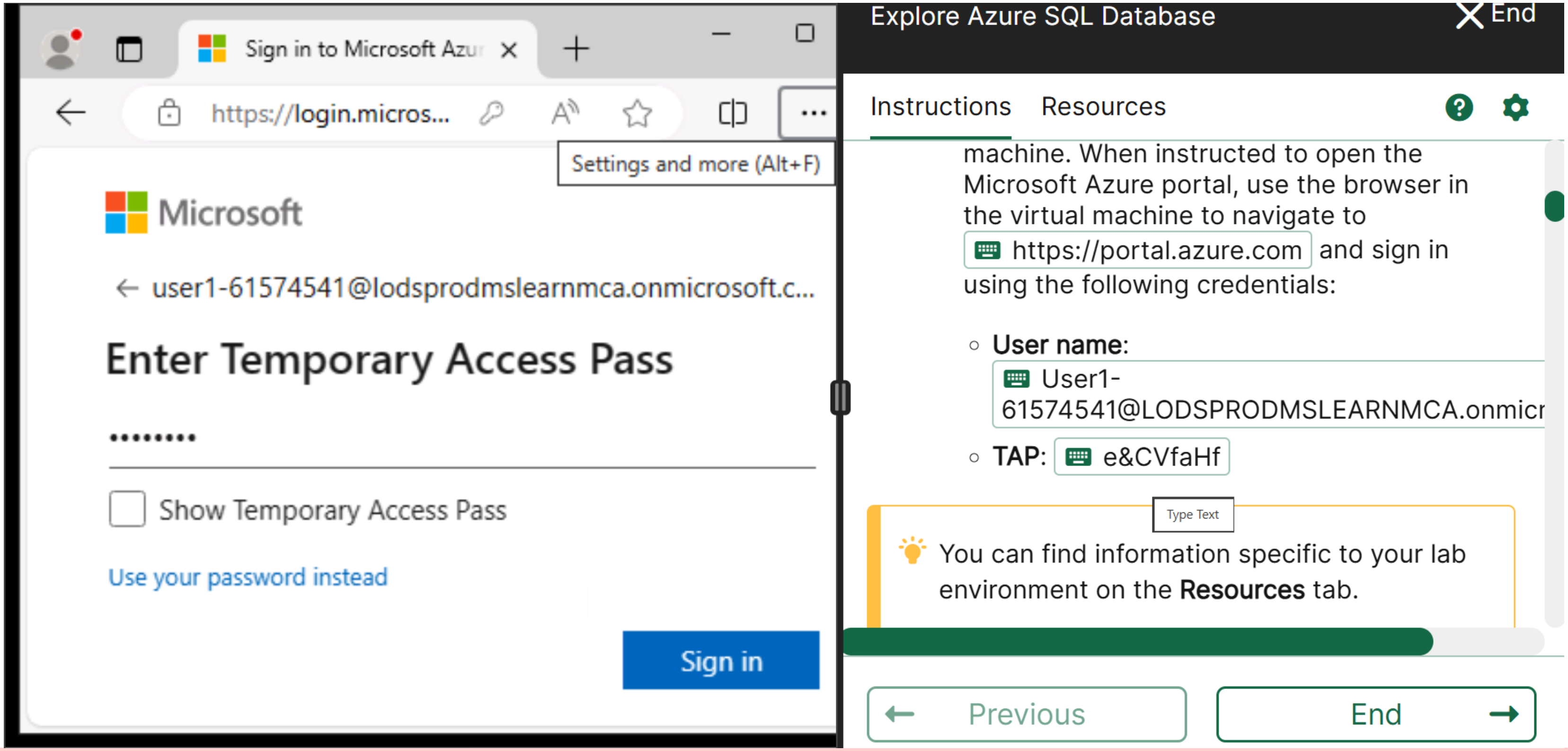
Press Enter; you will see this page



Important: Do not open Azure Portal on your browser, instead run the virtual machine and then open the Azure Portal on the virtual machine's browser

Week 8 Tutorial Activity (Step-by-Step)

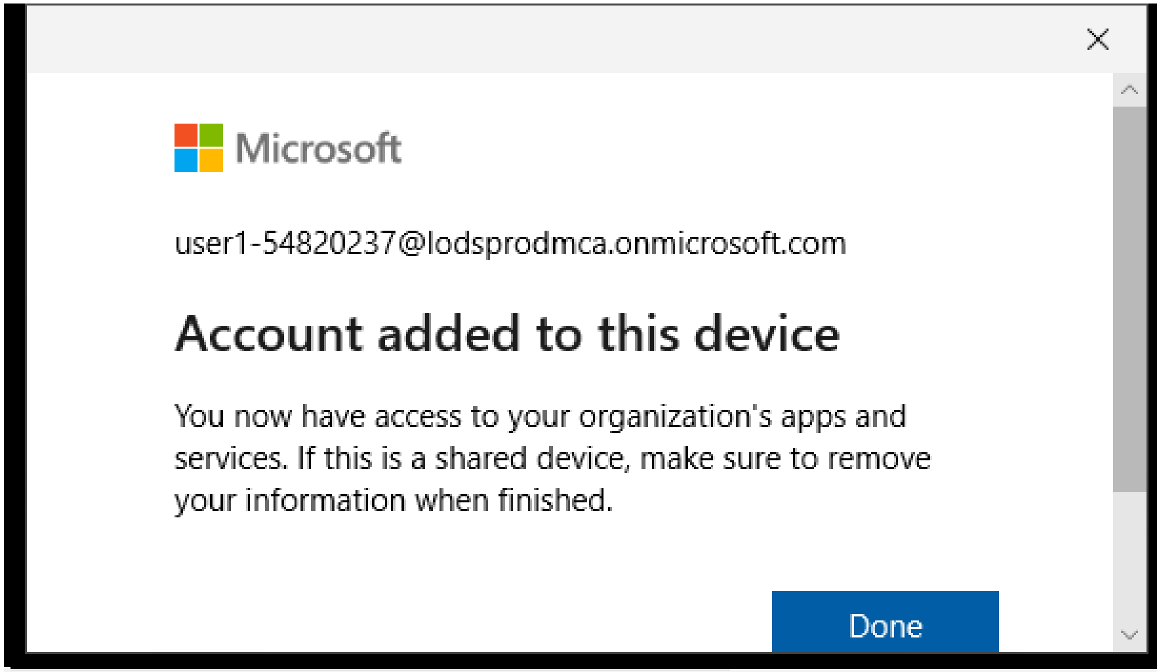
Click on User and Pass (on right sidebar) to log in



Important: Do not open Azure Portal on your browser, instead run the virtual machine and then open the Azure Portal on the virtual machine's browser

Week 8 Tutorial Activity (Step-by-Step)

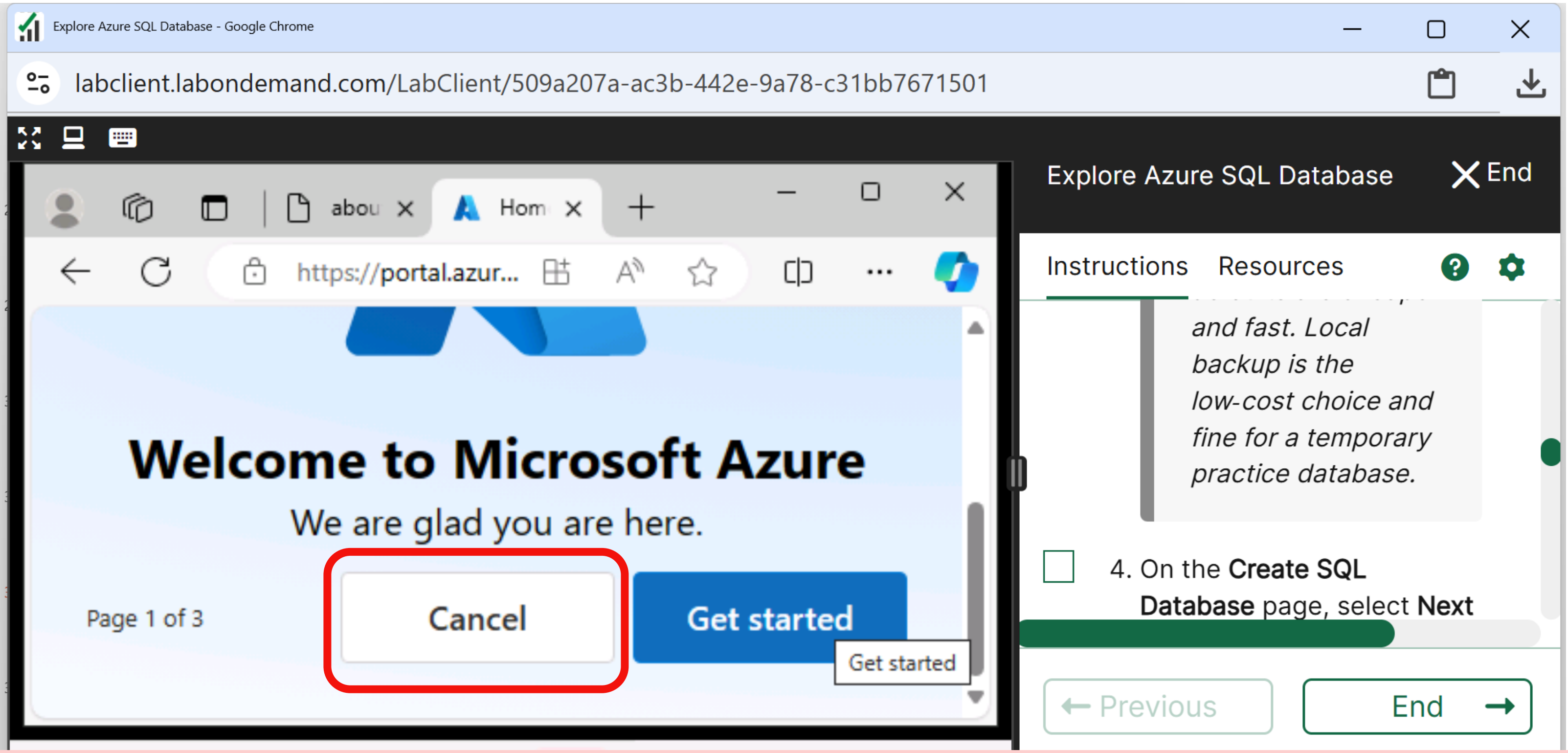
Click Yes then you see this



Important: Do not open Azure Portal on your browser, instead run the virtual machine and then open the Azure Portal on the virtual machine's browser

Week 8 Tutorial Activity (Step-by-Step)

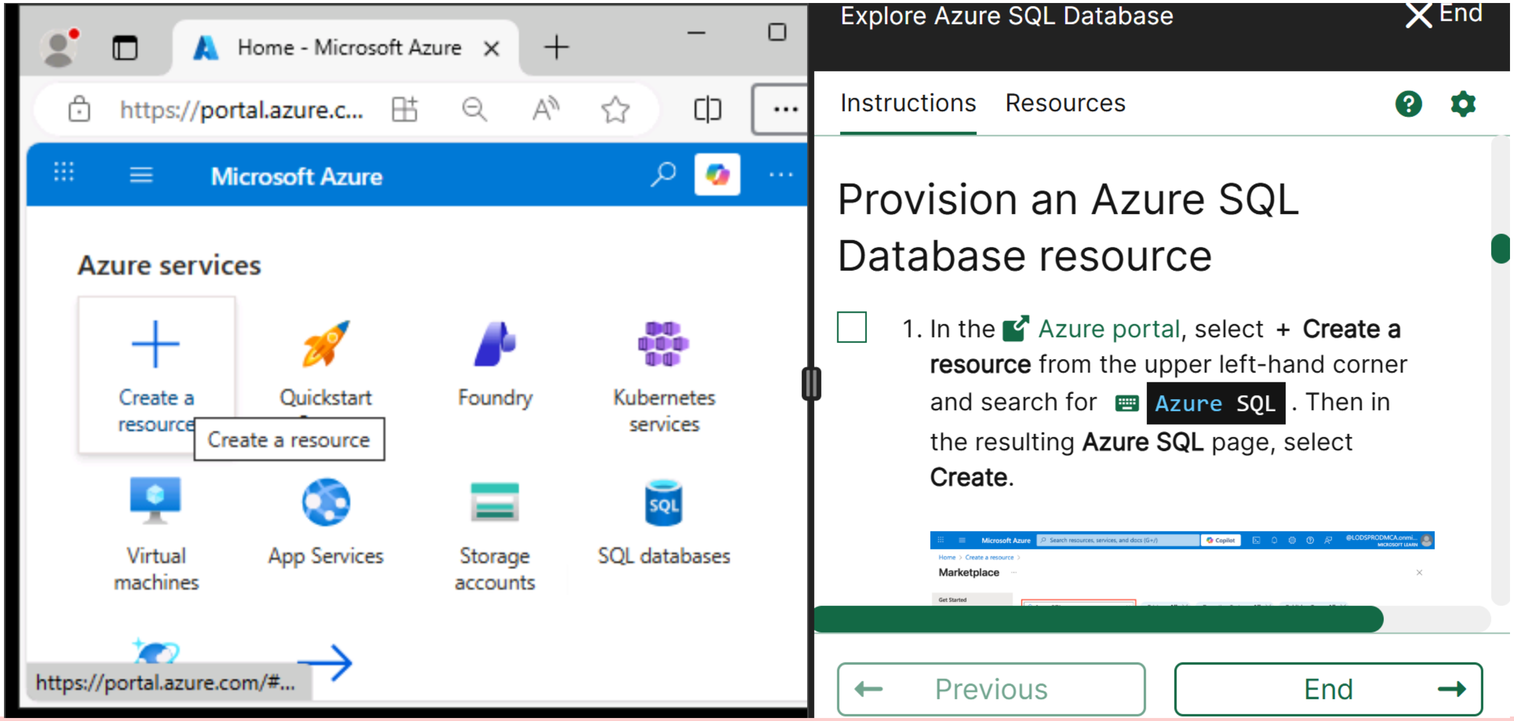
Cancel this for now



Important: Do not open Azure Portal on your browser, instead run the virtual machine and then open the Azure Portal on the virtual machine's browser

Week 8 Tutorial Activity (Step-by-Step)

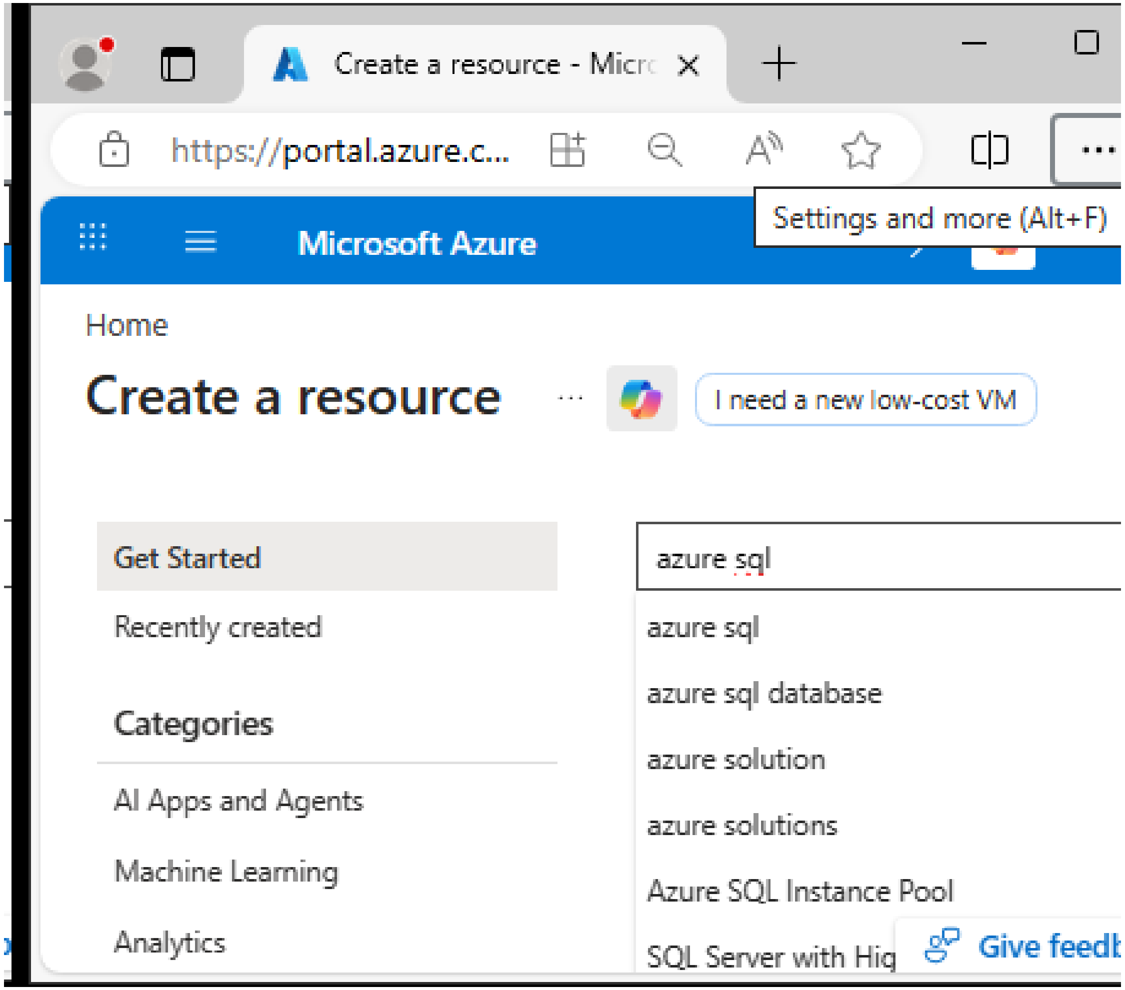
Zoom in/out to see Create a Resource and Click on it



Important: Do not open Azure Portal on your browser, instead run the virtual machine and then open the Azure Portal on the virtual machine's browser

Week 8 Tutorial Activity (Step-by-Step)

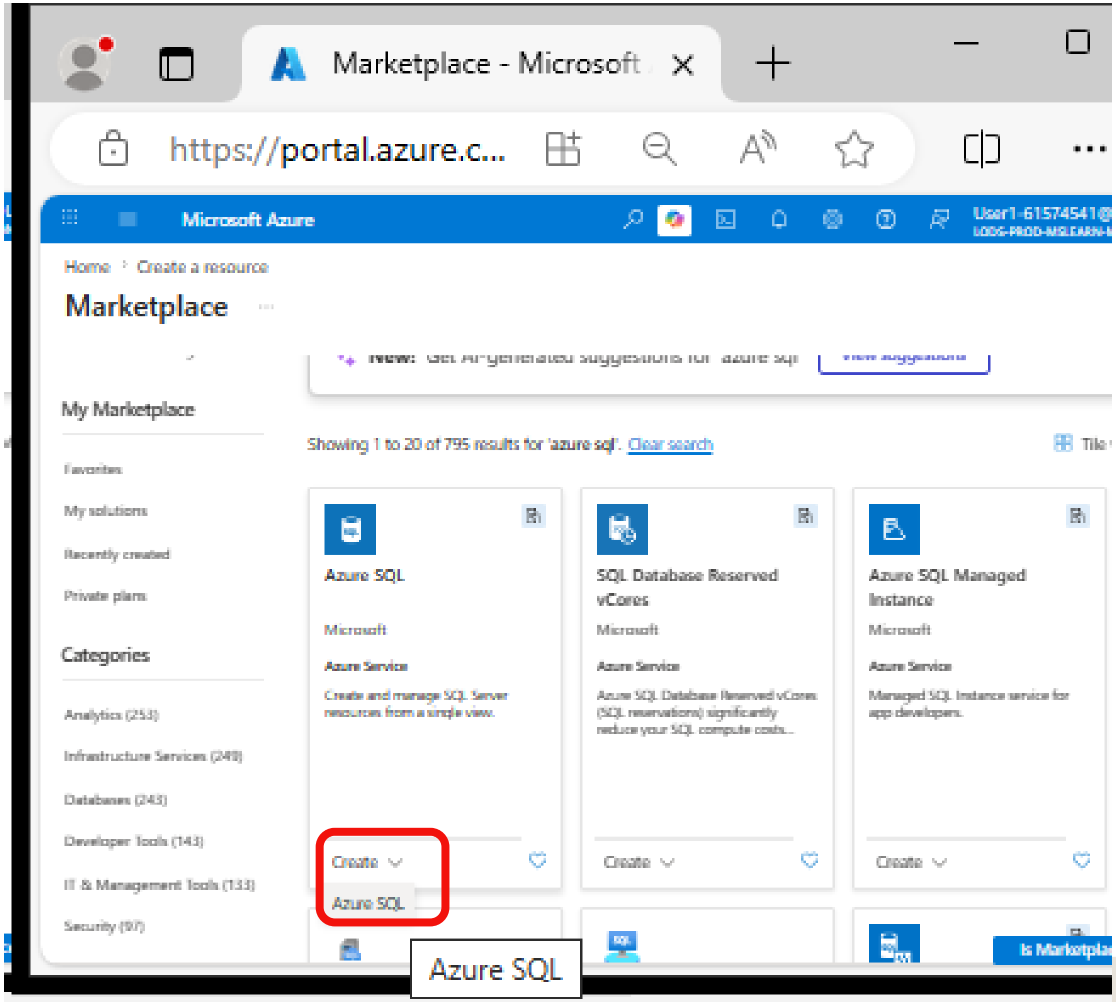
Search for Azure SQL



Week 8 Tutorial Activity (Step-by-Step)

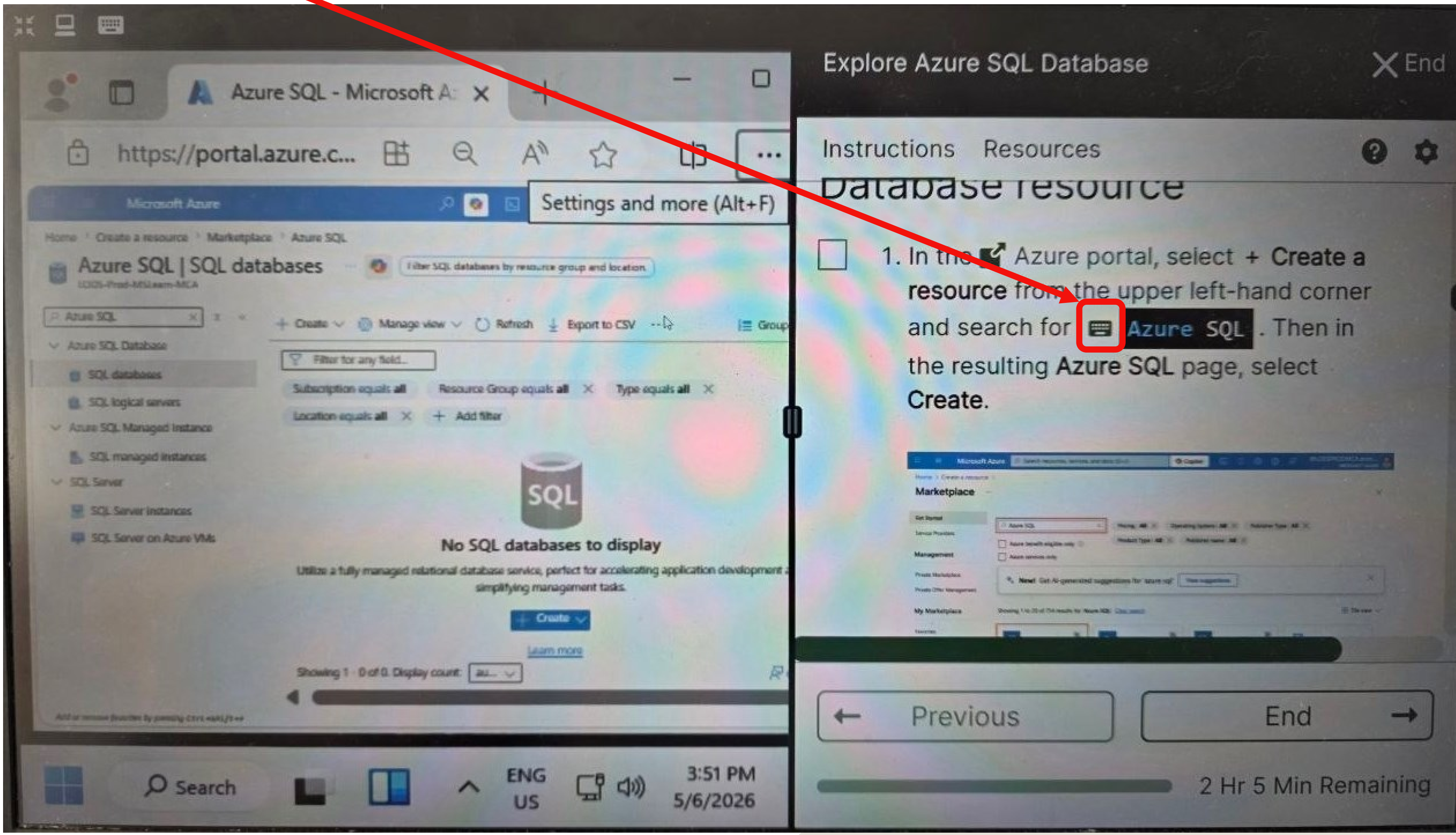
Zoom in/out to see Azure SQL

Click Azure SQL



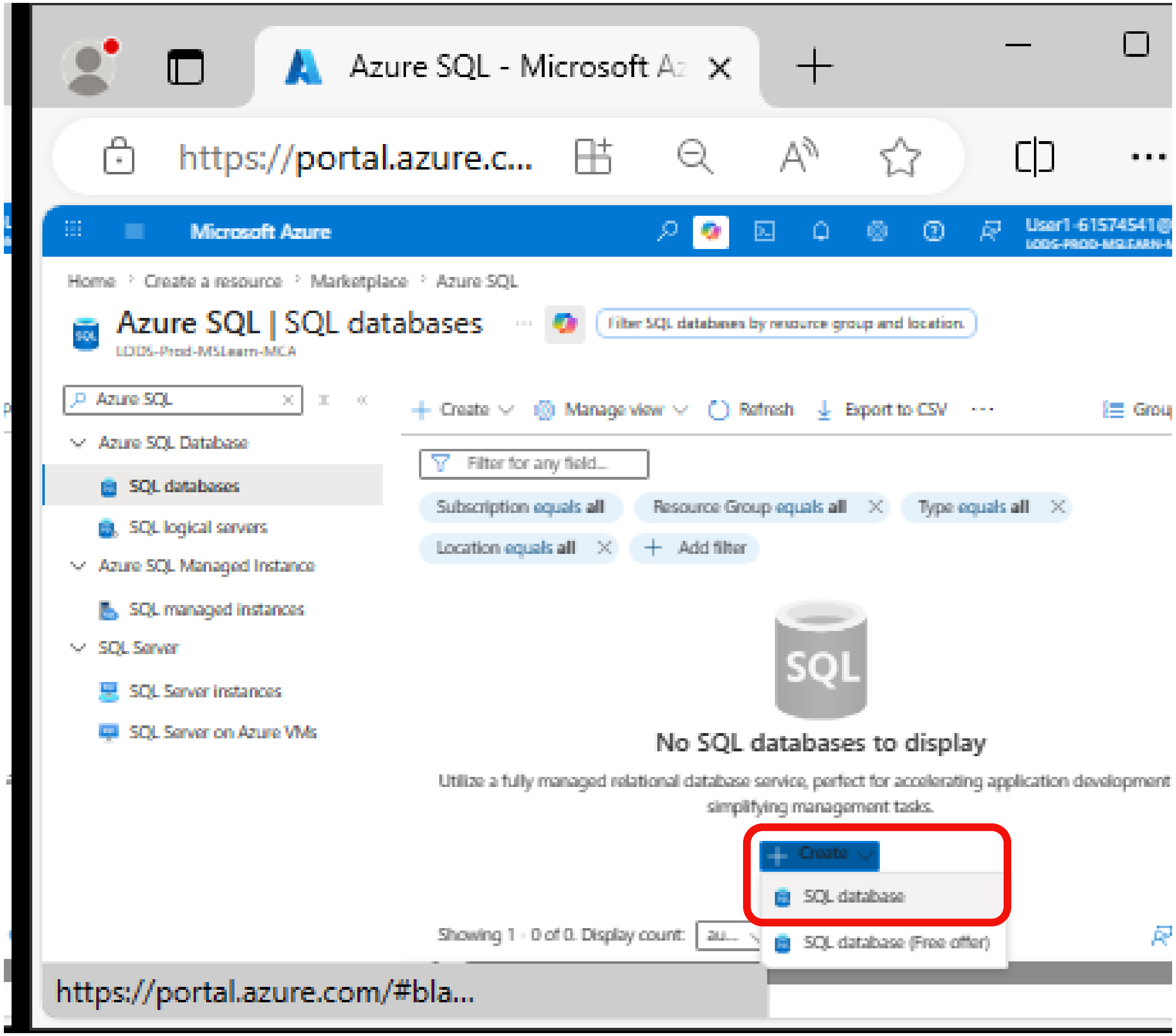
Week 8 Tutorial Activity (Step-by-Step)

Click on type on right side



Week 8 Tutorial Activity (Step-by-Step)

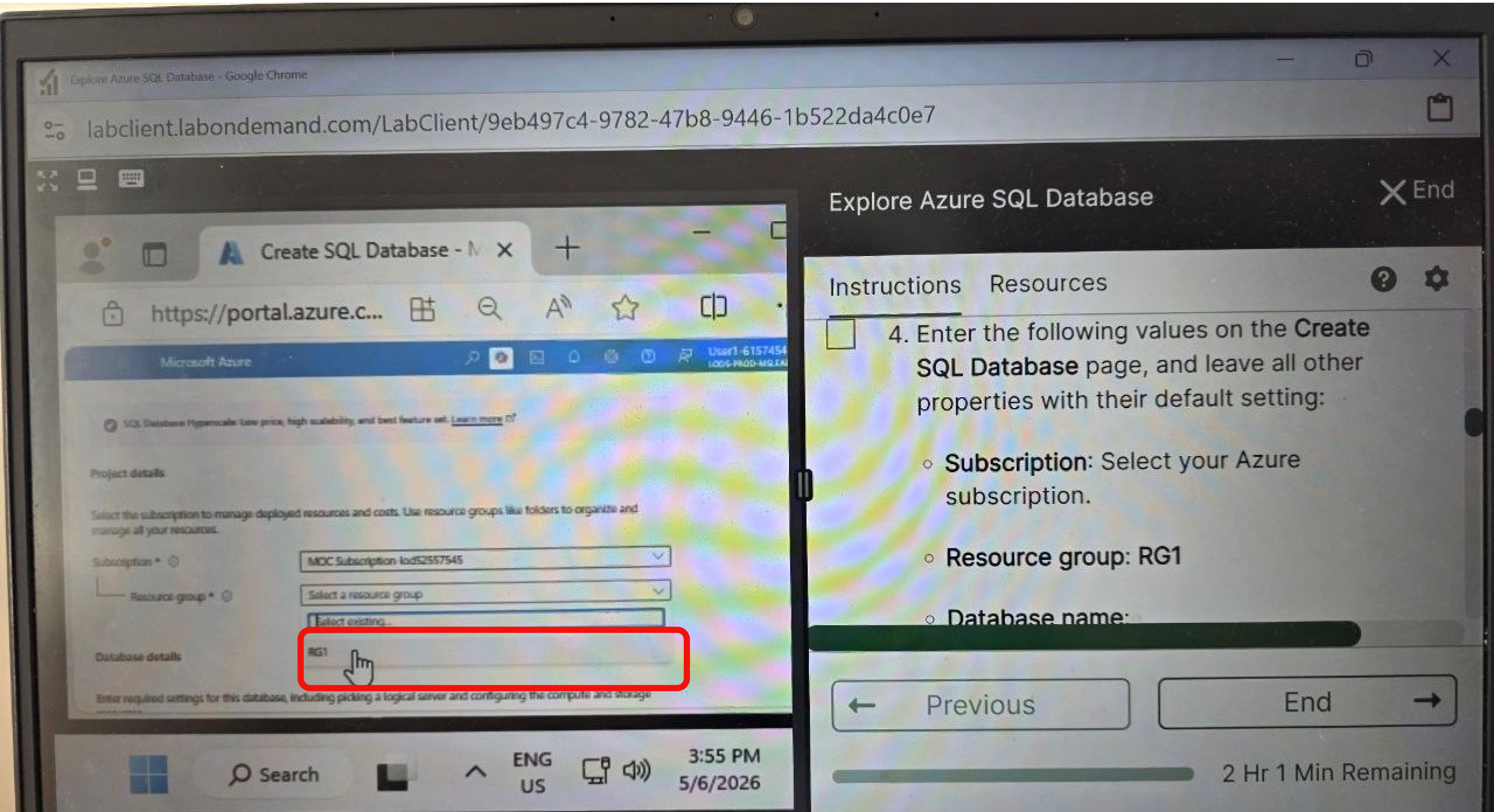
Click Create >> SQL database



Week 8 Tutorial Activity (Step-by-Step)

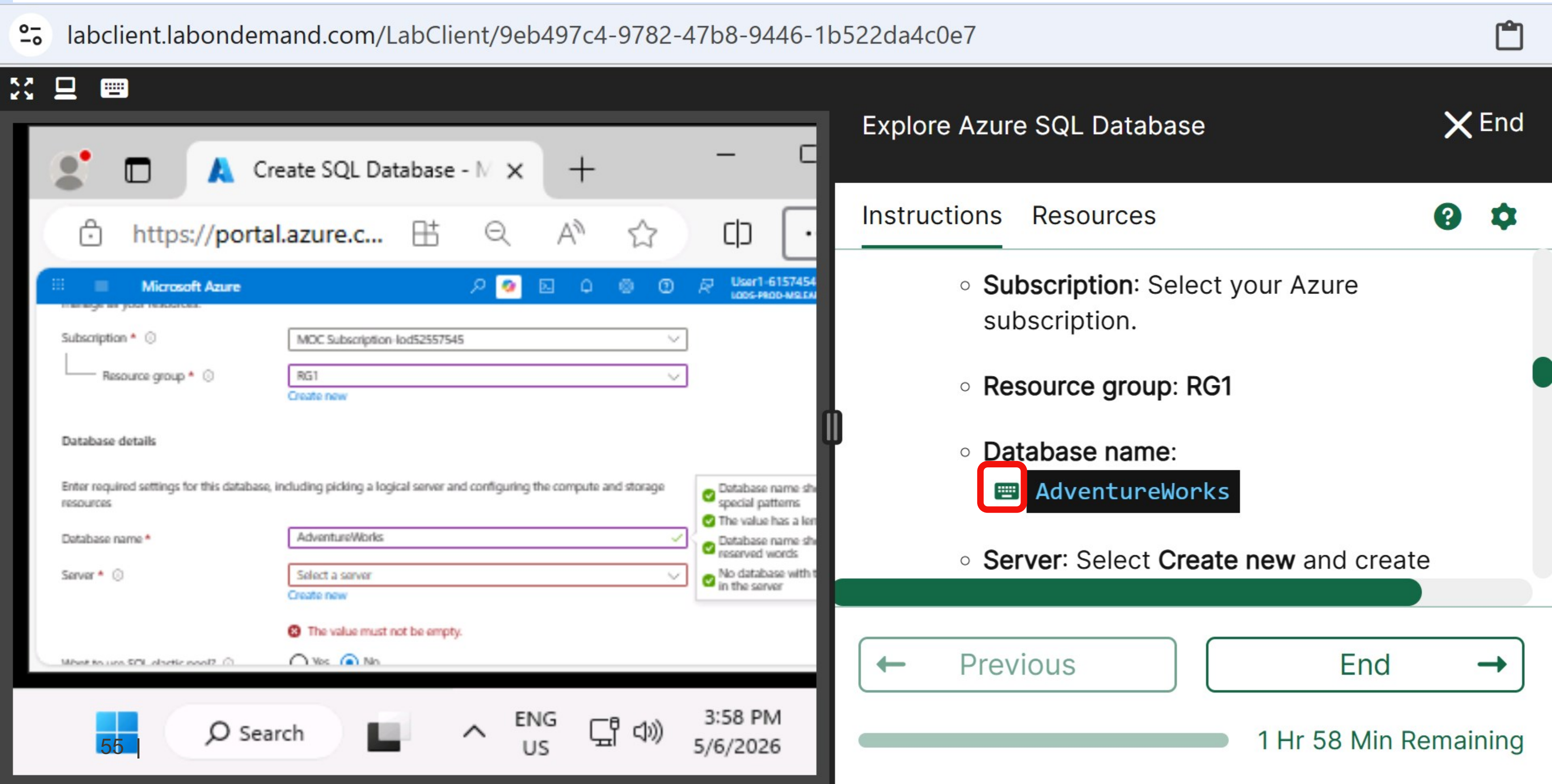
The subscription should be shown automatically

Choose Resource – Type: RG1



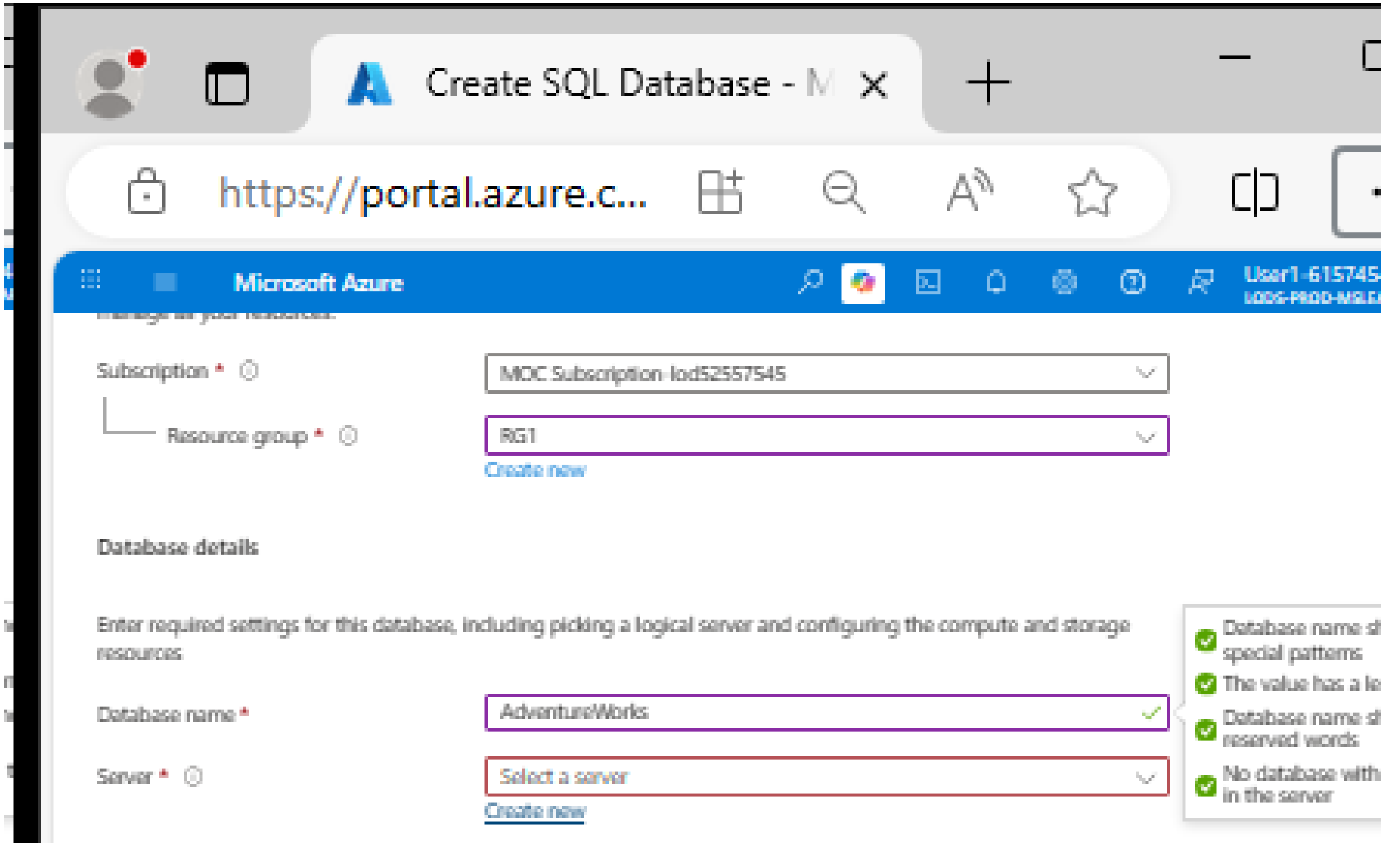
Week 8 Tutorial Activity (Step-by-Step)

- I chose default one (by clicking Type the database name automatically is written)
- You should follow Canvas: Student ID+name as the "database name"



Week 8 Tutorial Activity (Step-by-Step)

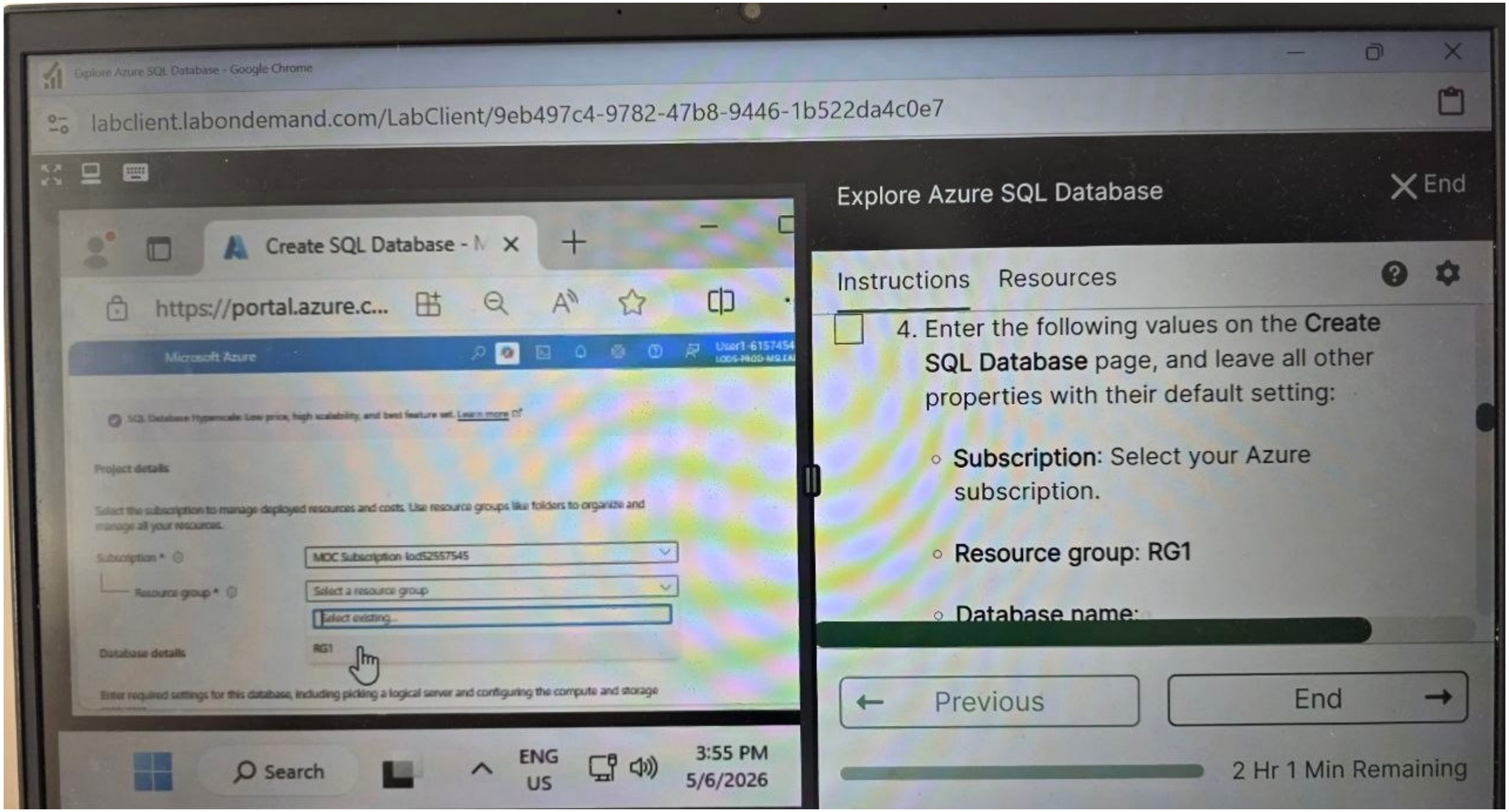
Server: Create new



Make sure to use the resource group/server name and region provided on the right-hand side window.

Week 8 Tutorial Activity (Step-by-Step)

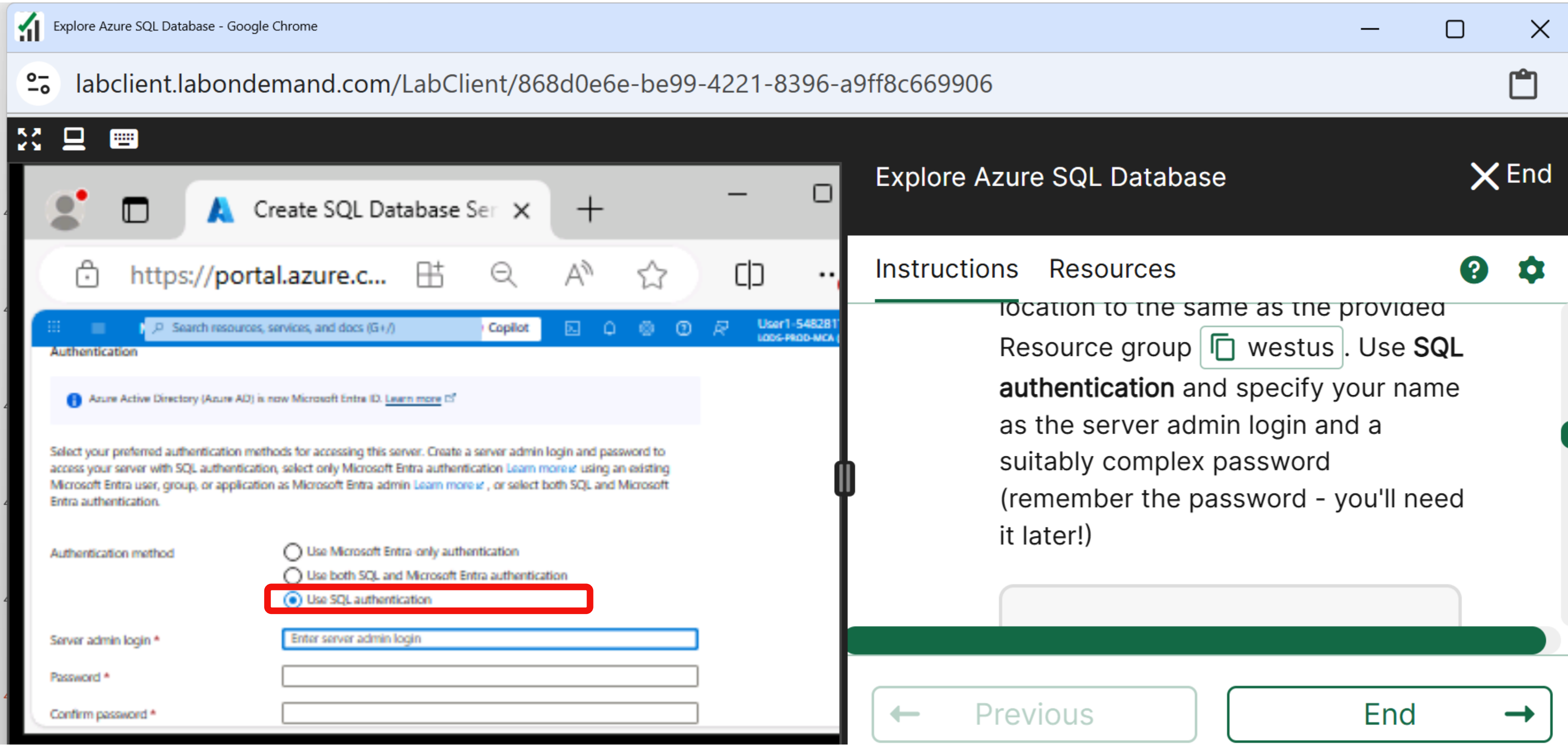
For the location, scroll down to find (US) West US



Do Not Choose West US2

Week 8 Tutorial Activity (Step-by-Step)

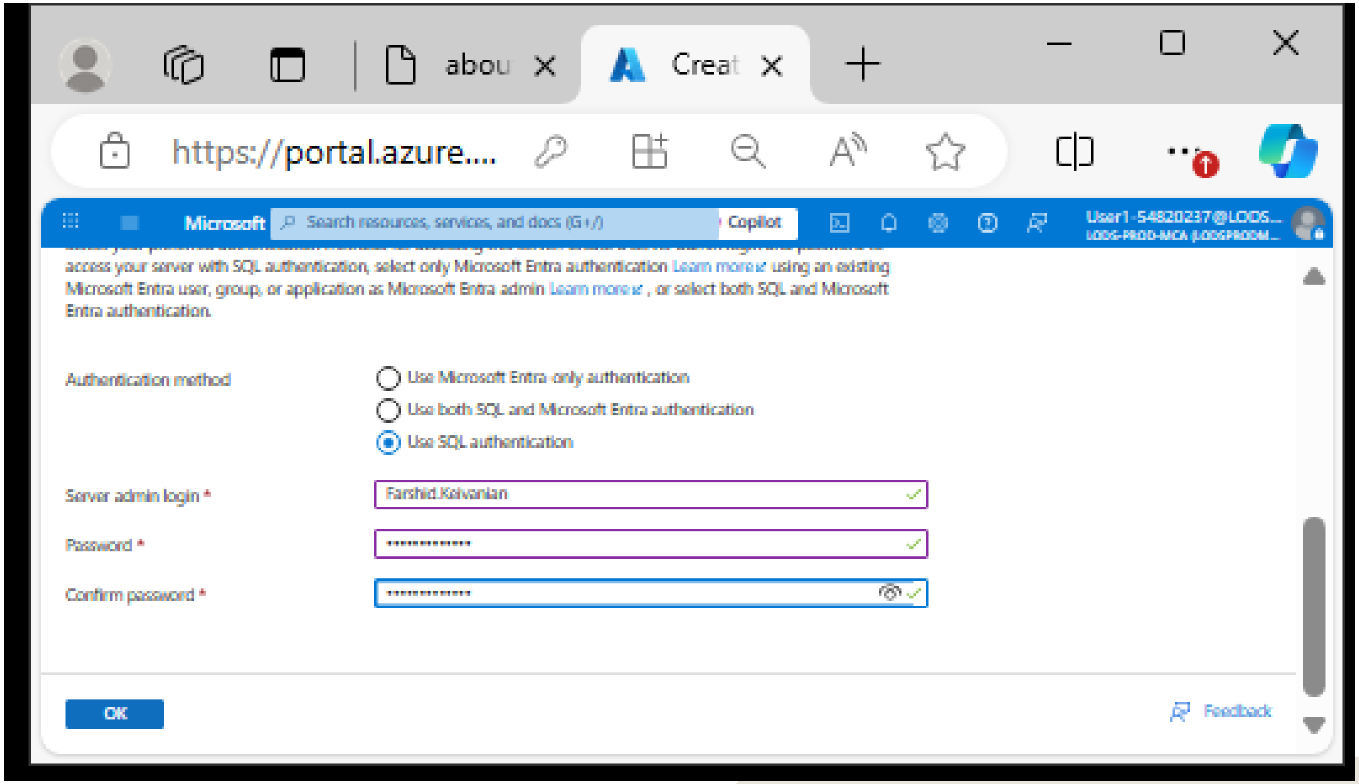
Use SQL Authentication



Week 8 Tutorial Activity (Step-by-Step)

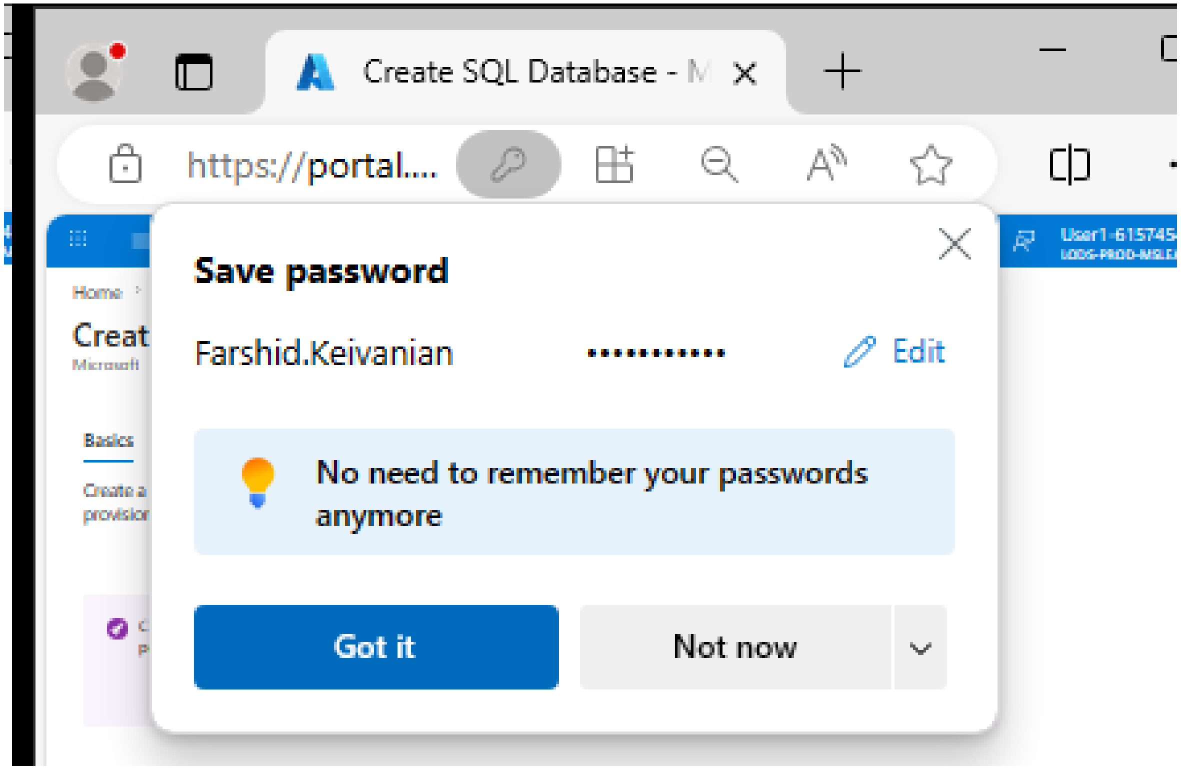
Enter Your name as Server name and set a password (You will need it later) e.g., I typed Farshid.Keivanian

Click OK (remember the password)



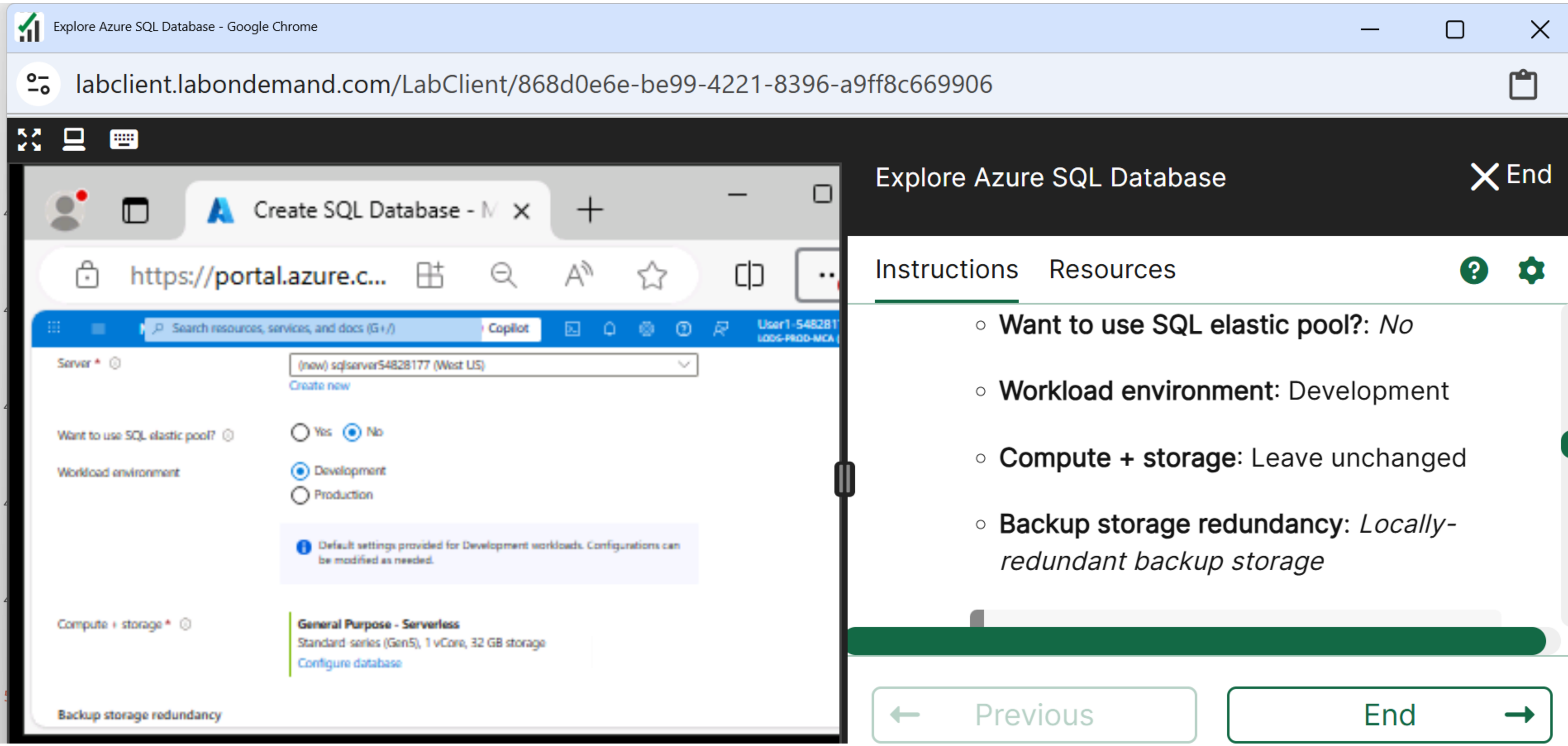
Week 8 Tutorial Activity (Step-by-Step)

Choose Got it



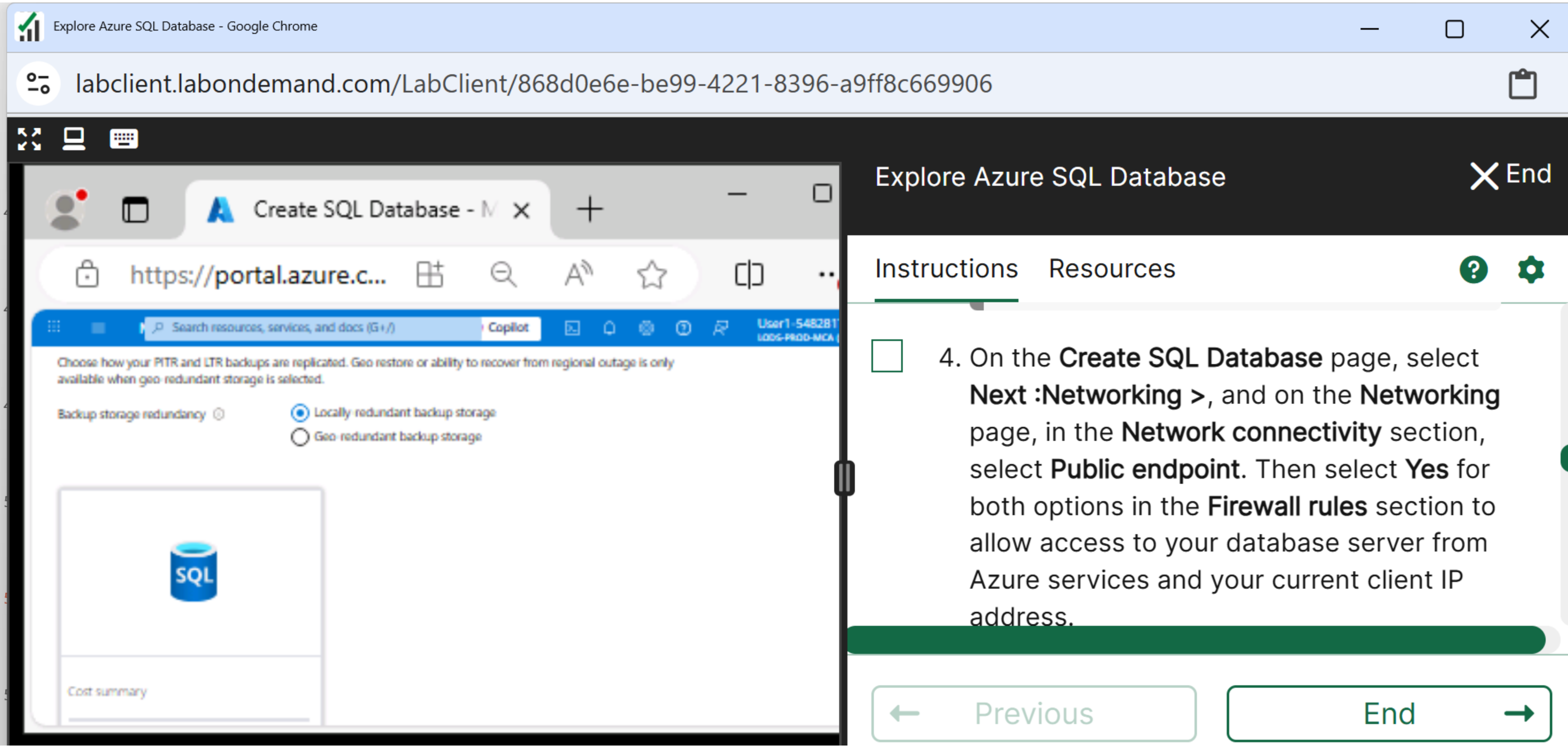
Week 8 Tutorial Activity (Step-by-Step)

Please follow the instructions on the right sidebar



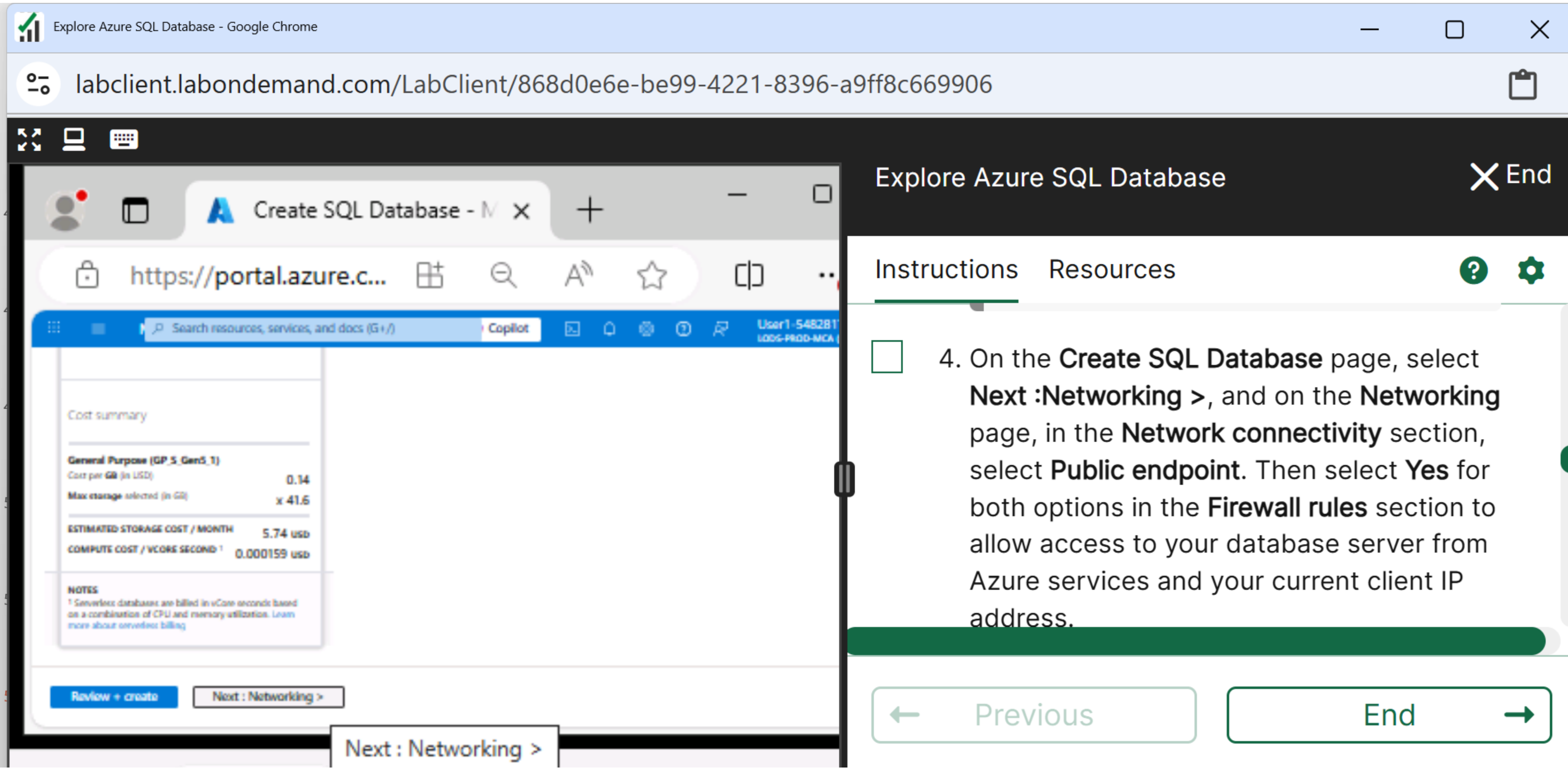
Week 8 Tutorial Activity (Step-by-Step)

Choose Locally-redundant backup storage



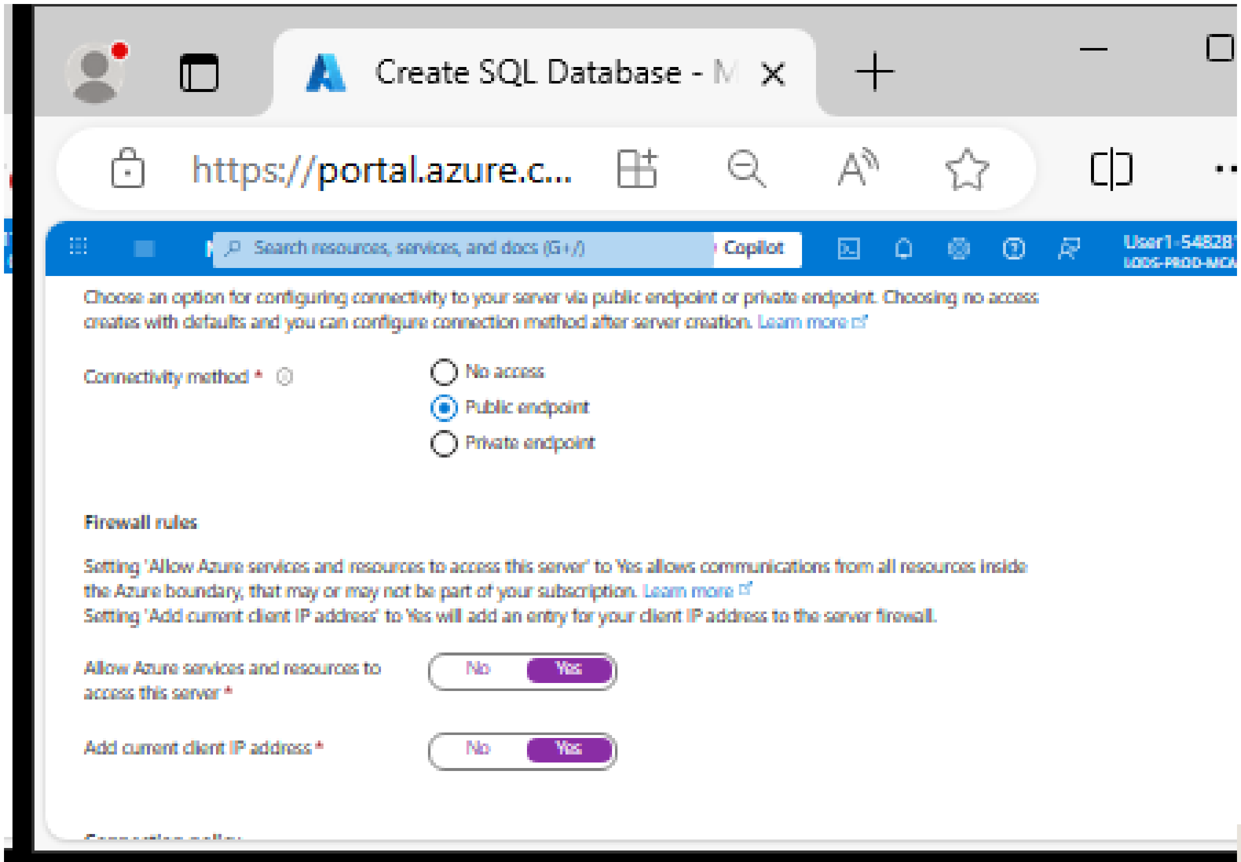
Week 8 Tutorial Activity (Step-by-Step)

Click Next: Networking



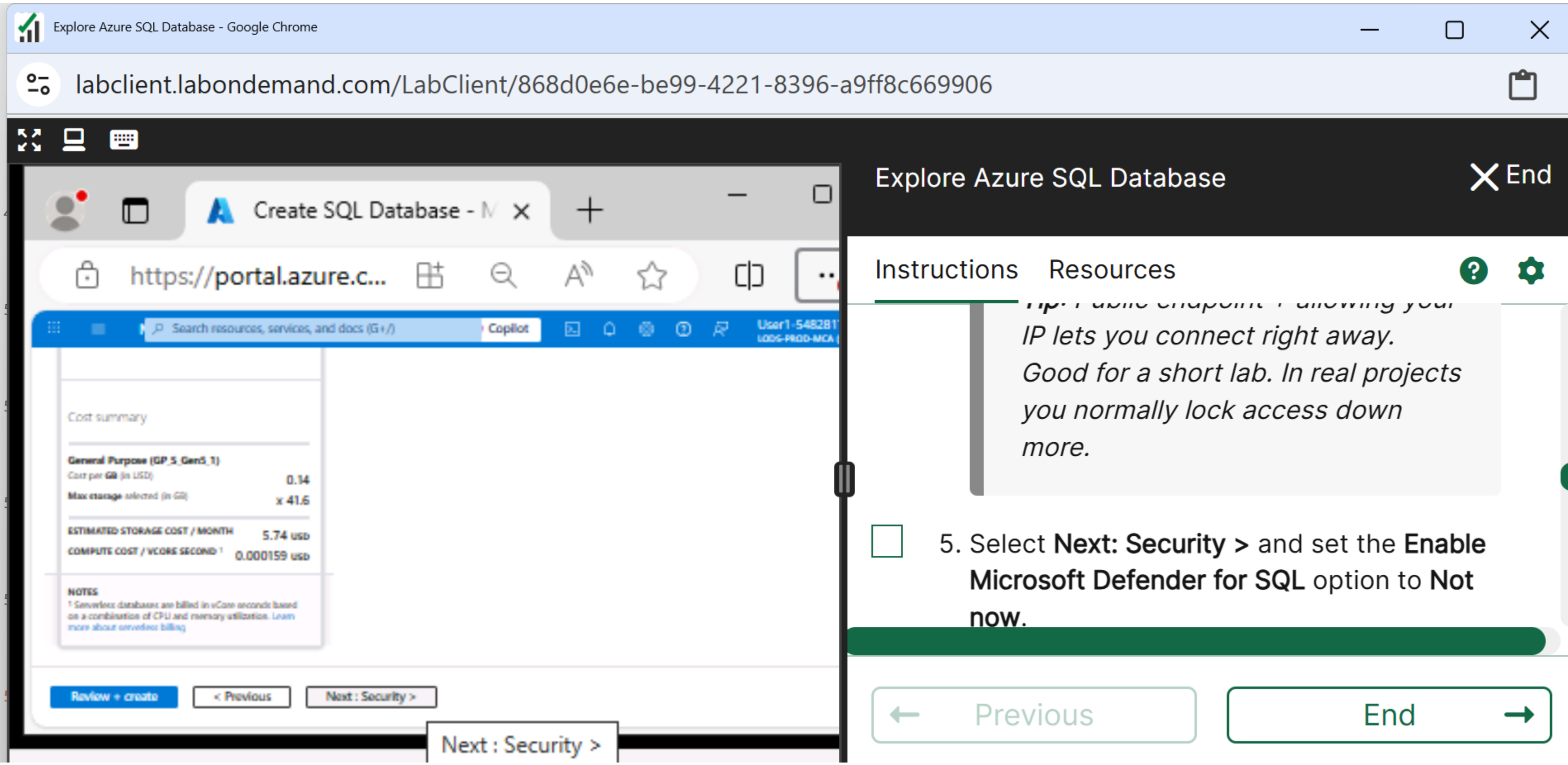
Week 8 Tutorial Activity (Step-by-Step)

Select Public endpoint, and Yes for both firewall options to allow access to your database server from Azure services and your current client IP address.



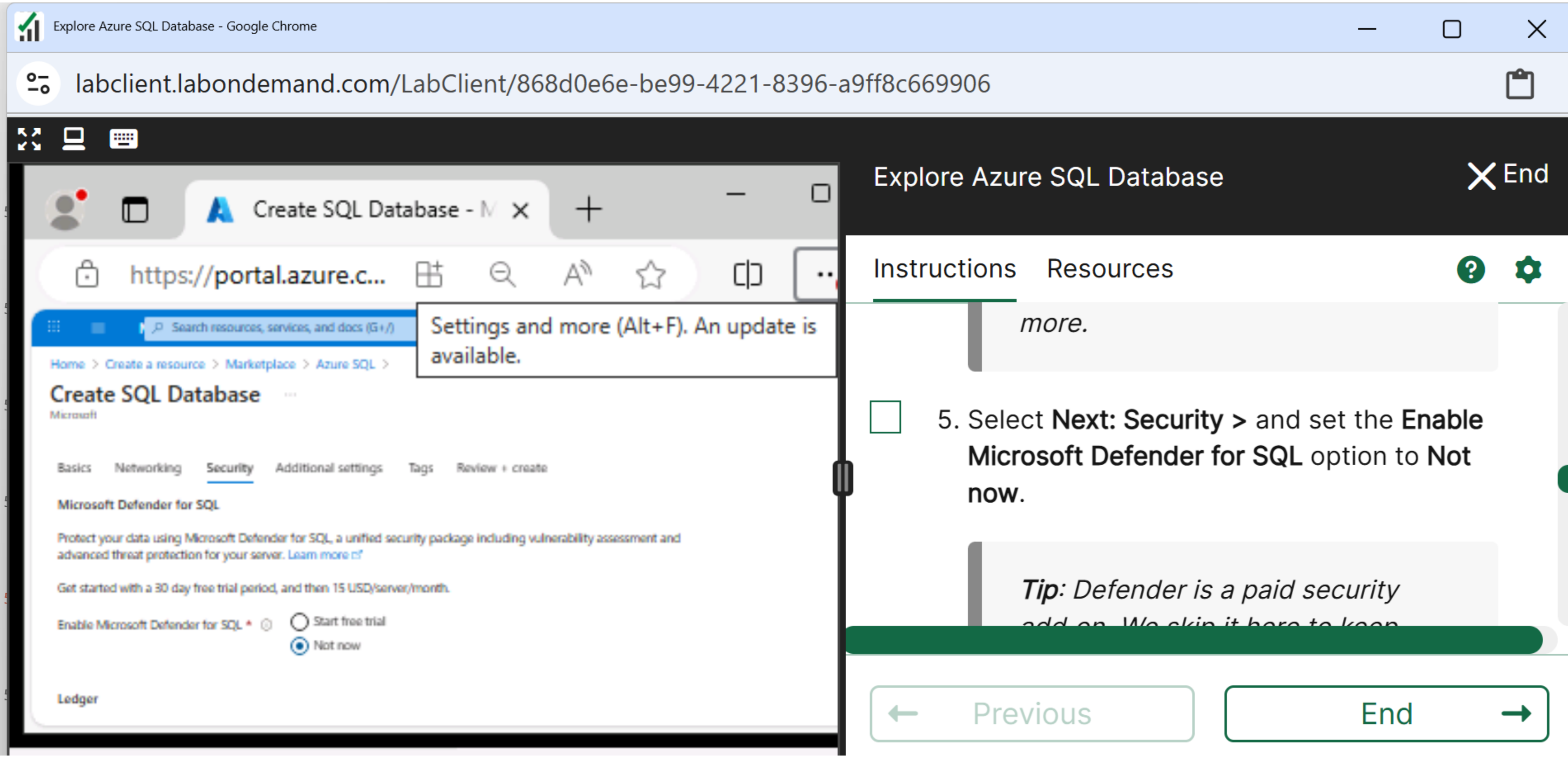
Week 8 Tutorial Activity (Step-by-Step)

Scroll Down and Click on Next: Security >



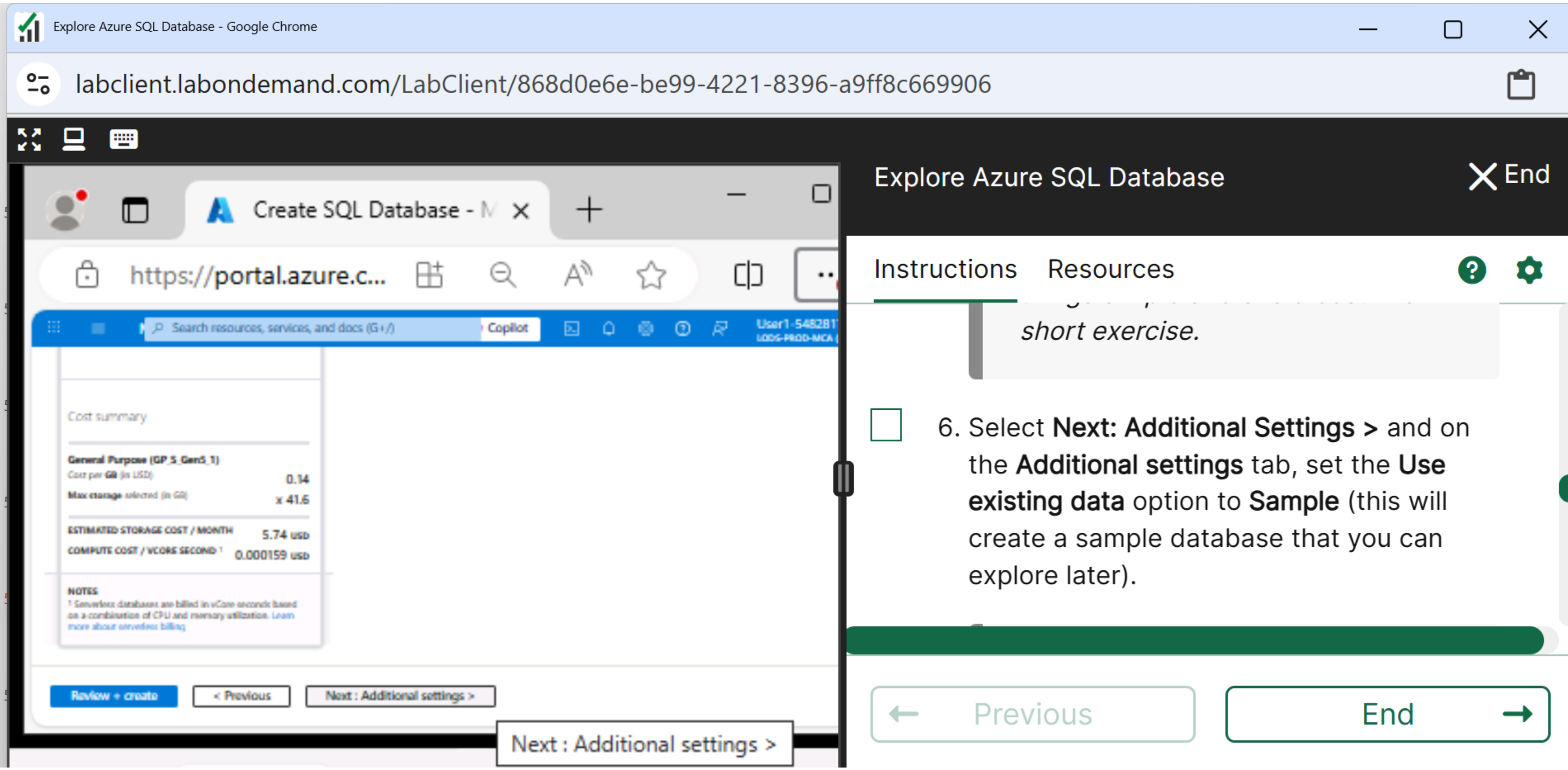
Week 8 Tutorial Activity (Step-by-Step)

Microsoft Defender: Not Now



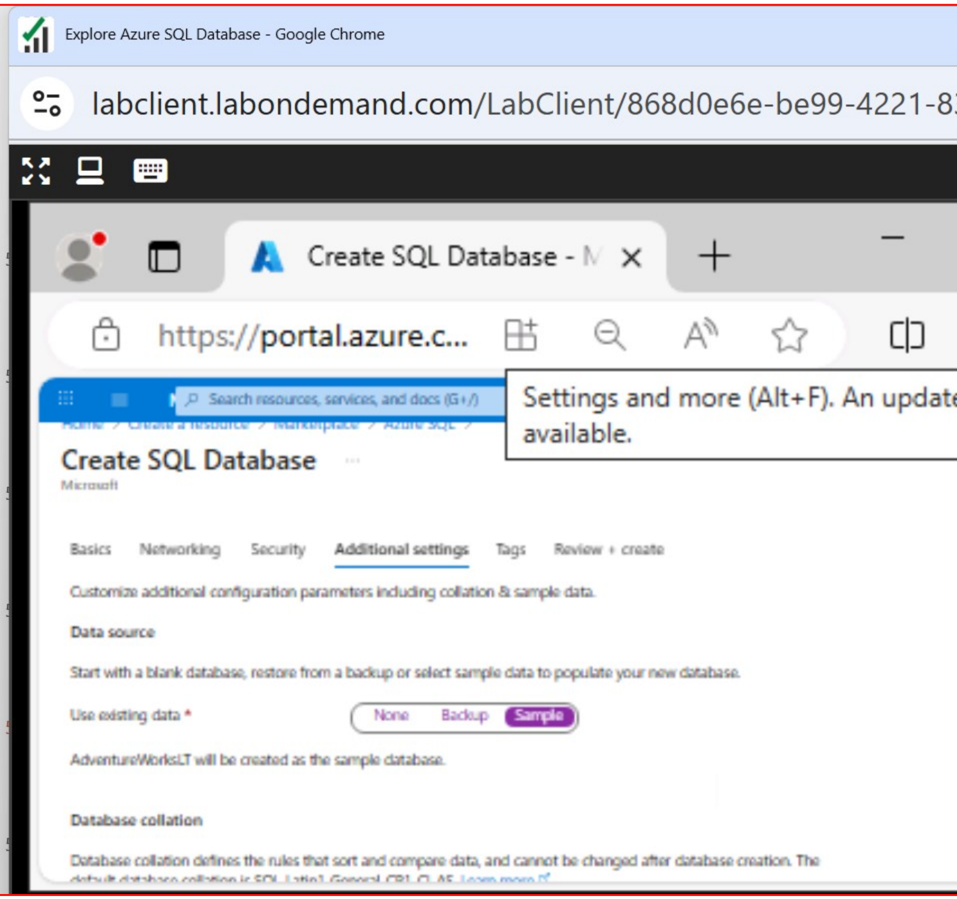
Week 8 Tutorial Activity (Step-by-Step)

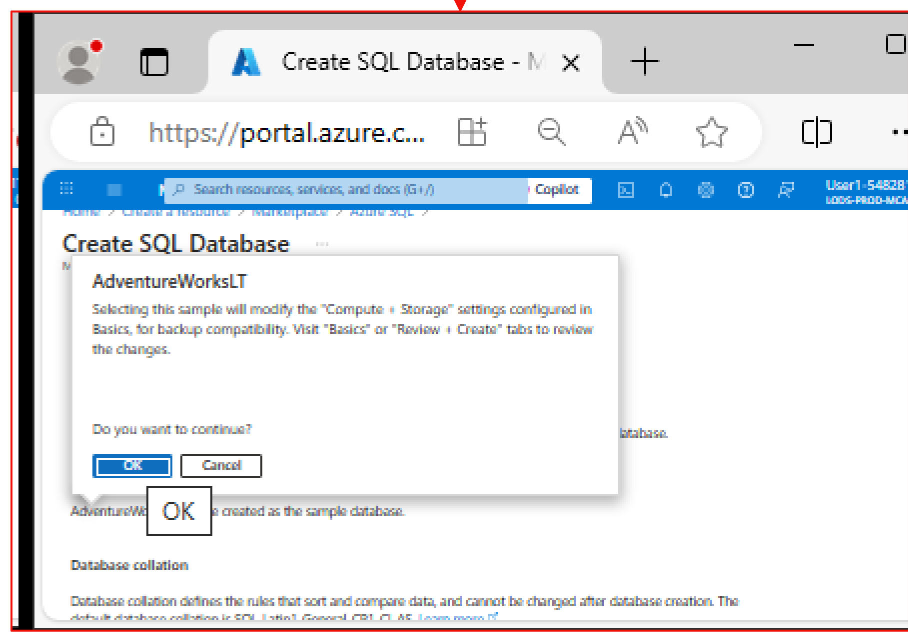
Next: Additional Settings



Week 8 Tutorial Activity (Step-by-Step)

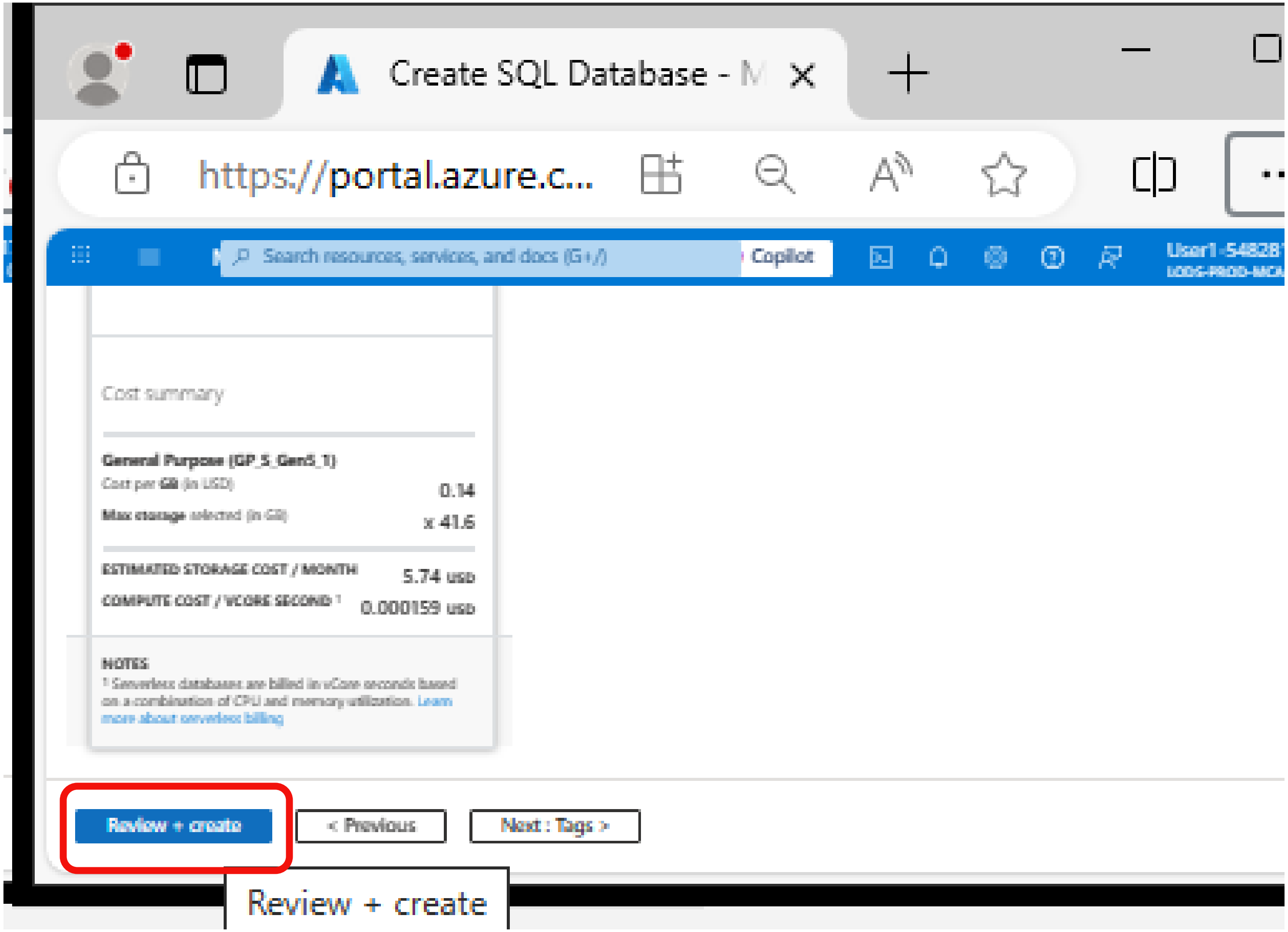
- Choose Sample
- Click Ok to continue





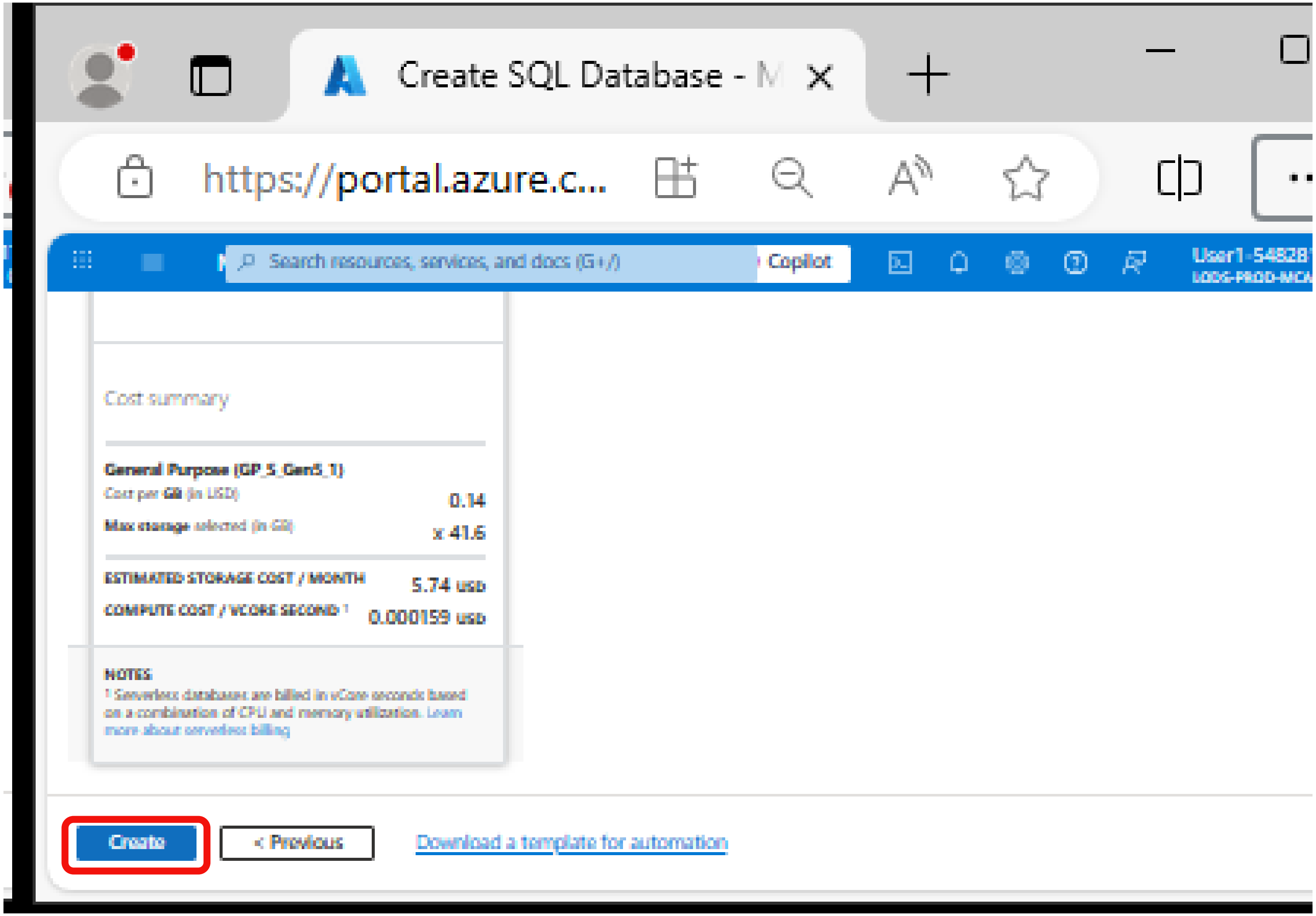
Week 8 Tutorial Activity (Step-by-Step)

Review and Create



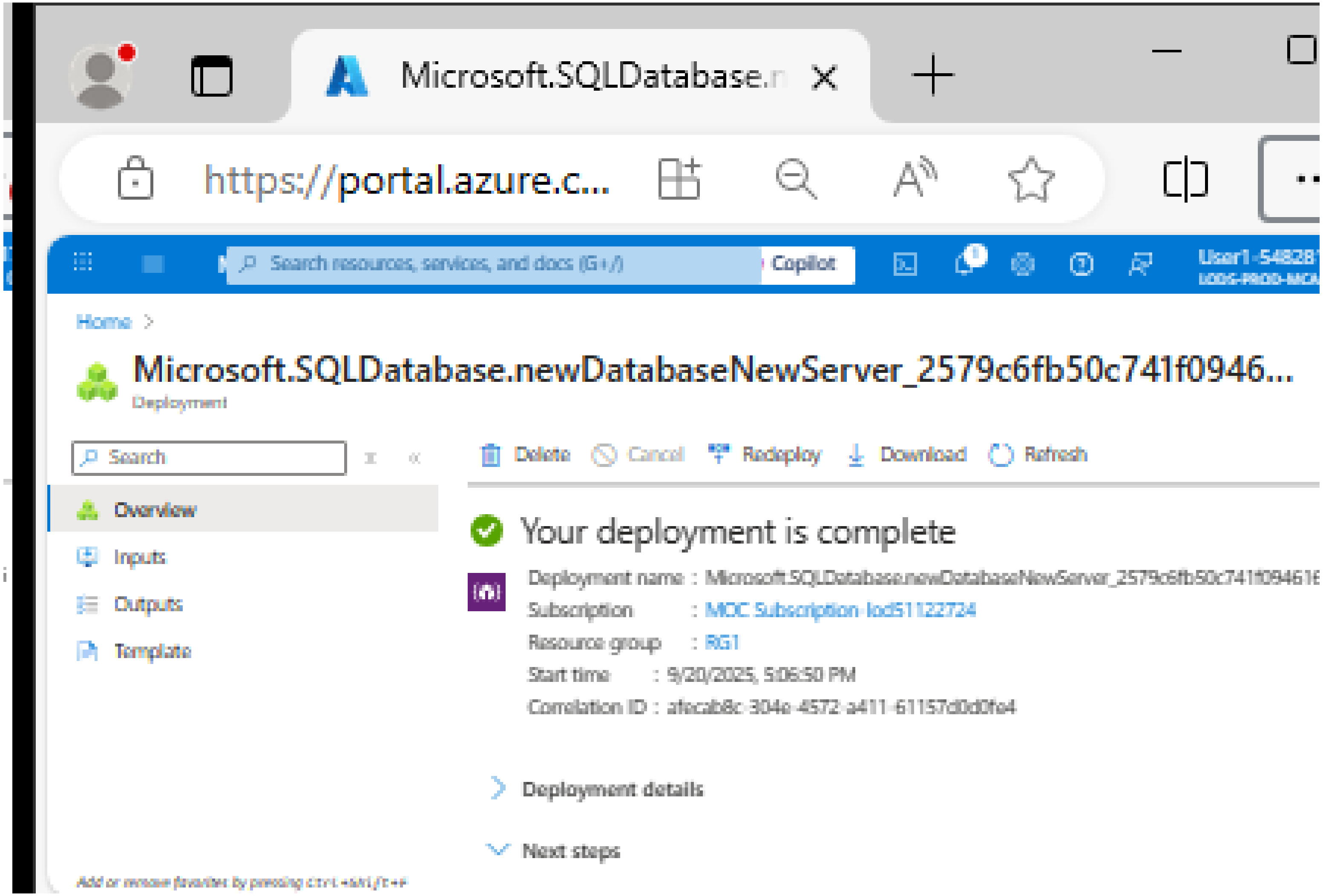
Week 8 Tutorial Activity (Step-by-Step)

Create



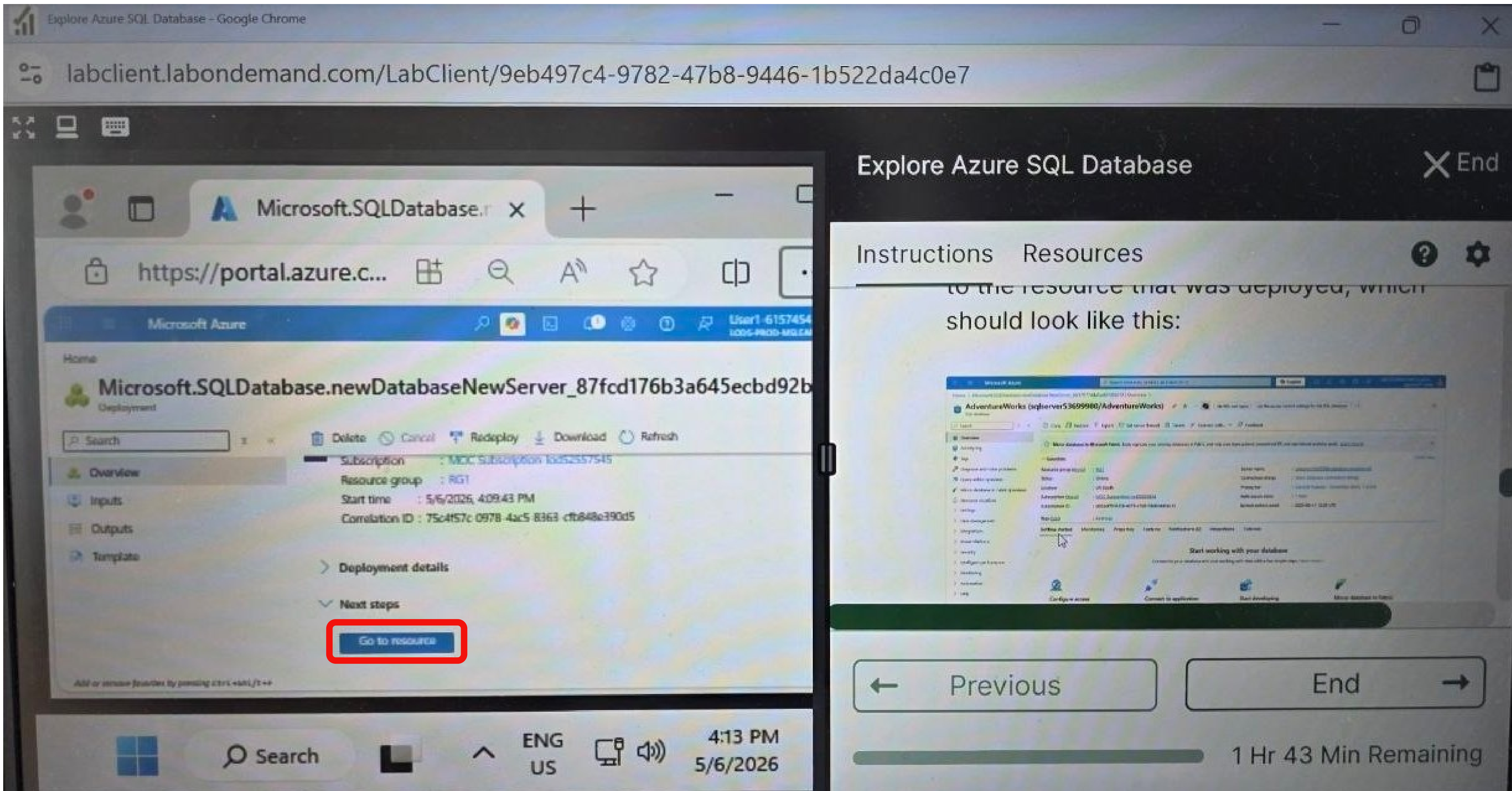
Week 8 Tutorial Activity (Step-by-Step)

Wait for deployment to complete



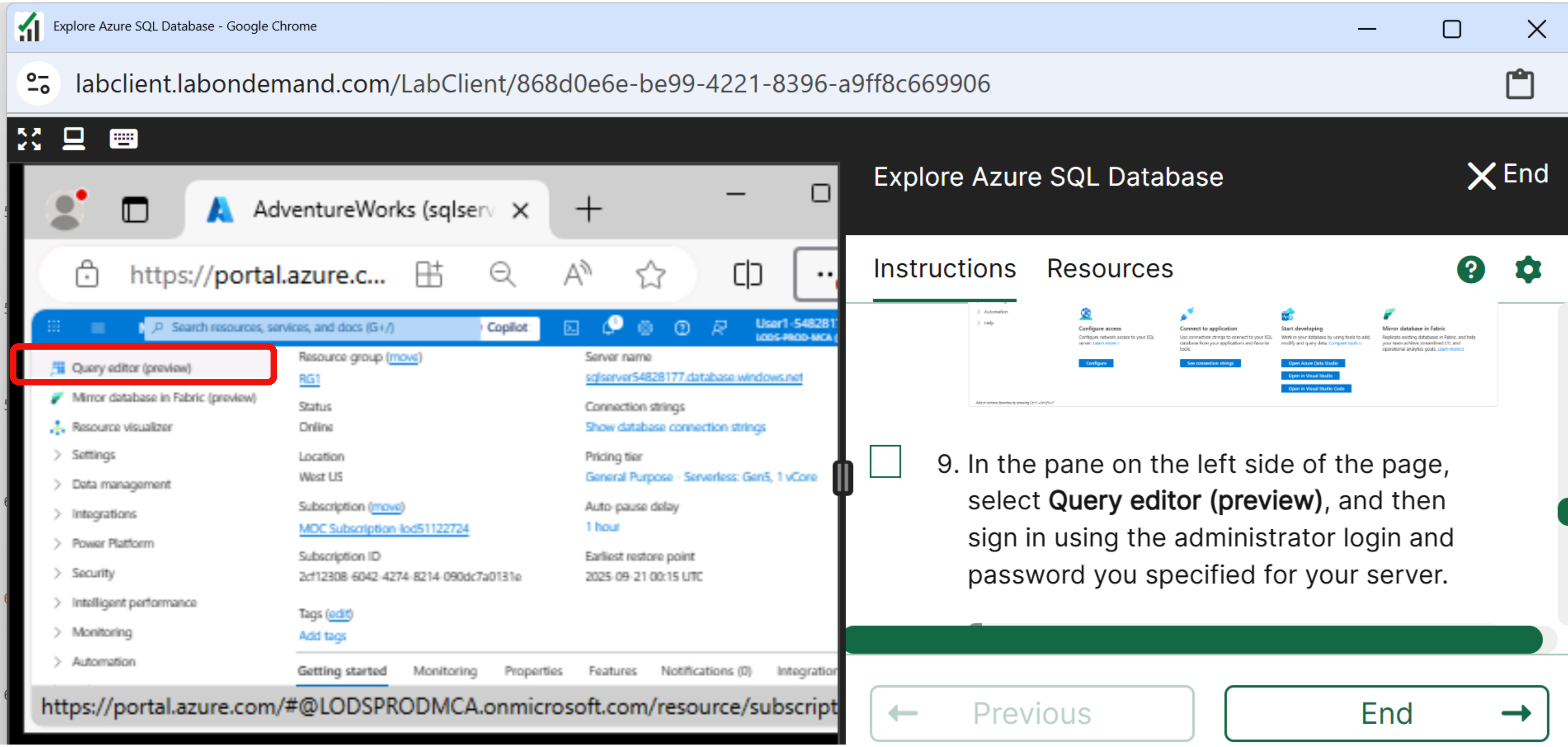
Week 8 Tutorial Activity (Step-by-Step)

Scroll Down and Go to the Resource that is now deployed



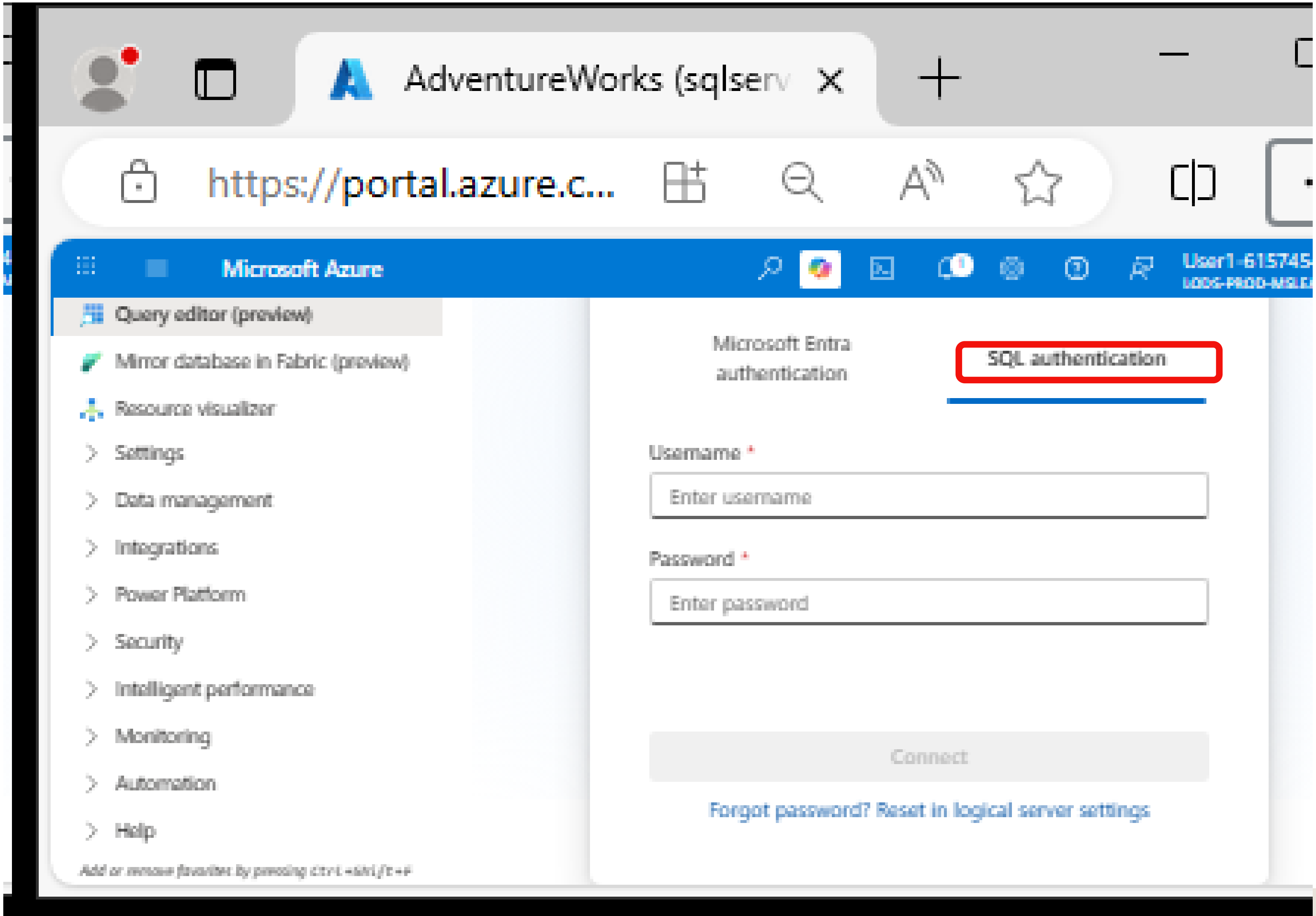
Week 8 Tutorial Activity (Step-by-Step)

Click on Query Editor (preview)



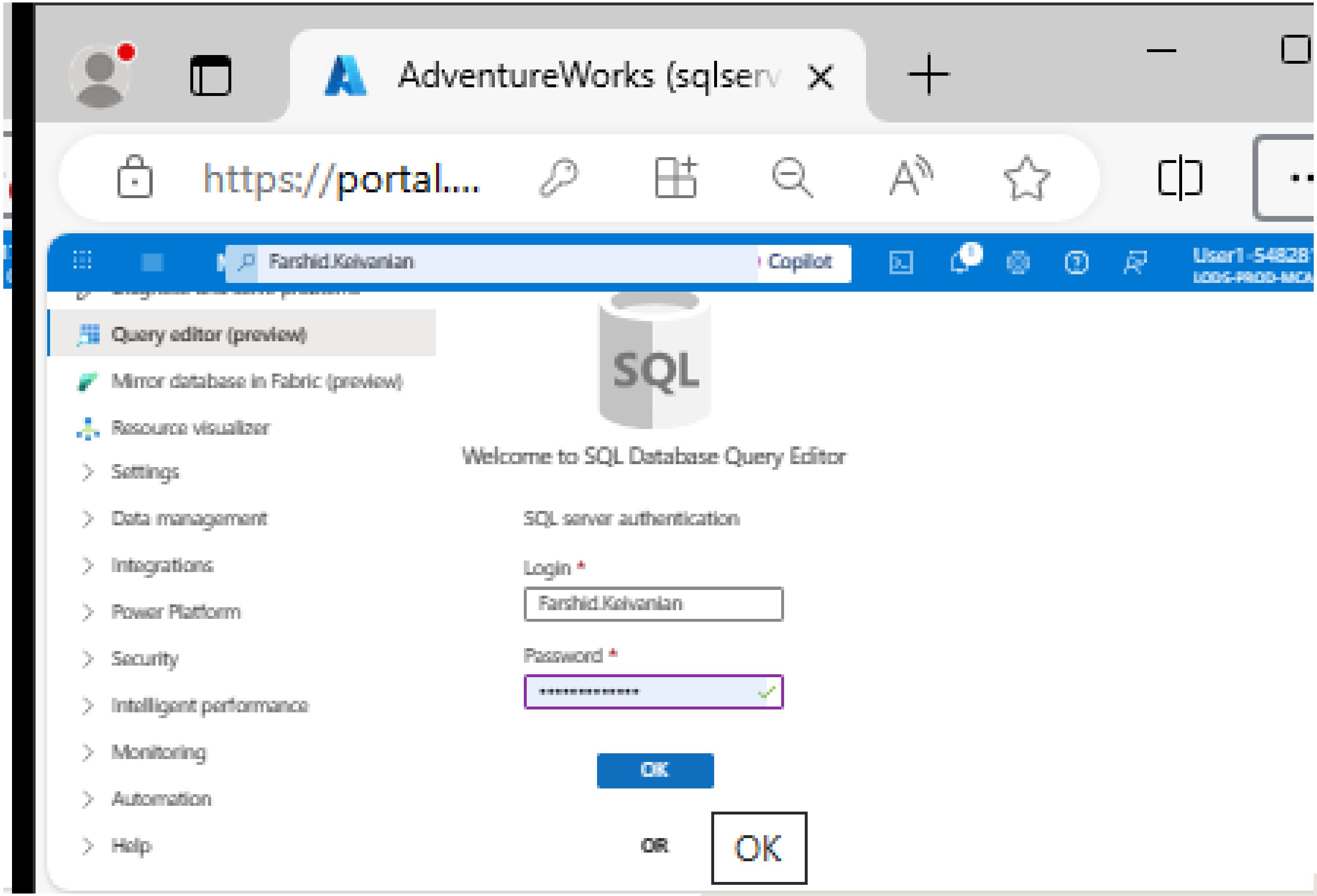
Week 8 Tutorial Activity (Step-by-Step)

Click SQL authentication



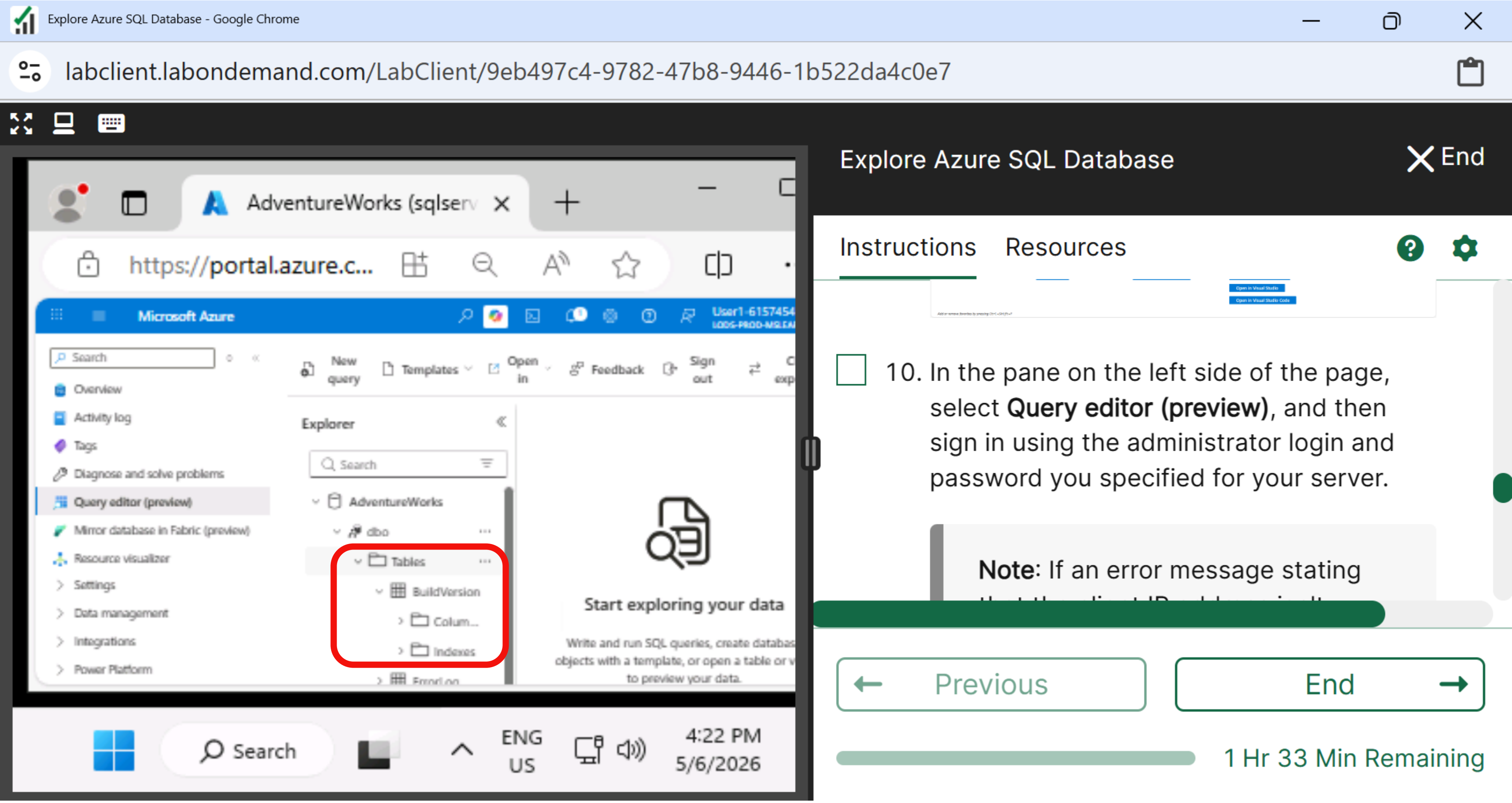
Week 8 Tutorial Activity (Step-by-Step)

Now enter your server administration log in details



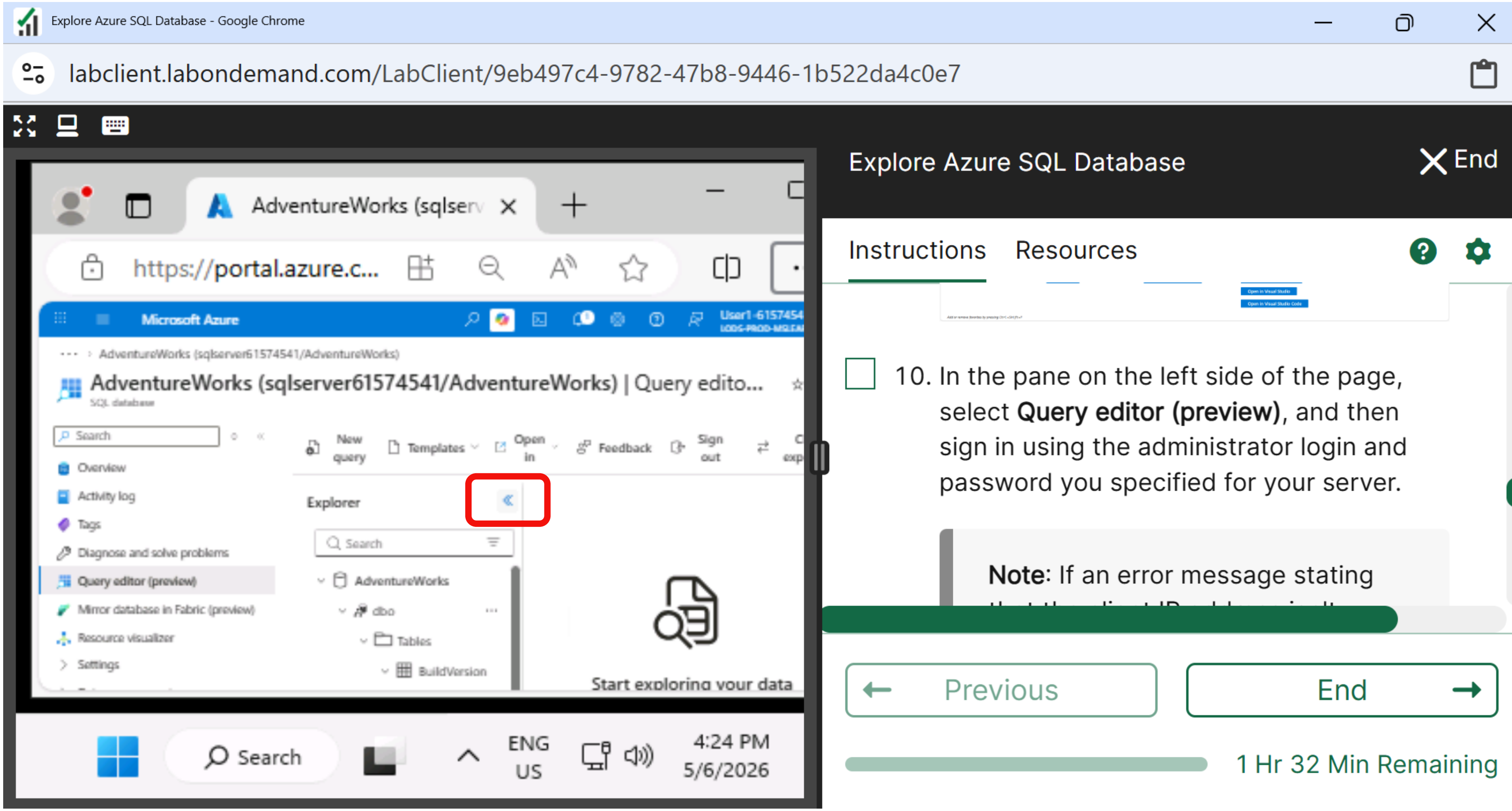
Week 8 Tutorial Activity (Step-by-Step)

You should see the Tables when you expand them



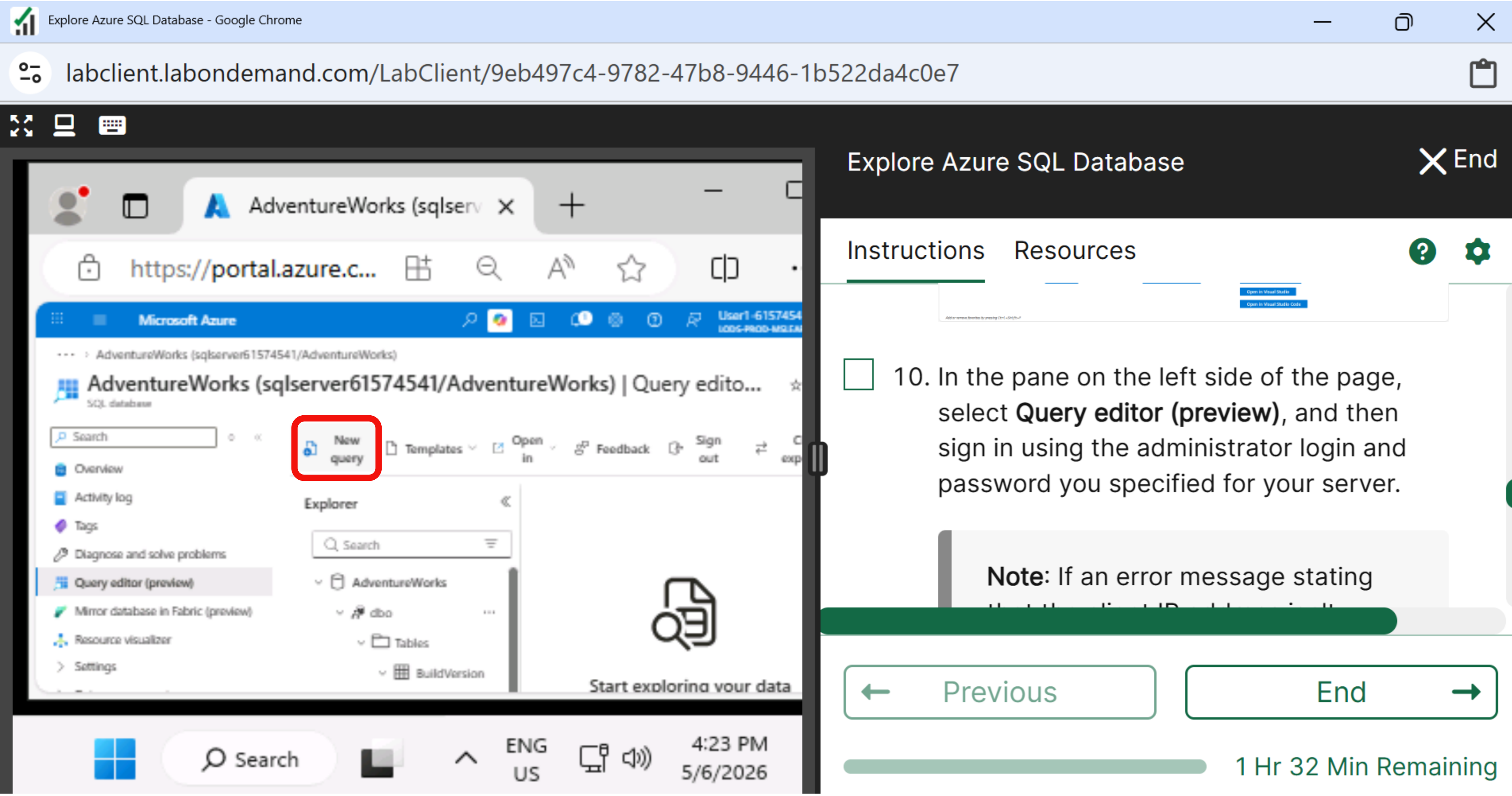
Week 8 Tutorial Activity (Step-by-Step)

Click on <<



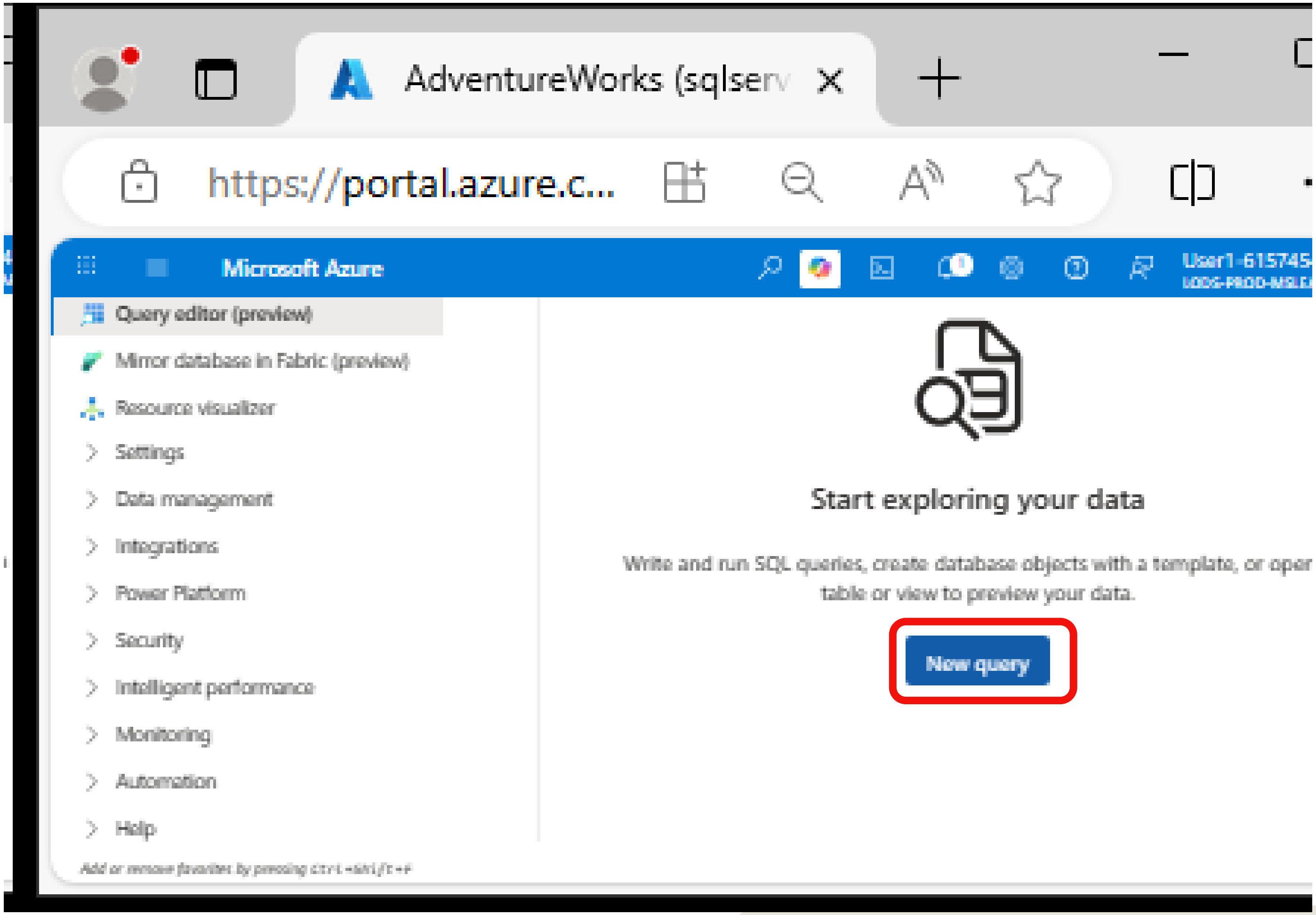
Week 8 Tutorial Activity (Step-by-Step)

Click New Query



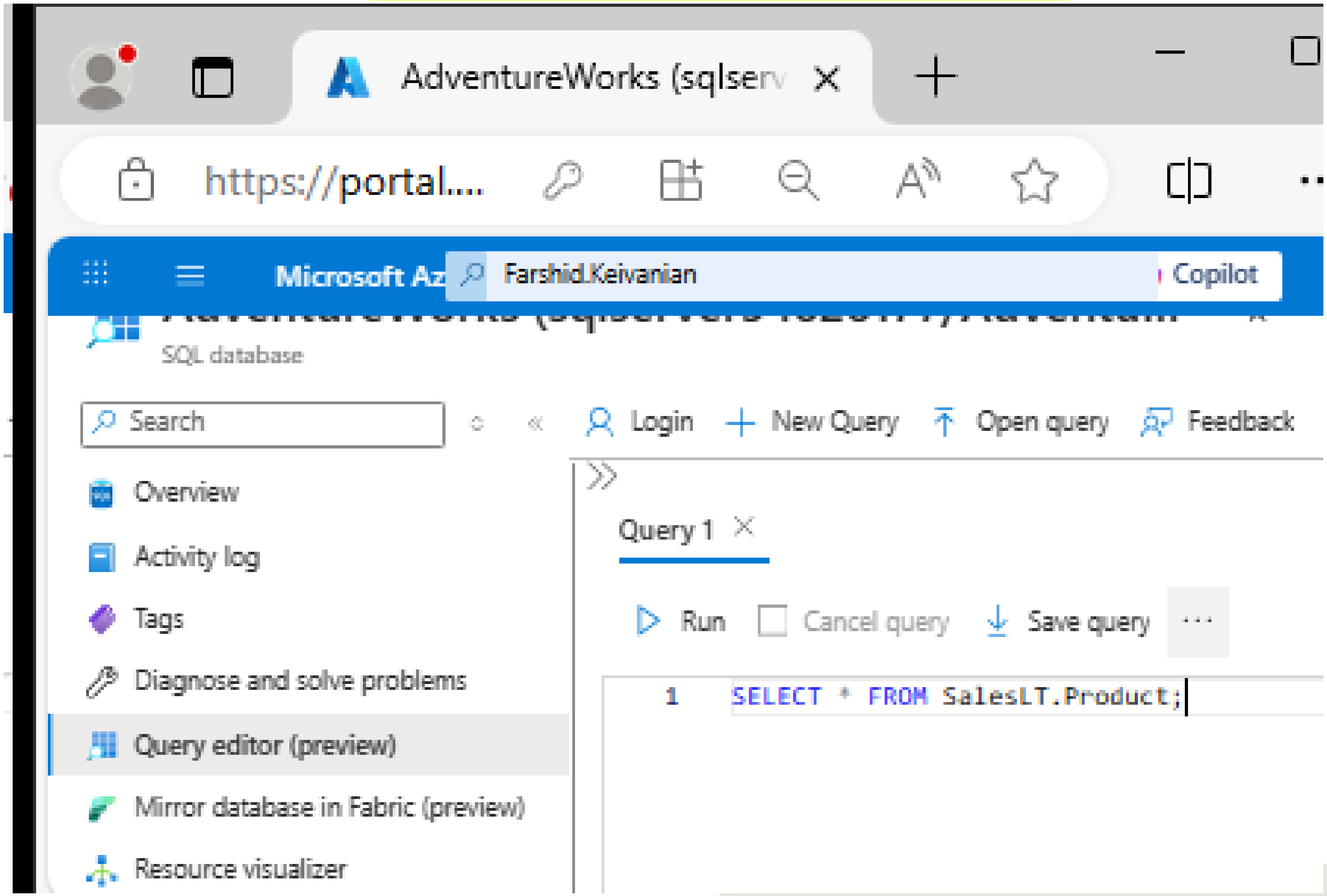
Week 8 Tutorial Activity (Step-by-Step)

Click New Query



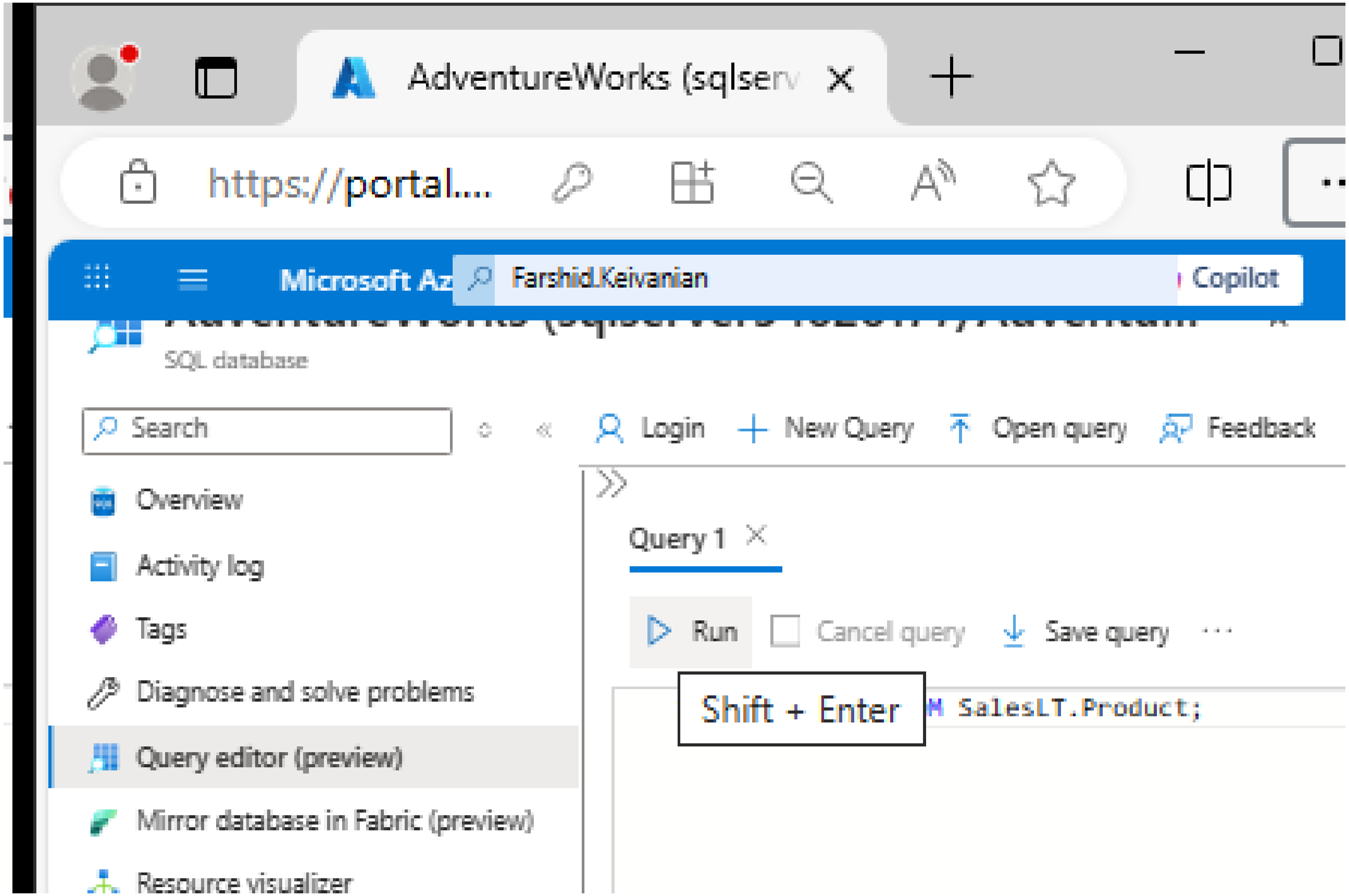
Week 8 Tutorial Activity (Step-by-Step)

Step 12 – Take a screenshot) In the Query 1, enter the following code: (Click on Type on right side panel)



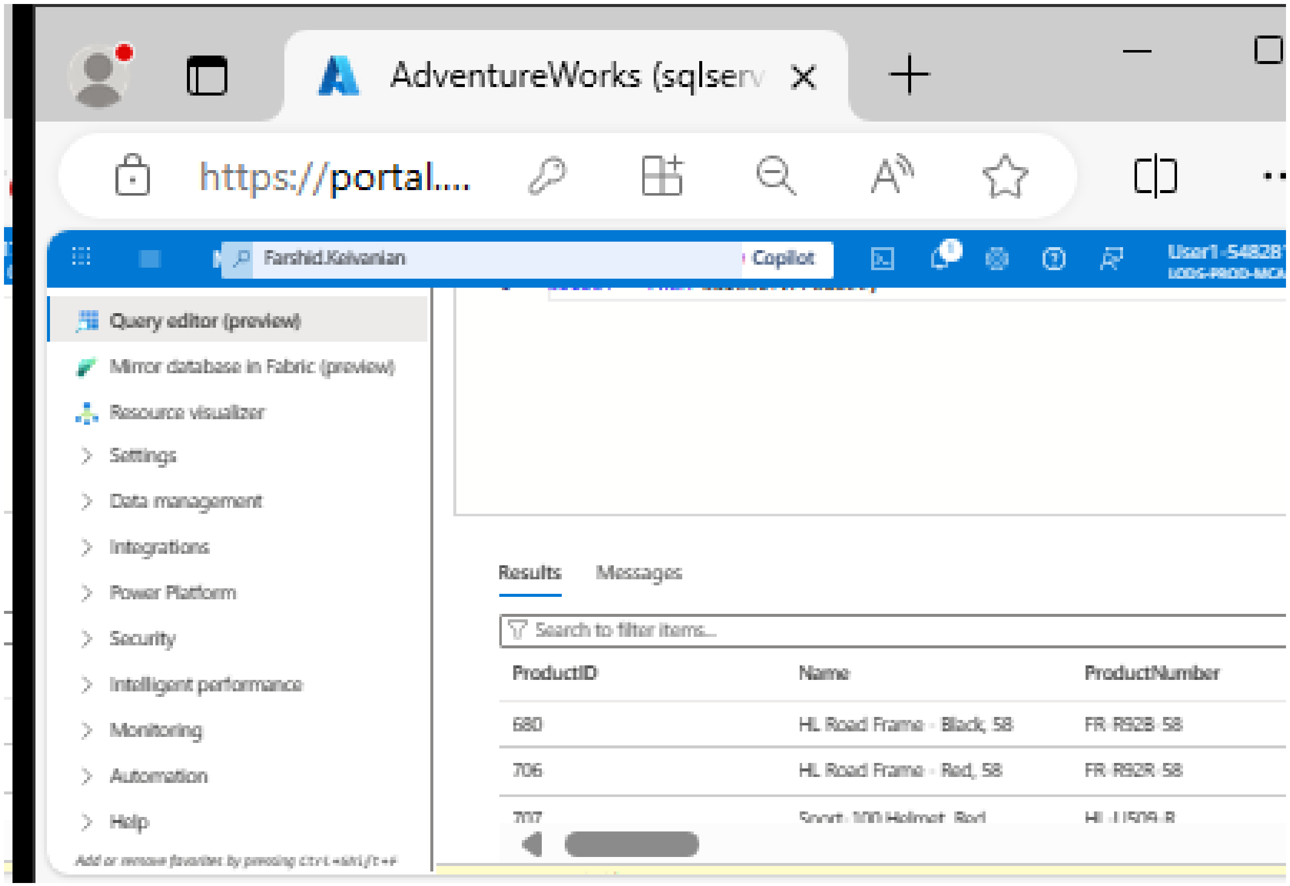
Week 8 Tutorial Activity (Step-by-Step)

Step 12 – Take a screenshot) Shift + Enter (Or click Run)



Week 8 Tutorial Activity (Step-by-Step)

Step 12 – Take a screenshot) It will show all columns for all rows



Week 8 Tutorial Activity (Step-by-Step)

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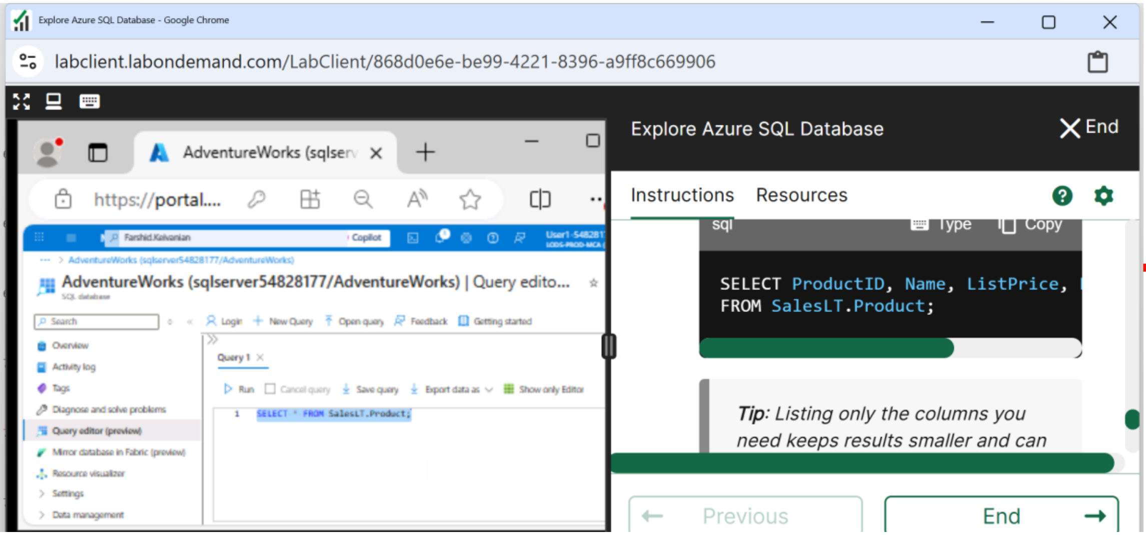
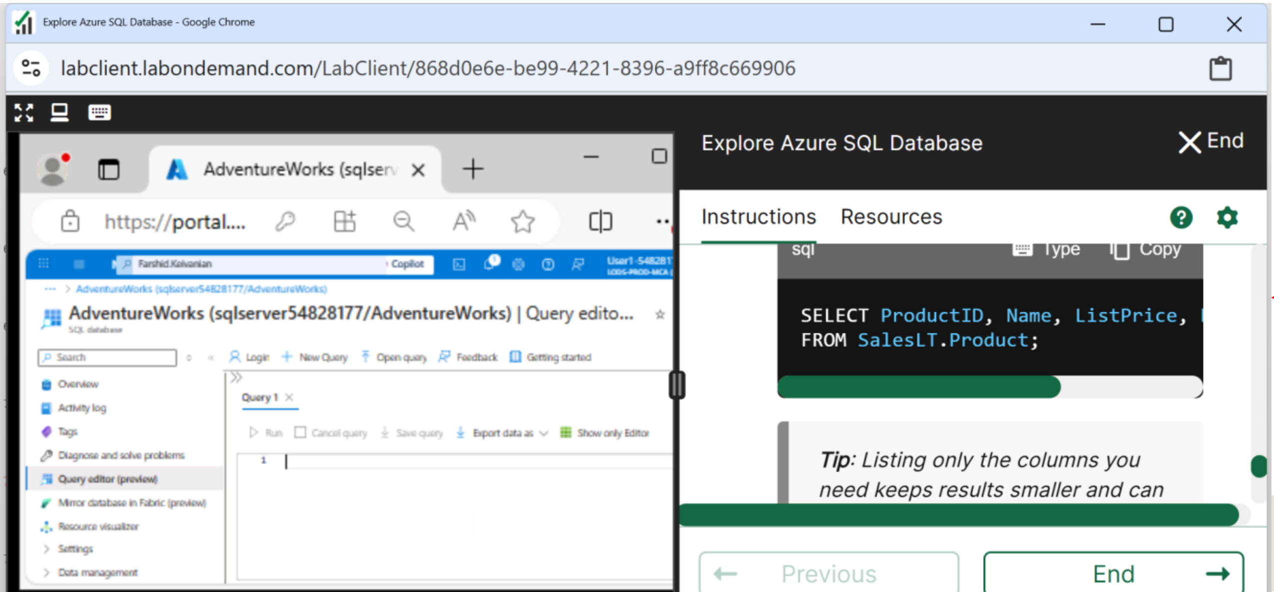
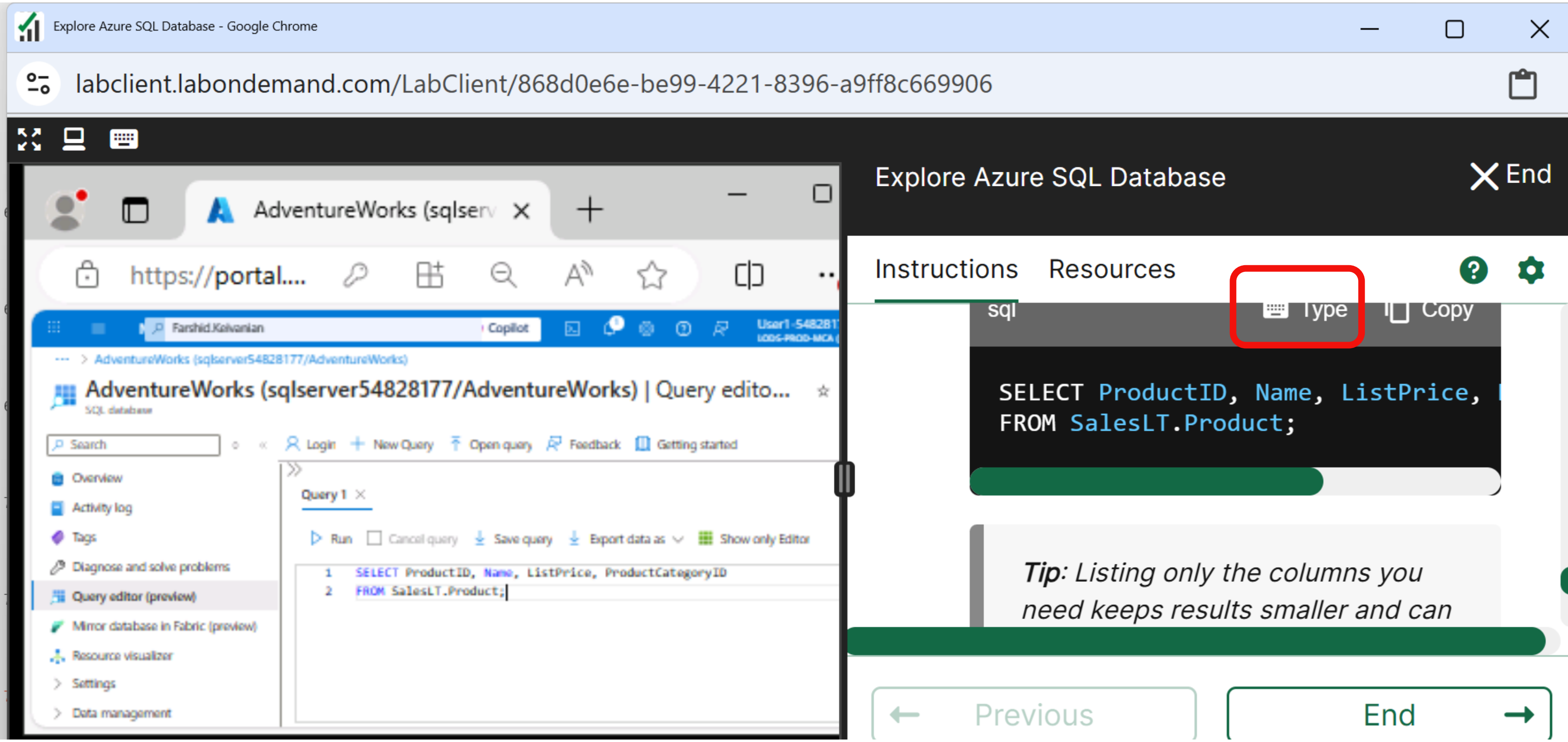
Select All and remove





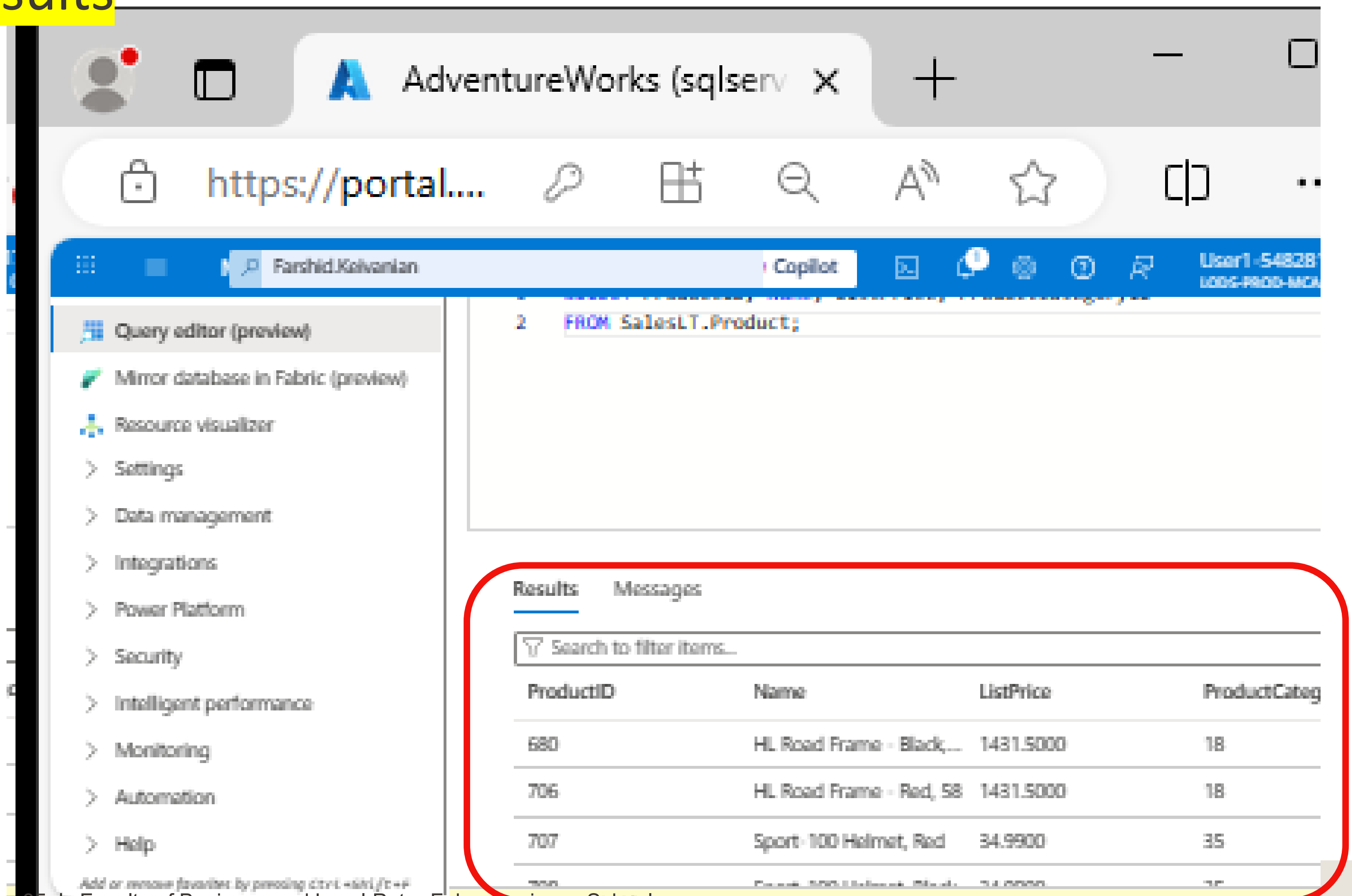
Week 8 Tutorial Activity (Step-by-Step)

Step 13 – Take a screenshot) Click on Type to enter this Query



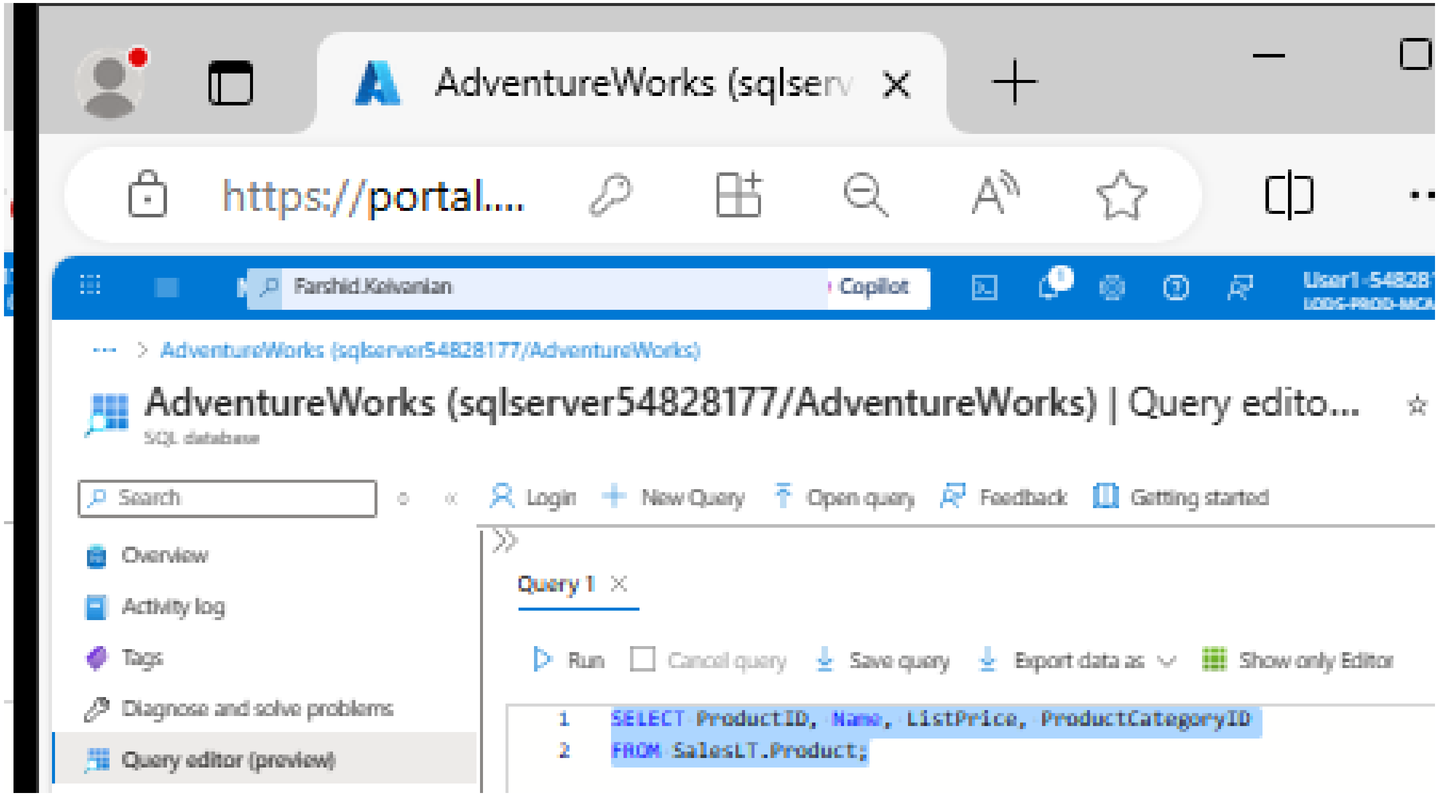
Week 8 Tutorial Activity (Step-by-Step)

Step 13 – Take a screenshot) You will see the results



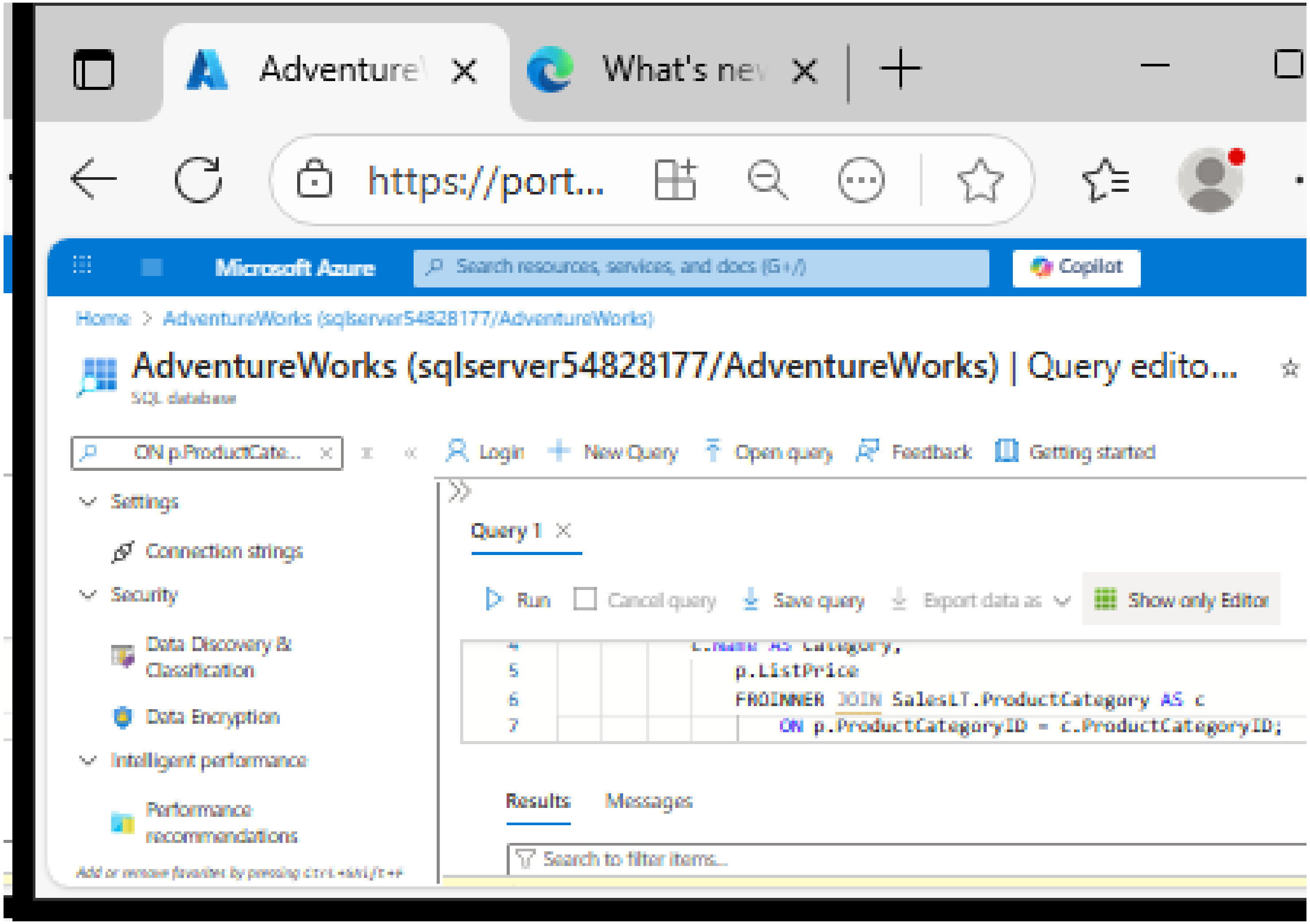
Week 8 Tutorial Activity (Step-by-Step)

Select all and remove them



Week 8 Tutorial Activity (Step-by-Step)

Step 14 – Take a screenshot) Click Type and Run



Week 8 Tutorial Activity (Step-by-Step)

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Step 14 – Take a screenshot) Click Type and Run

```
SELECT
    p.ProductID,
    p.Name AS ProductName,
    c.Name AS Category,
    p.ListPrice
FROM SalesLT.Product AS p
INNER JOIN SalesLT.ProductCategory AS c
    ON p.ProductCategoryID = c.ProductCategoryID;
```

What this query is doing

It is combining two tables to show product details along with their category names.

Week 8 Tutorial Activity (Step-by-Step)

Step 14 – Take a screenshot) Click Type and Run

The two tables involved

1. Product table (SalesLT.Product)

ProductID	Name	ProductCategoryID	ListPrice
1	Bike	10	500
2	Helmet	20	50
3	Gloves	20	25

Step 14 – Take a screenshot) Click Type and Run

2. Category table (SalesLT.ProductCategory)

ProductCategoryID	Name
10	Vehicles
20	Accessories

Week 8 Tutorial Activity (Step-by-Step)

Step 14 – Take a screenshot) Click Type and Run

What INNER JOIN does

It matches rows from both tables where:

p.ProductCategoryID = c.ProductCategoryID

In simple words:

- Find the category ID in the Product table
- Match it with the same ID in the Category table
- Combine the information into one row

Week 8 Tutorial Activity (Step-by-Step)

Step 14 – Take a screenshot) Click Type and Run

```sql
SELECT
    p.ProductID,
    p.Name AS ProductName,
    c.Name AS Category,
    p.ListPrice
FROM SalesLT.Product AS p
INNER JOIN SalesLT.ProductCategory AS c
    ON p.ProductCategoryID = c.ProductCategoryID;
```

Choose these columns:

- Product ID
- Product name (renamed as ProductName)
- Category name (from the other table)
- Price

Week 8 Tutorial Activity (Step-by-Step)

Step 14 – Take a screenshot) Click Type and Run

```
SELECT
    p.ProductID,
    p.Name AS ProductName,
    c.Name AS Category,
    p.ListPrice
FROM SalesLT.Product AS p
INNER JOIN SalesLT.ProductCategory AS c
    ON p.ProductCategoryID = c.ProductCategoryID;
```

We are asking the database to show these columns:

- p.ProductID → the product's ID number
- p.Name AS ProductName → the product's name (renamed to ProductName so it's clearer)
- c.Name AS Category → the category name (renamed to Category)
- p.ListPrice → the product's price

Week 8 Tutorial Activity (Step-by-Step)

```
SELECT
    p.ProductID,
    p.Name AS ProductName,
    c.Name AS Category,
    p.ListPrice
FROM SalesLT.Product AS p
INNER JOIN SalesLT.ProductCategory AS c
    ON p.ProductCategoryID = c.ProductCategoryID;
```

Start with the Product table

Give it a short name p (alias)

Look in the Product table (short name = p).

Week 8 Tutorial Activity (Step-by-Step)

```
SELECT
    p.ProductID,
    p.Name AS ProductName,
    c.Name AS Category,
    p.ListPrice
FROM SalesLT.Product AS p
INNER JOIN SalesLT.ProductCategory AS c
    ON p.ProductCategoryID = c.ProductCategoryID;
```

Bring in the Category table

Give it alias c

Week 8 Tutorial Activity (Step-by-Step)

```sql
SELECT
    p.ProductID,
    p.Name AS ProductName,
    c.Name AS Category,
    p.ListPrice
FROM SalesLT.Product AS p
INNER JOIN SalesLT.ProductCategory AS c
    ON p.ProductCategoryID = c.ProductCategoryID;
```

Match rows where both tables have the same category ID

Week 8 Tutorial Activity (Step-by-Step)

```
SELECT
    p.ProductID,
    p.Name AS ProductName,
    c.Name AS Category,
    p.ListPrice
FROM SalesLT.Product AS p
INNER JOIN SalesLT.ProductCategory AS c
    ON p.ProductCategoryID = c.ProductCategoryID;
```

Combine the Product table (p) with the ProductCategory table (c).

- INNER JOIN means: only show rows where the ProductCategoryID matches in both tables.
- This lets us see each product together with its category name.

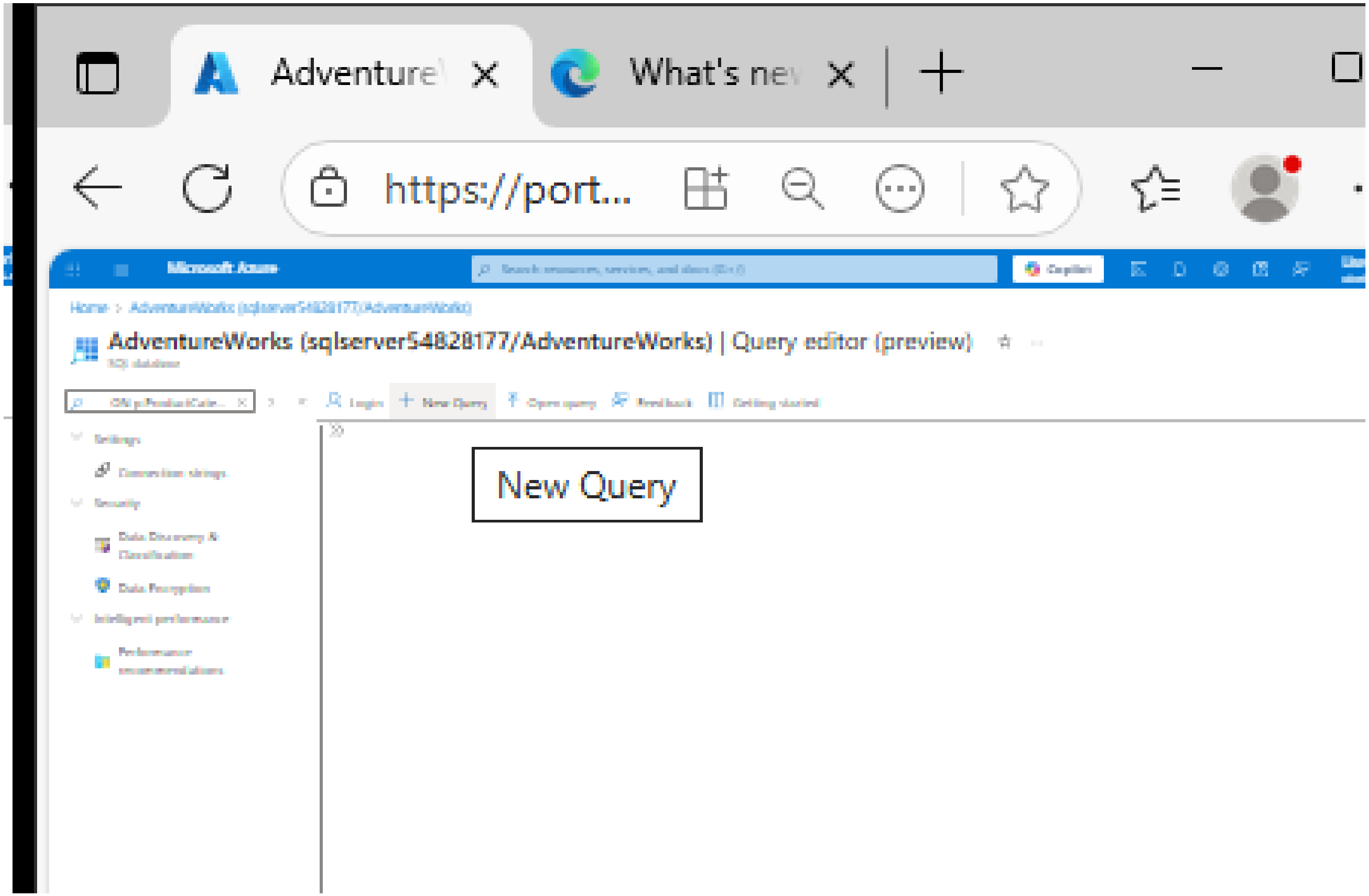
Week 8 Tutorial Activity (Step-by-Step)

```
SELECT
    p.ProductID,
    p.Name AS ProductName,
    c.Name AS Category,
    p.ListPrice
FROM SalesLT.Product AS p
INNER JOIN SalesLT.ProductCategory AS c
    ON p.ProductCategoryID = c.ProductCategoryID;
```

In summary, show me each product, its price, and the name of its category by matching the category IDs from both tables.

Week 8 Tutorial Activity (Step-by-Step)

New Query



Week 8 Tutorial Activity (Step-by-Step)

- Query 1: Write a new query to retrieve the CustomerID, Title, last name, and Company Name of Customers whose Customer ID is in the following list: (1, 3, 5, 20, 45), sorted by Customer last name in reverse alphabetical order.

- Scroll down to reach the "Explore Azure SQL Database" lab. Then, click on the 'Launch' button.

Explore Azure SQL Database (Expected Duration 2 hours, 15 minutes)
DP-900T00-A Introduction to Microsoft Azure Data [Cloud Slice Provided], Learning Path 02 (CSS)

Launch Cancel
10 of 10 launch attempts remaining

- The Virtual Lab will begin loading in a new window (Screenshot below). This will take 1-2 minutes. Follow the on-screen instructions in the right-hand side window of the virtual lab to log in and complete the tasks as instructed. **In step 3, use your Student ID+name as the "database name" .** **Make sure to use the resource group/server name and region provided on the right-hand side window.**
- Take a screenshot after completing steps 12, 13 and 14.
- Query 1: Write a new query to retrieve the CustomerID, Title, last name, and Company Name of Customers whose Customer ID is in the following list: (1, 3, 5, 20, 45), sorted by Customer last name in reverse alphabetical order.
- Query 2: Write a new query to retrieve a list of sales order revisions

Week 8 Tutorial Activity (Step-by-Step)

Query 1 – Take a screenshot) Type this

```
SELECT CustomerID, Title, LastName, CompanyName
FROM SalesLT.Customer
WHERE CustomerID IN (1, 3, 5, 20, 45)
ORDER BY LastName DESC;
```

SELECT

Choose these columns:

- CustomerID
- Title
- LastName
- CompanyName

Week 8 Tutorial Activity (Step-by-Step)

Query 1 – Take a screenshot) Type this

```
SELECT CustomerID, Title, LastName, CompanyName
FROM SalesLT.Customer
WHERE CustomerID IN (1, 3, 5, 20, 45)
ORDER BY LastName DESC;
```

WHERE CustomerID IN (1, 3, 5, 20, 45)

This means:

- Only include customers with these IDs
- Ignore all others

Week 8 Tutorial Activity (Step-by-Step)

Query 1 – Take a screenshot) Type this

```
SELECT CustomerID, Title, LastName, CompanyName
FROM SalesLT.Customer
WHERE CustomerID IN (1, 3, 5, 20, 45)
ORDER BY LastName DESC;
```

Selected rows with CustomerID: 1, 3, 5, 20, 45

ORDER BY LastName DESC;

Sort by LastName

- DESC = descending (Z → A)

Week 8 Tutorial Activity (Step-by-Step)

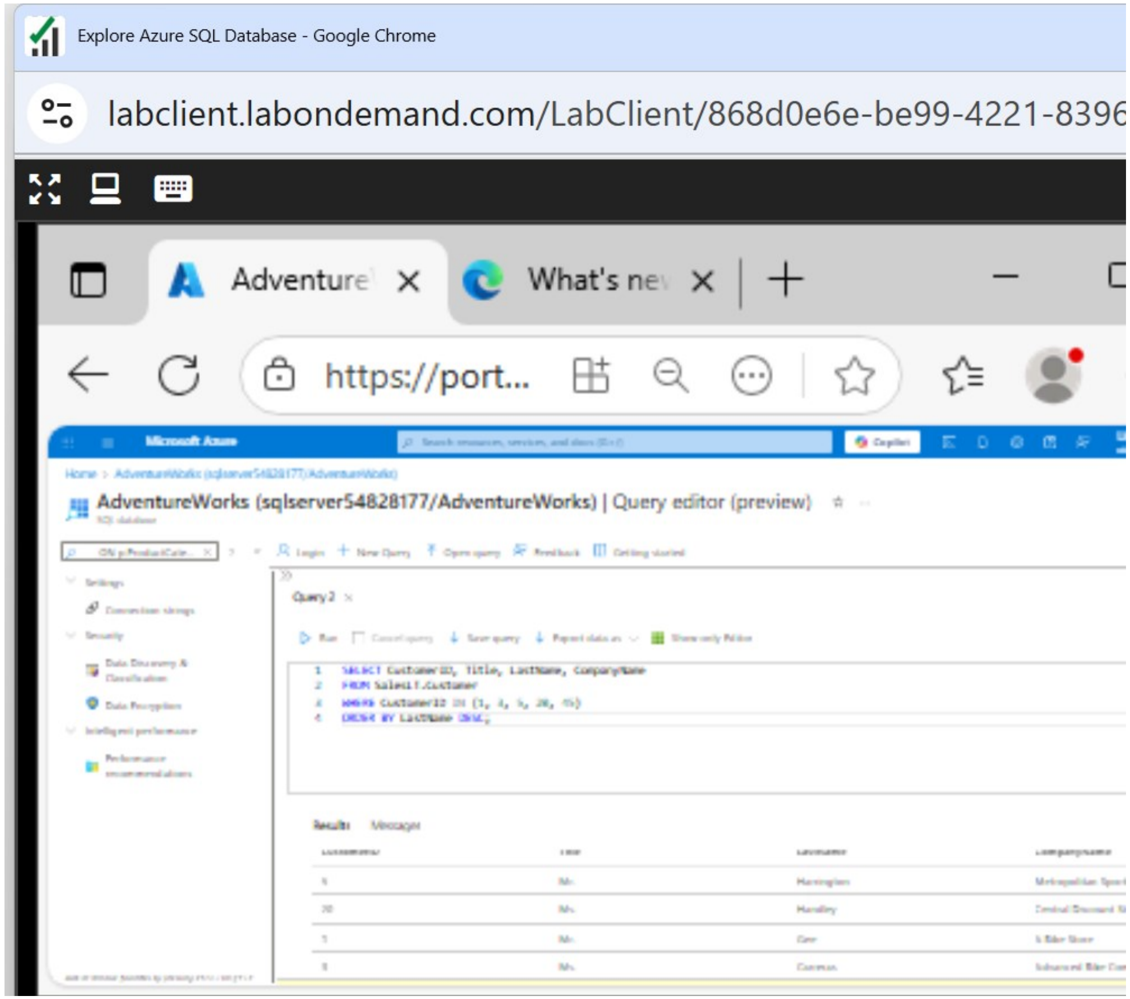
Query 1 – Take a screenshot) Type this

```
SELECT CustomerID, Title, LastName, CompanyName
FROM SalesLT.Customer
WHERE CustomerID IN (1, 3, 5, 20, 45)
ORDER BY LastName DESC;
```

In summary, show specific customers (IDs 1, 3, 5, 20, 45) and sort them by last name in reverse alphabetical order.

Week 8 Tutorial Activity (Step-by-Step)

Query 1 – Take a screenshot) Run and See the results



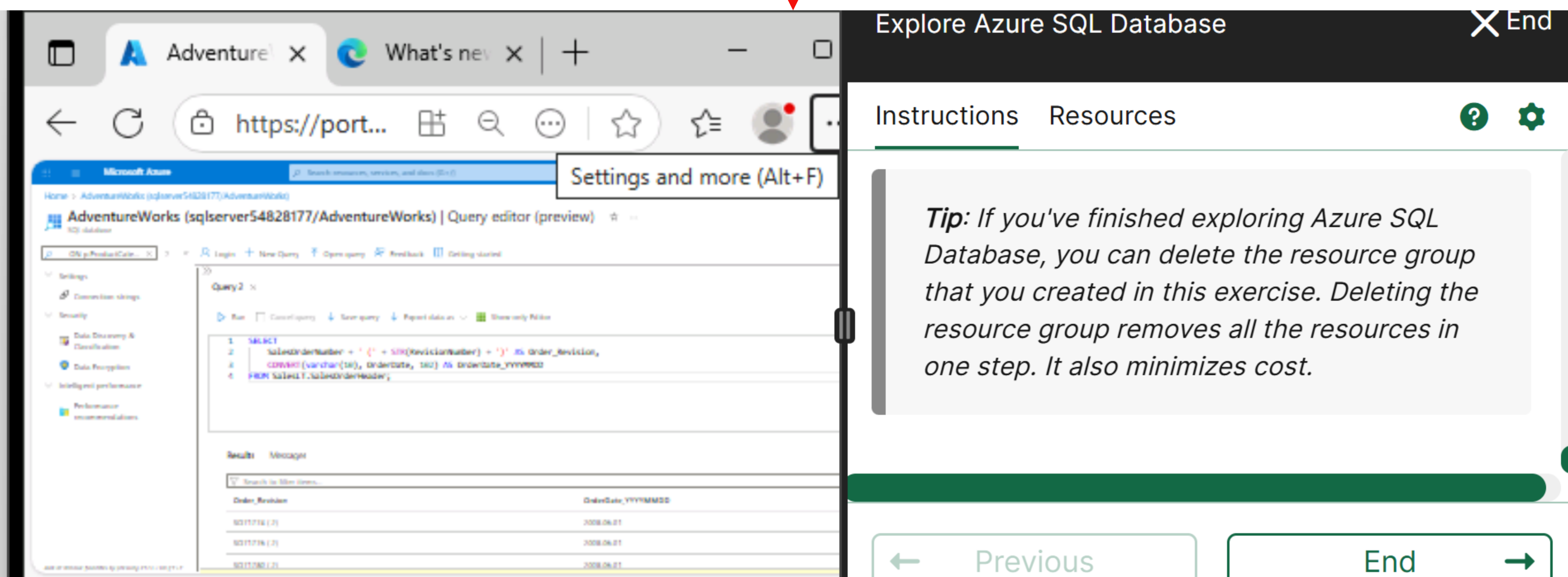
Week 8 Tutorial Activity (Step-by-Step)

Query 2: Write a new query to retrieve a list of sales order revisions

- The SalesLT.SalesOrderHeader table contains records of sales orders. You have been asked to retrieve data for a report that shows:
  - The sales order number and revision number in the format () – for example, SO71774 (2).
  - The order date converted to ANSI standard *102* format (*yyyy.mm.dd* – for example, *2015.01.31*).
  - Hint: Use the STR function to convert the number data type to String.
  - Take a screenshot showing both the SQL code of the query and its results.

Week 8 Tutorial Activity (Step-by-Step)

```sql
SELECT
    SalesOrderNumber + ' (' + STR(RevisionNumber) + ')' AS Order_Revision,
    CONVERT(varchar(10), OrderDate, 102) AS OrderDate_YYYYMMDD
FROM SalesLT.SalesOrderHeader;
```



Week 8 Tutorial Activity (Step-by-Step)

```
SELECT
    SalesOrderNumber + ' (' + STR(RevisionNumber) + ')' AS Order_Revision,
    CONVERT(varchar(10), OrderDate, 102) AS OrderDate_YYYYMMDD
FROM SalesLT.SalesOrderHeader;
```

What this query does

It:

1. Combines order number + revision number into one column
2. Formats the date into a specific style (YYYY.MM.DD)

Week 8 Tutorial Activity (Step-by-Step)

```
SELECT
    SalesOrderNumber + ' (' + STR(RevisionNumber) + ')' AS Order_Revision,
    CONVERT(varchar(10), OrderDate, 102) AS OrderDate_YYYYMMDD
FROM SalesLT.SalesOrderHeader;
```

Step 1: Creating Order + Revision

SalesOrderNumber + ' (' + STR(RevisionNumber) + ')' AS Order_Revision

What's happening:

- Take SalesOrderNumber → e.g. SO123
- Add " ("

Week 8 Tutorial Activity (Step-by-Step)

```sql
SELECT
    SalesOrderNumber + ' (' + STR(RevisionNumber) + ')' AS Order_Revision,
    CONVERT(varchar(10), OrderDate, 102) AS OrderDate_YYYYMMDD
FROM SalesLT.SalesOrderHeader;
```

- Convert RevisionNumber (number → text) using STR()
- Add ")"

Week 8 Tutorial Activity (Step-by-Step)

```
SELECT
    SalesOrderNumber + ' (' + STR(RevisionNumber) + ')' AS Order_Revision,
    CONVERT(varchar(10), OrderDate, 102) AS OrderDate_YYYYMMDD
FROM SalesLT.SalesOrderHeader;
```

Step 2: Formatting the date

CONVERT(varchar(10), OrderDate, 102) AS

OrderDate_YYYYMMDD

This means:

- Convert the date into text (varchar)
- Use format 102

Week 8 Tutorial Activity (Step-by-Step)

```
SELECT
    SalesOrderNumber + ' (' + STR(RevisionNumber) + ')' AS Order_Revision,
    CONVERT(varchar(10), OrderDate, 102) AS OrderDate_YYYYMMDD
FROM SalesLT.SalesOrderHeader;
```

Format 102 =

YYYY.MM.DD

Example:

- 2026-04-20 → 2026.04.20

Week 8 Tutorial Activity (Step-by-Step)

```
SELECT
    SalesOrderNumber + ' (' + STR(RevisionNumber) + ')' AS Order_Revision,
    CONVERT(varchar(10), OrderDate, 102) AS OrderDate_YYYYMMDD
FROM SalesLT.SalesOrderHeader;
```

First column

Join order number and revision number into one string

Second column

Change the date into a readable text format (YYYY.MM.DD)

FROM

Get data from the SalesOrderHeader table

Week 8 Tutorial Activity (Step-by-Step)

```
SELECT
    SalesOrderNumber + ' (' + STR(RevisionNumber) + ')' AS Order_Revision,
    CONVERT(varchar(10), OrderDate, 102) AS OrderDate_YYYYMMDD
FROM SalesLT.SalesOrderHeader;
```

In summary, show each order as ‘OrderNumber (Revision)’ and display the date in YYYY.MM.DD format.

Week 8 Tutorial Activity (Step-by-Step)

```
SELECT
    SalesOrderNumber + ' (' + STR(RevisionNumber) + ')' AS Order_Revision,
    CONVERT(varchar(10), OrderDate, 102) AS OrderDate_YYYYMMDD
FROM SalesLT.SalesOrderHeader;
```

- **SalesOrderNumber + ' (' + STR(RevisionNumber) + ')':**
  - Joins the order number with the revision number.
  - Example: SO71774 (2).
  - STR() changes the RevisionNumber (a number) into text so it can be joined.
  - Renamed as Order_Revision.

Week 8 Tutorial Activity (Step-by-Step)

```
SELECT
    SalesOrderNumber + ' (' + STR(RevisionNumber) + ')' AS Order_Revision,
    CONVERT(varchar(10), OrderDate, 102) AS OrderDate_YYYYMMDD
FROM SalesLT.SalesOrderHeader;
```

- **CONVERT(varchar(10), OrderDate, 102):**
  - Converts the date into the format yyyy.mm.dd.
  - Example: 2015.01.31.
  - Renamed as OrderDate_YYYYMMDD.

Week 8 Tutorial Activity (Step-by-Step)

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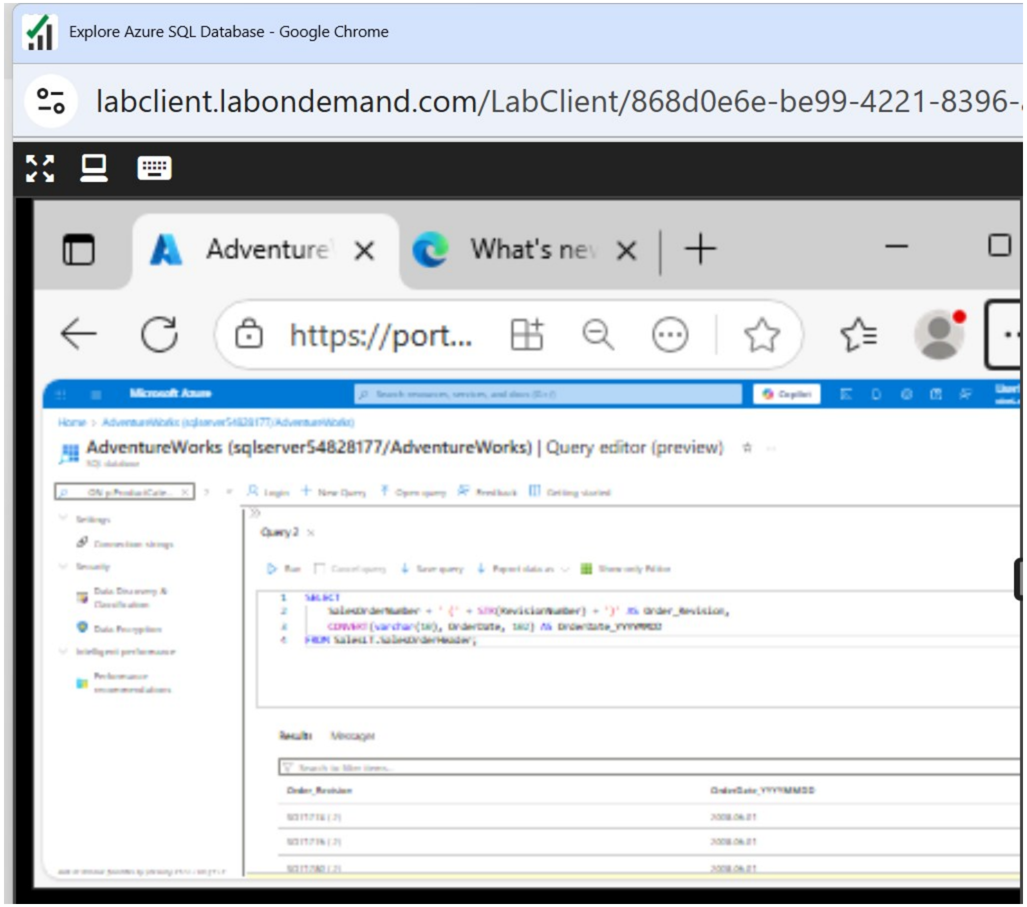
```
SELECT
    SalesOrderNumber + ' (' + STR(RevisionNumber) + ')' AS Order_Revision,
    CONVERT(varchar(10), OrderDate, 102) AS OrderDate_YYYYMMDD
FROM SalesLT.SalesOrderHeader;
```

- **FROM SalesLT.SalesOrderHeader:**
  - Tells SQL to pull the data from the SalesLT.SalesOrderHeader table.

Week 8 Tutorial Activity (Step-by-Step)



The result should be two columns:

1. _YYYYMOrder_Revision (e.g., SO71774 (2))
2. OrderDateMDD (e.g., 2015.01.31)

Week 8 Tutorial Activity (Step-by-Step)

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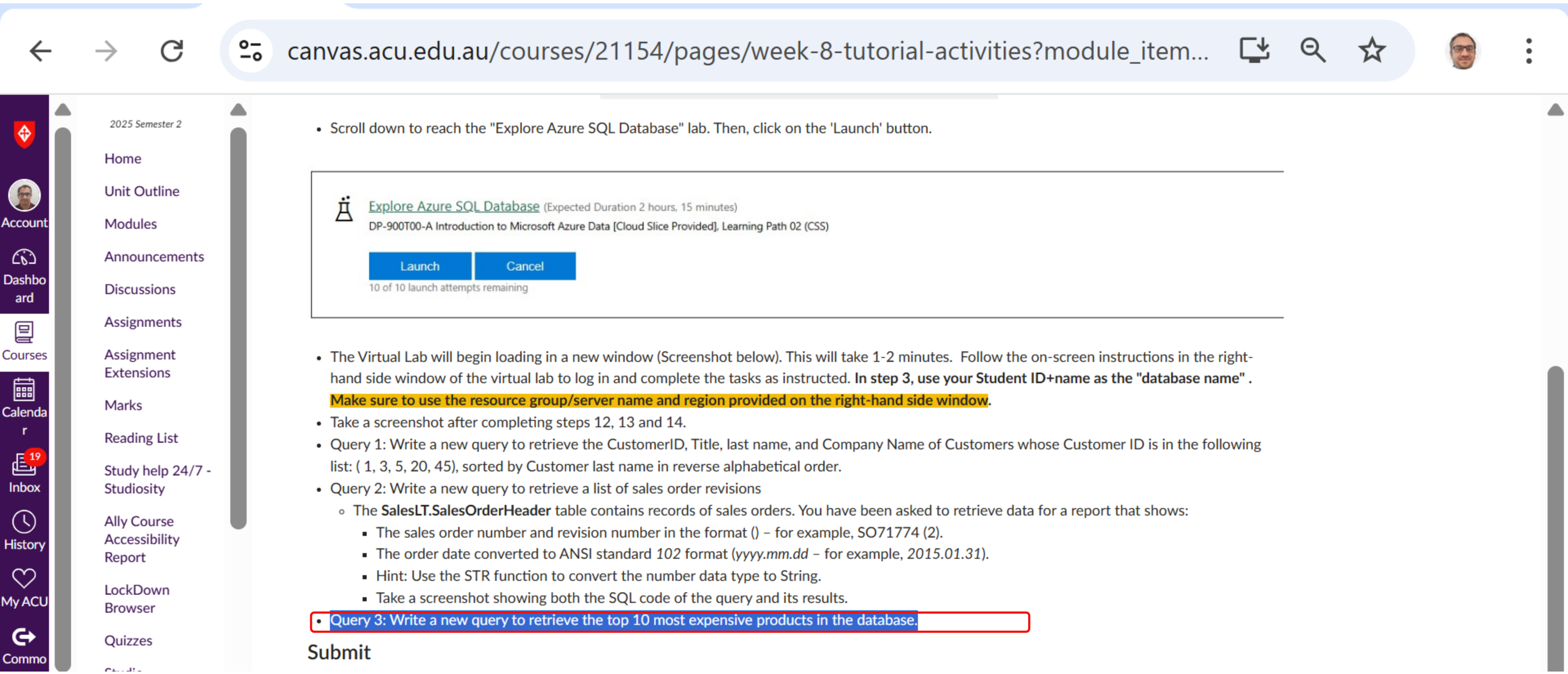
Query 3: Write a new query to retrieve the top 10 most expensive products in the database.



Week 8 Tutorial Activity (Step-by-Step)

Query 3 – Take a screenshot: Write a new query to retrieve the top 10 most expensive products in the database.

```
SELECT TOP (10)
    ProductID,
    Name,
    ListPrice
FROM SalesLT.Product
WHERE ListPrice IS NOT NULL
ORDER BY ListPrice DESC;
```

What this query does

It shows the 10 most expensive products (that actually have a price).

Week 8 Tutorial Activity (Step-by-Step)

Query 3 – Take a screenshot: Write a new query to retrieve the top 10 most expensive products in the database.

```
SELECT TOP (10)
    ProductID,
    Name,
    ListPrice
FROM SalesLT.Product
WHERE ListPrice IS NOT NULL
ORDER BY ListPrice DESC;
```

Step 1: WHERE clause

WHERE ListPrice IS NOT NULL

This means:

- Ignore products with no price (NULL)
- Only keep products that have a price

Week 8 Tutorial Activity (Step-by-Step)

Query 3 – Take a screenshot: Write a new query to retrieve the top 10 most expensive products in the database.

```
SELECT TOP (10)
    ProductID,
    Name,
    ListPrice
FROM SalesLT.Product
WHERE ListPrice IS NOT NULL
ORDER BY ListPrice DESC;
```

Step 2: ORDER BY

ORDER BY ListPrice DESC;

Sort by price:

- DESC = highest → lowest

Week 8 Tutorial Activity (Step-by-Step)

Query 3 – Take a screenshot: Write a new query to retrieve the top 10 most expensive products in the database.

```
SELECT TOP (10)
    ProductID,
    Name,
    ListPrice
FROM SalesLT.Product
WHERE ListPrice IS NOT NULL
ORDER BY ListPrice DESC;
```

Step 3: TOP (10)

SELECT TOP (10)

- Only show the first 10 rows
- Since we sorted highest first, this gives:
- Top 10 most expensive products

Week 8 Tutorial Activity (Step-by-Step)

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Query 3 – Take a screenshot: Write a new query to retrieve the top 10 most expensive products

```
SELECT TOP (10)
    ProductID,
    Name,
    ListPrice
FROM SalesLT.Product
WHERE ListPrice IS NOT NULL
ORDER BY ListPrice DESC;
```

SELECT TOP (10)

Only show the first 10 results

WHERE

Ignore products with no price

ORDER BY DESC

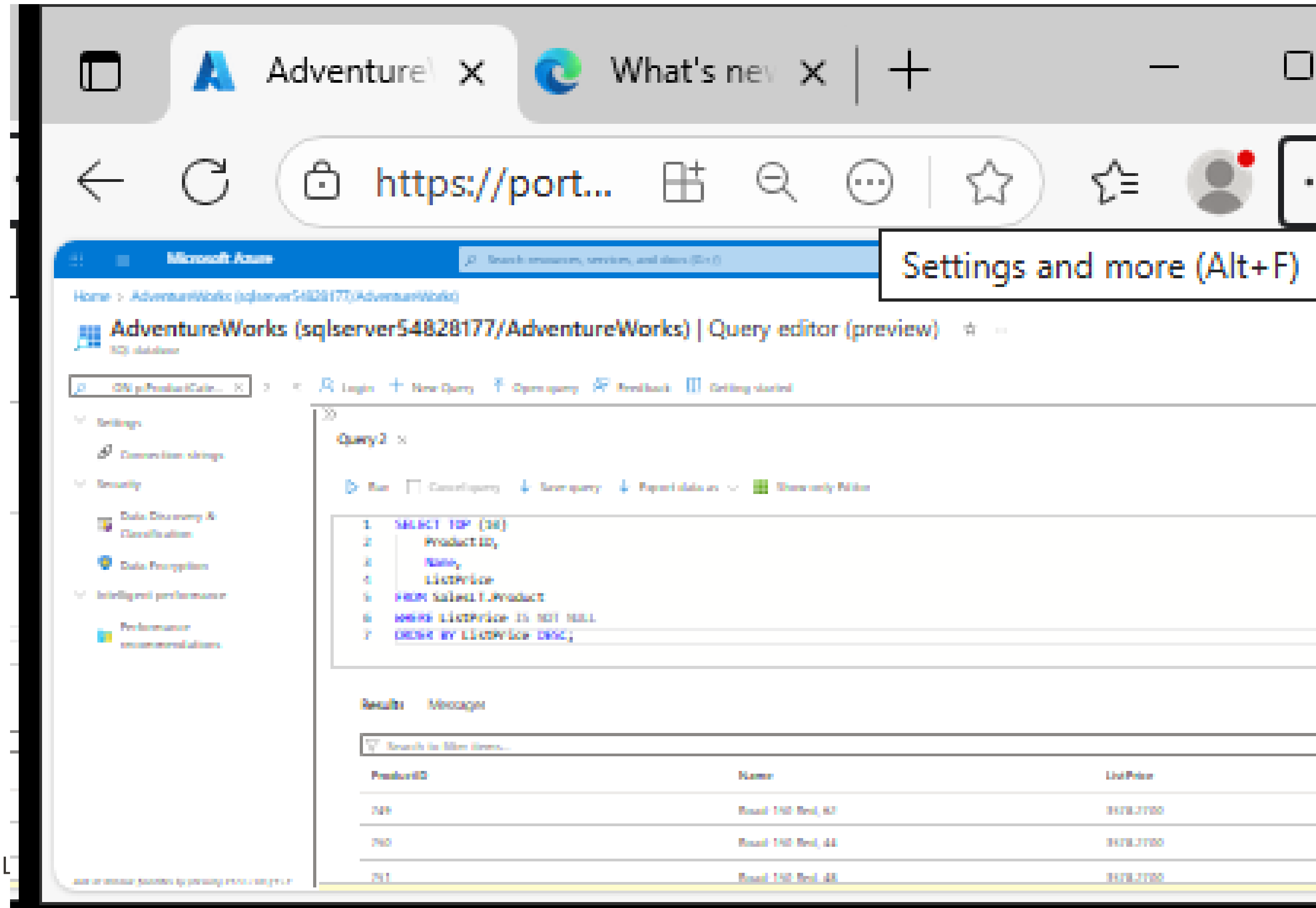
Sort from most expensive to cheapest

One-line summary

Show the 10 most expensive products that have a price.

Week 8 Tutorial Activity (Step-by-Step)

The results lists the 10 most expensive products in the database, starting with the highest price at the top. (Take a screenshot)



Week 8 Tutorial Activity (Step-by-Step)

Submit

Paste the screenshots of completing steps 12, 13 and 14, Query 1, Query 2, and Query 3 (6 screenshots) in a single Word/pdf file and submit using the assignment module 8 submission link. The screenshots must show both the SQL code of the query and its results.

Week 8 Tutorial Activity (Step-by-Step)

Quick fixes (common issues)

- "Customers table not found" → Use the right name: SalesLT.Customer (not Customers)
- Login fails in Query editor → Add your client IP in Networking
- No sample tables → You likely missed AdventureWorksLT; use SELECT 1; or create a simple table to test

Week 8 Tutorial Activity (Step-by-Step)

What the answers should show (in words)

- Q1: Only those 5 CustomerIDs, sorted by LastName descending
- Q2: Orders with their RevisionNumber (and a neat date), or a count per revision
- Q3: The 10 highest ListPrice products, from most to less expensive

Review & Reflection

- **What did we learn?**
  - Difference: Access vs Azure SQL.
  - Provision DBs in the cloud.
  - Write SQL queries.
- **Why important?**
  - Skills → Azure Fundamentals Certification.
  - Real jobs in data engineering, DB admin.

Discussion & Role-Play Questions

1. **Q:** Why can't Woolworths or Qantas run their operations on MS Access?

   A: Access can't handle thousands of concurrent users & terabytes of data.

2. **Q:** Debate: Should Australian universities keep data on campus servers or move to Azure Cloud?

   For: Cloud = secure, scalable, cheaper long-term.

   Against: Privacy risks, ongoing cost, internet dependency.

Discussion & Role-Play Questions

3. **Q:** If Azure SQL is PaaS, what control do we lose vs IaaS?

   A: We lose full control of the server (patching, OS-level settings).

Thank You

- *Have a Great Learning Day!*
- Feel free to reach out with any questions!
- Dr. Farshid Keivanian